Student Hub Portal

Practical File 1 to 6

HTML CSS JavaScript JSON and supporting source code

Questions answers analysis interpretation and complete code

Practical No. 1 - Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

Question: Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure, README file, and GitHub repository.

Key Questions

Q: What is URL and parts of URL?

Answer: A URL is the address used to open a resource on the internet. Its main parts are the protocol such as https, domain name, path, query string, and fragment.

Q: How does HTML file processed in web browser?

Answer: The browser reads the HTML file, builds the DOM tree, loads linked CSS and JavaScript files, and then displays the final page on the screen.

Q: How will page navigation flow managed among all HTML pages?

Answer: Navigation is managed with anchor tags and relative links such as index.html, about.html, events.html, and register.html. A common navigation bar keeps the flow consistent.

Q: How will GitHub commits be maintained after each practical?

Answer: After completing and testing each practical, the changed files should be added, committed with a clear message, and pushed to the GitHub repository.

Analysis

A: The Student Hub project is divided into clear pages for home, information, events, support, authentication, student details, and administration. Separate HTML, CSS, and JavaScript files keep the work organized.

Interpretation

I: A sitemap and wireframe give the project a clear direction before coding starts. This reduces confusion about page connections and makes later GitHub commits easier to understand.

Project File Structure

```
index.html
about.html
events.html
faq.html
contact.html
login.html
register.html
dashboard.html
profile.html
admin.html
style.css
script.js
```

Practical No. 2 - Semantic HTML5 Pages with Accessibility-Ready Structure

Question: Develop static HTML5 skeletons for at least 10 StudentHub pages such as Home, About, Register, Login, Dashboard, Events, Profile, Contact, Admin, FAQ, and Feedback. Use semantic tags and accessibility-friendly structure.

Key Questions

Q: Are semantic tags such as header, nav, main, section, article, aside, and footer used properly?

Answer: Yes. The pages use header for branding, nav for links, main for page content, sections or cards for grouped information, and footer for common bottom links.

Q: Are form labels, headings, image alt text, and navigation links accessible?

Answer: The form fields have labels, the pages use headings, the logo has alt text, and navigation is provided through normal links. These features help keyboard and screen-reader users understand the page.

Q: Is the page structure consistent across all pages?

Answer: Yes. The pages follow a common pattern with a header, navigation menu, main content area, footer, and linked CSS and JavaScript files.

Analysis

A: The project has ten linked HTML pages with a common layout. Semantic elements make the structure easier for browsers, developers, and assistive technologies to understand.

Interpretation

I: The pages form a consistent static website foundation. A user can move from the home page to information, events, registration, login, and support pages without losing the common site structure.

HTML Code - index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Home</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- 1. Notification Banner -->
  <div id="notification-banner" class="notification-banner">
    <span>📢 <strong>Notice:</strong> Tech Fest 2026 registration is now
open!</span>
    <button class="banner-close" onclick="closeBanner()">&times;</button>
  </div>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
```

```
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <!-- 2. Light / Dark Theme Switcher -->
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <!-- 3. Hamburger Menu & Navigation -->
  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main>
    <div class="main">
      <h1>Student Hub Portal</h1>
      <p>Manage Students &amp;amp; College Events Easily</p>
      <br>
      <!-- 4. Modal Popup Trigger -->
      <button onclick="openModal('welcomeModal')" class="btn-announcement">📢 View
Announcement</button>
    </div>

    <!-- 5. Content / Announcement Slider -->
    <div class="slider-container">
      <div class="slide">
        <h3>🎉 Welcome New Students!</h3>
        <p>Orientation program starts this Monday at 10:00 AM in the
Auditorium.</p>
      </div>
      <div class="slide">
        <h3>⏳ Annual Sports Meet 2026</h3>
        <p>Register your sports teams before October 20th to participate.</p>
      </div>
      <div class="slide">
        <h3>💡 Hackathon Submissions Open</h3>
        <p>Submit your software project ideas on the dashboard before Friday.</p>
      </div>
      <div class="slider-controls">
        <button class="slider-btn" onclick="changeSlide(-1)">❮ Prev</button>
        <button class="slider-btn" onclick="changeSlide(1)">Next ❯</button>
      </div>
    </div>

    <br>

    <div style="display: flex; gap: 20px;" class="boxes-container">
      <div class="box">
        <h3 style="font-size:24px; text-decoration: underline;">Latest Notice</h3>
        <ul>
          <li class="t1">Semester Exam Schedule Released</li>
```

```
          <li class="t1">Holiday on Friday</li>
          <li class="t1">Library Books Return Date</li>
        </ul>
      </div>

      <div class="box">
        <h3 style="font-size:24px; text-decoration: underline;">Upcoming
Events</h3>
        <ul>
          <li class="t1">Tech Fest - Oct 25</li>
          <li class="t1">AI Workshop - Nov 10</li>
          <li class="t1">Career Seminar - Dec 02</li>
        </ul>
      </div>

      <div class="box">
        <h3 style="font-size:24px; text-decoration: underline;">Quick Links</h3>
        <ul>
          <li class="t1links"><a href="dashboard.html">Student Portal</a></li>
          <li class="t1links"><a href="events.html">Events</a></li>
          <li class="t1links"><a href="faq.html">FAQs</a></li>
        </ul>
      </div>
    </div>
  </main>

  <!-- 6. Modal Popup Structure -->
  <div id="welcomeModal" class="modal">
    <div class="modal-box">
      <button class="modal-close"
onclick="closeModal('welcomeModal')">&times;</button>
      <h2 style="color: #EC9F05; margin-bottom: 10px;">Important Announcement</h2>
      <p style="margin-bottom: 15px;">Welcome to the Student Hub Portal! Check the
Events section for upcoming workshops and register for campus activities.</p>
      <button onclick="closeModal('welcomeModal')">Got it!</button>
    </div>
  </div>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <!-- JavaScript File -->
  <script src="script.js"></script>
</body>

</html>
```

HTML Code - about.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Student Hub Portal - About</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>
  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main style="max-width: 900px; margin: 0 auto; text-align: center;">
    <h1 style="font-size: 32px; margin-bottom: 20px;">About Student Hub</h1>

    <div style="display: flex; gap: 20px;" class="boxes-container">
      <div class="boxes2">
        <p style="font-size: 26px; font-weight: bold;">Our Mission</p>
        <p style="font-size: 18px;">To eliminate communication gaps across campus
by delivering real-time updates and effortless management tools for every
student.</p>
      </div>
      <div class="boxes2">
        <p style="font-size: 26px; font-weight: bold;">Our Vision</p>
        <p style="font-size: 18px;">To build a modern, fully digital campus
ecosystem where information is accessible anywhere, anytime with zero operational
friction.</p>
      </div>
    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>
```

```
</html>
```

HTML Code - events.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Events (Practical 6)</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🌓 Theme</button>
      <button><a href="login.html">Logout</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html" style="background-color: rgba(255, 255, 255, 0.2);
color: #DAF264; font-weight: bold;">Events</a>
      <a href="dashboard.html">Dashboard</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main style="max-width: 1150px; width: 100%; margin: 20px auto; padding: 10px;">
    <h1 style="font-size: 32px; text-align: center; margin-bottom: 6px;">College
Events</h1>
    <p style="text-align: center; color: #CBD5E1; margin-bottom: 24px; font-size:
15px;">
      Explore upcoming campus workshops, sports competitions, and cultural
festivals (Fetched dynamically from external JSON)
    </p>

    <!-- Practical 6: Search, Category Filter, and Sorting Controls Toolbar -->
    <div class="table1" id="events-toolbar-card" style="padding: 18px 24px; margin-
bottom: 24px;">
      <div class="events-toolbar">

        <!-- Live Search Input -->
        <div class="search-box-wrap">
          <label for="search" style="font-weight: bold; color: #EC9F05; white-
space: nowrap;">Search:</label>
```

```
          <input type="text" id="search" oninput="applyEventFilters()"
placeholder="Search by title, venue, category...">
        </div>

        <!-- Filter, Sort and Reset Controls -->
        <div class="controls-group">
          <!-- Category Filter -->
          <label for="categoryFilter" style="font-weight: bold; color: #EC9F05;
font-size: 14px; white-space: nowrap;">Category:</label>
          <select id="categoryFilter" onchange="applyEventFilters()">
            <option value="ALL">All Categories</option>
            <option value="SPORTS">Sports</option>
            <option value="TECH">Technology</option>
            <option value="WORKSHOP">Workshops</option>
            <option value="CULTURAL">Cultural</option>
            <option value="ACADEMIC">Academic</option>
          </select>

          <!-- Sorting Dropdown -->
          <label for="sortOrder" style="font-weight: bold; color: #EC9F05; font-
size: 14px; white-space: nowrap;">Sort by:</label>
          <select id="sortOrder" onchange="applyEventFilters()">
            <option value="date-asc">Date (Earliest First)</option>
            <option value="date-desc">Date (Latest First)</option>
            <option value="name-asc">Name (A &rarr; Z)</option>
            <option value="name-desc">Name (Z &rarr; A)</option>
          </select>

          <!-- Reset Filter Button -->
          <button onclick="resetEventFilters()" style="padding: 9px 18px; font-
size: 14px;">Reset</button>
        </div>

      </div>
    </div>

    <!-- Dynamic Events List Container (Populated via Fetch API) -->
    <div id="events-list">
      <!-- Dynamically generated event cards appear here -->
    </div>

    <!-- Practical 6 Pagination Controls -->
    <div id="pagination-controls" class="pagination-container">
      <!-- Dynamically generated page buttons appear here -->
    </div>

  </main>

  <!-- Universal Event Registration Modal -->
  <div id="dynamicEventModal" class="modal">
    <div class="modal-box">
      <button class="modal-close"
onclick="closeModal('dynamicEventModal')">&times;</button>
      <h2 id="modalEventTitle" style="color: #EC9F05; margin-bottom: 12px; font-
size: 22px;">Register for Event</h2>
      <p id="modalEventDetails" style="margin-bottom: 20px; color: #CBD5E1; font-
size: 15px;"></p>
      <button onclick="confirmEventRegistration()" style="padding: 10px 28px; font-
size: 15px;">Confirm Registration</button>
    </div>
  </div>
```

```
  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

HTML Code - faq.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - FAQ</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="dashboard.html">Dashboard</a>
      <a href="faq.html" style="background-color: rgba(255, 255, 255, 0.2); color:
#DAF264; font-weight: bold;">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main style="max-width: 1150px; width: 100%; margin: 20px auto; padding: 10px;">
    <h1 style="font-size: 32px; text-align: center; margin-bottom: 6px;">Frequently
Asked Questions</h1>
    <p style="text-align: center; color: #CBD5E1; margin-bottom: 24px; font-size:
15px;">
```

```
      Quick answers to common questions about Student Hub Portal, events, and
academics
    </p>

    <!-- 2-Column FAQ Layout -->
    <div class="dash-cards-flex" style="align-items: stretch; gap: 24px;">

      <!-- Category 1: Portal & Student Account -->
      <div class="table1" style="flex: 1; padding: 26px;">
        <h2 style="font-size: 22px; color: #EC9F05; margin-bottom: 12px; border-
bottom: 1.5px solid #6B4EFD; padding-bottom: 8px;">
          🎓 Portal &amp;amp; Student Account
        </h2>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. How do I register as a student?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>Click on the 'Register' button in the top menu, fill in your details
(Name, Email, Course, Semester), accept the terms, and submit your
registration.</p>
          </div>
        </div>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. How do I log in to my dashboard?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>Go to the Login page, enter your registered email address and
password, then click the 'Login' button to open your student dashboard.</p>
          </div>
        </div>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. What is my Student ID format?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>For 3rd Semester Computer Science students, IDs follow the format
<strong>25DCS200</strong>, visible in your profile and dashboard.</p>
          </div>
        </div>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. Where can I see my enrolled subjects?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>Your Dashboard features a dedicated Sem 3 section listing your 5
registered subjects: WDF, FCN, DSA, Discrete Maths, and Java OOP.</p>
          </div>
        </div>
      </div>

      <!-- Category 2: Events & Campus Life -->
```

```
      <div class="table1" style="flex: 1; padding: 26px;">
        <h2 style="font-size: 22px; color: #EC9F05; margin-bottom: 12px; border-
bottom: 1.5px solid #6B4EFD; padding-bottom: 8px;">
          🖩 Events &amp;amp; Campus Notices
        </h2>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. How can I join college events?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>Navigate to the Events page, browse available workshops or
competitions, and click 'Register' for your selected event to reserve your entry
slot.</p>
          </div>
        </div>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. Where do I find exam dates and notices?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>Official circulars, exam dates, and assignment deadlines are
published on the Home page notice board and on your Dashboard.</p>
          </div>
        </div>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. Are workshop seats limited?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>Yes, hands-on technical workshops (like Cyber Security, AI &amp;
Robotics) have limited lab capacity and are allocated first-come, first-served.</p>
          </div>
        </div>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. How do I contact the college admin?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>You can use the Contact page form or send an email to
<strong>support@studenthub.com</strong> to get in touch directly with the
administration.</p>
          </div>
        </div>
      </div>

    </div>

    <!-- Bottom Help Card -->
    <div class="table1" style="padding: 22px; text-align: center; margin-top:
24px;">
      <h3 style="color: #EC9F05; margin-bottom: 6px;">Still have questions?</h3>
      <p style="color: #CBD5E1; margin-bottom: 16px; font-size: 15px;">
```

```
        Can't find the answer you are looking for? Reach out to our campus support
team.
      </p>
      <a href="contact.html" style="display: inline-block; background-color:
#EC9F05; color: white; padding: 10px 24px; border-radius: 20px; font-weight: bold;
text-decoration: none;">Contact Support &rarr;</a>
    </div>

  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

HTML Code - contact.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Contact &amp; Feedback</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html" style="background-color: rgba(255, 255, 255, 0.2);
color: #DAF264; font-weight: bold;">Contact</a>
```

```
    </nav>
  </div>

  <main style="max-width: 1150px; width: 100%; margin: 20px auto; padding: 10px;">
    <h1 style="font-size: 32px; text-align: center; margin-bottom: 24px;">Contact
Us &amp; Feedback</h1>

    <div class="dash-cards-flex" style="align-items: stretch; gap: 24px;">

      <!-- Contact Info Card -->
      <div class="table1" style="flex: 1; padding: 30px; display: flex; flex-
direction: column; justify-content: space-between;">
        <div>
          <h2 style="font-size: 24px; color: #EC9F05; margin-bottom: 10px;">Get in
Touch</h2>
          <p style="color: #CBD5E1; font-size: 15px; margin-bottom: 20px; line-
height: 1.5;">
            Have questions about college events, your student profile, or academic
schedules? Reach out to our campus help desk directly.
          </p>
          <hr style="border-color: rgba(107, 78, 253, 0.4); margin: 15px 0 20px
0;">

          <div style="display: flex; flex-direction: column; gap: 16px; font-size:
15px;">
            <p>
              <strong style="color: #EC9F05;">📧 Email :</strong><br>
              <span style="color: #CBD5E1;">support@studenthub.com</span>
            </p>
            <p>
              <strong style="color: #EC9F05;">📞 Phone :</strong><br>
              <span style="color: #CBD5E1;">+1 (234) 567-8900 / +1 (987) 654-
3210</span>
            </p>
            <p>
              <strong style="color: #EC9F05;">📍 Address :</strong><br>
              <span style="color: #CBD5E1;">Campus Center, Student Block 4, Tech
Campus</span>
            </p>
            <p>
              <strong style="color: #EC9F05;">🕒 Office Hours :</strong><br>
              <span style="color: #CBD5E1;">Monday &ndash; Friday: 9:00 AM &ndash;
5:00 PM</span>
            </p>
          </div>
        </div>

        <div style="margin-top: 25px; padding: 14px; background-color: #141E2E;
border-radius: 12px; border-left: 4px solid #EC9F05; font-size: 13px; color:
#CBD5E1;">
          💡 <strong>Tip:</strong> For urgent exam or admit card queries, please
contact the Examination Branch in Block 2.
        </div>
      </div>

      <!-- Feedback Form Card -->
      <div class="table1" id="feedback" style="flex: 1.2; padding: 30px;">
        <h2 style="font-size: 24px; color: #EC9F05; margin-bottom: 10px;">Send a
Message</h2>
        <p style="color: #CBD5E1; font-size: 15px; margin-bottom: 20px;">
```

```
          Fill out the form below and our portal support team will respond
promptly.
        </p>

        <form onsubmit="event.preventDefault(); alert('Thank you! Your message has
been submitted successfully.'); this.reset();">
          <div class="form-group">
            <label for="fb-name">Your Full Name</label>
            <input type="text" id="fb-name" placeholder="e.g. John Doe" required>
          </div>

          <div class="form-group">
            <label for="fb-email">Your Email Address</label>
            <input type="email" id="fb-email" placeholder="e.g.
student@college.edu" required>
          </div>

          <div class="form-group">
            <label for="fb-category">Category</label>
            <select id="fb-category">
              <option value="general">General Inquiry</option>
              <option value="events">College Events &amp;amp; Workshops</option>
              <option value="academic">Academic &amp;amp; Exam Schedule</option>
              <option value="technical">Portal Technical Feedback</option>
            </select>
          </div>

          <div class="form-group">
            <label for="fb-message">Message / Feedback</label>
            <textarea id="fb-message" rows="4" placeholder="Write your message
here..." required></textarea>
          </div>

          <div style="margin-top: 20px; text-align: center;">
            <button type="submit" style="padding: 12px 35px; font-size: 16px; font-
weight: bold;">Send Message 🚀</button>
          </div>
        </form>
      </div>

    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

HTML Code - login.html

```
<!DOCTYPE html>
```

```
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Login</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="register.html">Register</a>
    </nav>
  </div>

  <main>
    <!-- Login Box -->
    <div class="auth-card">
      <h2>LOGIN</h2>

      <div id="login-error-msg" style="color: #FF5252; font-size: 13px; margin-
bottom: 14px; text-align: center; display: none; background: rgba(255, 82, 82,
0.15); padding: 8px; border-radius: 8px;"></div>

      <form onsubmit="handlePortalLogin(event)">
        <div class="form-group">
          <label for="email">Student ID or Email</label>
          <input type="text" id="email" required placeholder="e.g. 25DCS200 or
admin@studenthub.com">
        </div>

        <div class="form-group">
          <label for="password">Password</label>
          <input type="password" id="password" required placeholder="Enter your
password">
        </div>

        <div class="form-options">
          <label for="remember">
            <input type="checkbox" id="remember"> Remember Me
```

```
          </label>
          <a href="#">Forgot Password?</a>
        </div>

        <button type="submit" class="auth-btn">Login &rarr;</button>

        <!-- Credentials Helper Box -->
        <div style="margin-top: 18px; padding: 12px 14px; background-color:
#141E2E; border-radius: 12px; border-left: 3px solid #EC9F05; font-size: 12px;
color: #CBD5E1; text-align: left;">
          <strong style="color: #EC9F05; font-size: 13px;">🔑 Login
Credentials:</strong>

          <div style="display: flex; justify-content: space-between; align-items:
center; margin-top: 8px;">
            <span>&bull; <strong>Student:</strong> <code>25DCS200</code> /
<code>student123</code></span>
            <button type="button" onclick="fillCredentials('25DCS200',
'student123')" style="padding: 3px 8px; font-size: 11px; border-radius: 6px;
background-color: #6B4EFD;">Fill</button>
          </div>

          <div style="display: flex; justify-content: space-between; align-items:
center; margin-top: 6px;">
            <span>&bull; <strong>Admin:</strong> <code>admin@studenthub.com</code>
/ <code>admin123</code></span>
            <button type="button" onclick="fillCredentials('admin@studenthub.com',
'admin123')" style="padding: 3px 8px; font-size: 11px; border-radius: 6px;
background-color: #6B4EFD;">Fill</button>
          </div>
        </div>

        <p class="auth-footer-text">
          Don't have an account? <a href="register.html">Register here</a>
        </p>
      </form>
    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

HTML Code - register.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Student Registration</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🌓 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="login.html">Login</a>
    </nav>
  </div>

  <main>
    <!-- Perfect Student Registration Box -->
    <div class="auth-card register-card">
      <h2>STUDENT REGISTRATION</h2>

      <form id="registrationForm" onsubmit="return validateRegistration(event)">
        <div class="form-grid">
          <!-- 1. Full Name -->
          <div class="form-group">
            <label for="name">Full Name</label>
            <input type="text" id="name" placeholder="e.g. John Doe">
            <small class="error-text" id="nameError"></small>
          </div>

          <!-- 2. Email -->
          <div class="form-group">
            <label for="email">Email Address</label>
            <input type="email" id="email" placeholder="student@college.edu">
            <small class="error-text" id="emailError"></small>
          </div>

          <!-- 3. Mobile Number -->
          <div class="form-group">
            <label for="mobile">Mobile Number</label>
            <input type="tel" id="mobile" placeholder="10-digit mobile number">
            <small class="error-text" id="mobileError"></small>
          </div>

          <!-- 4. Course -->
```

```
          <div class="form-group">
            <label for="course">Course</label>
            <select id="course">
              <option value="">-- Select Course --</option>
              <option value="B.Tech Computer Science">B.Tech Computer
Science</option>
              <option value="B.Tech Information Technology">B.Tech Information
Technology</option>
              <option value="BCA">BCA</option>
              <option value="MCA">MCA</option>
              <option value="B.Sc IT">B.Sc IT</option>
            </select>
            <small class="error-text" id="courseError"></small>
          </div>

          <!-- 5. Year -->
          <div class="form-group">
            <label for="year">Year of Study</label>
            <input type="number" id="year" min="1" max="4" placeholder="Enter Year
(1 - 4)">
            <small class="error-text" id="yearError"></small>
          </div>

          <!-- 6. Gender -->
          <div class="form-group">
            <label>Gender</label>
            <div class="radio-group">
              <label for="male">
                <input type="radio" name="gender" id="male" value="male"> Male
              </label>
              <label for="female">
                <input type="radio" name="gender" id="female" value="female">
Female
              </label>
              <label for="other">
                <input type="radio" name="gender" id="other" value="other"> Other
              </label>
            </div>
            <small class="error-text" id="genderError"></small>
          </div>

          <!-- 7. Password -->
          <div class="form-group">
            <label for="password">Password</label>
            <input type="password" id="password" placeholder="Minimum 6 chars
(letters & numbers)">
            <small class="error-text" id="passwordError"></small>
          </div>

          <!-- 8. Confirm Password -->
          <div class="form-group">
            <label for="confirmPassword">Confirm Password</label>
            <input type="password" id="confirmPassword" placeholder="Re-type
password">
            <small class="error-text" id="confirmPasswordError"></small>
          </div>
        </div>

        <!-- 9. Terms Acceptance -->
        <div class="form-options center-terms">
          <label for="terms">
```

```
            <input type="checkbox" id="terms"> I accept the Terms &amp; Conditions
          </label>
          <small class="error-text" id="termsError"></small>
        </div>

        <button type="submit" class="auth-btn">Register</button>

        <p class="auth-footer-text">
          Already registered? <a href="login.html">Login here</a>
        </p>
      </form>
    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

HTML Code - dashboard.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Dashboard</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- Top Notification Banner -->
  <div id="notification-banner" class="notification-banner">
    <span>📢 <strong>Notice:</strong> Semester Examination Schedule has been
published!</span>
    <button class="banner-close" onclick="closeBanner()">&times;</button>
  </div>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>

    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button onclick="handleLogout()">Logout</button>
```

```
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="dashboard.html" style="background-color: rgba(255, 255, 255, 0.2);
color: #DAF264; font-weight: bold;">Dashboard</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main class="dash-wrapper">
    <div class="dash-main-flex">

      <!-- Sidebar Navigation Column -->
      <div class="table1 dash-sidebar-col">
        <!-- Student Info Header -->
        <div style="text-align: center; padding-bottom: 15px; border-bottom: 1px
solid rgba(107, 78, 253, 0.4);">
          <div style="width: 60px; height: 60px; border-radius: 50%; background-
color: #6B4EFD; color: white; display: flex; align-items: center; justify-content:
center; margin: 0 auto 10px auto; font-size: 20px; font-weight: bold; border: 2px
solid #EC9F05;">
            JD
          </div>
          <h3 style="font-size: 19px; margin-bottom: 3px;">John Doe</h3>
          <p style="font-size: 13px; color: #CBD5E1;">B.Tech CSE - 3rd Sem</p>
          <p style="font-size: 13px; color: #EC9F05; font-weight: bold; margin-top:
2px;">ID: 25DCS200</p>
        </div>

        <h4 style="color: #EC9F05; margin: 16px 0 8px 0; text-align: center; text-
transform: uppercase; font-size: 12px; letter-spacing: 1px;">Navigation</h4>
        <div class="dash-sidebar-menu">
          <a href="index.html" class="dash-nav-btn">Home</a>
          <a href="profile.html" class="dash-nav-btn">My Profile</a>
          <a href="events.html" class="dash-nav-btn">Events</a>
          <a href="dashboard.html" class="dash-nav-btn active">Dashboard</a>
          <a href="contact.html" class="dash-nav-btn">Contact Admin</a>
          <button onclick="handleLogout()" class="dash-nav-btn logout-btn"
style="cursor: pointer; width: 100%;">Logout</button>
        </div>

        <div style="margin-top: 20px; padding-top: 15px; border-top: 1px solid
rgba(107, 78, 253, 0.4); text-align: center; font-size: 13px; color: #CBD5E1;">
          <p>Academic Year 2026-27</p>
          <p style="color: #10B981; font-weight: bold; margin-top: 3px;">● Active
Student</p>
        </div>
      </div>

      <!-- Main Dashboard Content Column -->
      <div class="dash-content-col">

        <!-- Administrator Mode Indicator (Shown only when logged in as admin) -->
```

```
        <div id="admin-access-indicator" style="display: none; background-color:
#6B4EFD; color: white; padding: 10px 18px; border-radius: 12px; font-size: 14px;
text-align: center; border: 1.5px solid #EC9F05;">
          🌓 <strong>Administrator Mode:</strong> Viewing student dashboard &bull;
<a href="admin.html" style="color: #DAF264; font-weight: bold; text-decoration:
underline;">Switch to Admin Panel &rarr;</a>
        </div>

        <!-- Welcome Title Card -->
        <div class="table1" style="padding: 20px 24px;">
          <h2 style="font-size: 26px; margin-bottom: 4px;">Welcome back, John
Doe</h2>
          <p style="color: #CBD5E1; font-size: 15px;">B.Tech Computer Science &amp;
Engineering &ndash; Semester 3 | Student Hub Portal</p>
        </div>

        <!-- 3 Clean Stat Cards (No Notices, No Attendance, No Credits) -->
        <div class="dash-stats-flex">
          <div class="table1" style="flex: 1; text-align: center; padding: 18px
10px;">
            <p class="stat-label">Total Events</p>
            <h1 class="stat-num">20</h1>
            <p style="font-size: 12px; color: #CBD5E1;">Campus Activities</p>
          </div>
          <div class="table1" style="flex: 1; text-align: center; padding: 18px
10px;">
            <p class="stat-label">Registered Events</p>
            <h1 class="stat-num">5</h1>
            <p style="font-size: 12px; color: #CBD5E1;">Confirmed Slots</p>
          </div>
          <div class="table1" style="flex: 1; text-align: center; padding: 18px
10px;">
            <p class="stat-label">Enrolled Subjects</p>
            <h1 class="stat-num">5</h1>
            <p style="font-size: 12px; color: #10B981; font-weight: bold;">Sem 3
Regular</p>
          </div>
        </div>

        <!-- Full-Width Registered Events Table Card (No Notices Card) -->
        <div class="table1" style="width: 100%; padding: 24px;">
          <div style="display: flex; justify-content: space-between; align-items:
center; margin-bottom: 14px; flex-wrap: wrap; gap: 8px;">
            <h3 style="color: #EC9F05;">My Registered Events</h3>
            <a href="events.html" style="font-size: 14px;">Browse All Events
&rarr;</a>
          </div>

          <table class="dash-table">
            <thead>
              <tr>
                <th>Event Name</th>
                <th>Category</th>
                <th>Date</th>
                <th>Venue</th>
                <th>Status</th>
              </tr>
            </thead>
            <tbody>
              <tr>
                <td><strong>WDF Full-Stack Workshop</strong></td>
                <td><span class="dash-tag enrolled">TECH</span></td>
```

```
                <td>Nov 15, 2026</td>
                <td>Lab 3</td>
                <td><span class="dash-tag confirmed">Confirmed</span></td>
              </tr>
              <tr>
                <td><strong>DSA Coding Challenge</strong></td>
                <td><span class="dash-tag notice-tag">ACADEMIC</span></td>
                <td>Oct 28, 2026</td>
                <td>Lab 1</td>
                <td><span class="dash-tag confirmed">Confirmed</span></td>
              </tr>
              <tr>
                <td><strong>Java OOP Software Seminar</strong></td>
                <td><span class="dash-tag enrolled">TECH</span></td>
                <td>Dec 05, 2026</td>
                <td>Audi B</td>
                <td><span class="dash-tag enrolled">Enrolled</span></td>
              </tr>
              <tr>
                <td><strong>SPOURAL - Sports Championship</strong></td>
                <td><span class="dash-tag notice-tag">SPORTS</span></td>
                <td>Oct 25, 2026</td>
                <td>Main Ground</td>
                <td><span class="dash-tag confirmed">Confirmed</span></td>
              </tr>
            </tbody>
          </table>
        </div>

        <!-- Academic Overview Card - Sem 3 CS Subjects -->
        <div class="table1" style="padding: 22px;">
          <div style="display: flex; justify-content: space-between; align-items:
center; margin-bottom: 14px; flex-wrap: wrap; gap: 8px;">
            <h3 style="color: #EC9F05;">Semester 3 Enrolled Subjects (Computer
Science)</h3>
            <span style="font-size: 13px; color: #10B981; font-weight: bold;">5
Subjects Enrolled</span>
          </div>

          <div style="display: flex; gap: 14px; flex-wrap: wrap;">
            <!-- 1. WDF -->
            <div style="flex: 1; min-width: 170px; background-color: #141E2E;
padding: 14px; border-radius: 12px; border: 1.5px solid #6B4EFD;">
              <p style="color: #EC9F05; font-weight: bold; font-size:
13px;">ITUE203</p>
              <h4 style="font-size: 16px; margin: 4px 0 2px 0;">WDF</h4>
              <p style="font-size: 13px; color: #FFFFFF;">Web Development
Frameworks</p>
              <p style="font-size: 12px; color: #CBD5E1; margin-top: 6px;">Faculty:
Prof. Sharma</p>
              <p style="font-size: 12px; color: #10B981; margin-top: 4px;">Theory +
Lab</p>
            </div>

            <!-- 2. FCN -->
            <div style="flex: 1; min-width: 170px; background-color: #141E2E;
padding: 14px; border-radius: 12px; border: 1.5px solid #6B4EFD;">
              <p style="color: #EC9F05; font-weight: bold; font-size:
13px;">ITUE204</p>
              <h4 style="font-size: 16px; margin: 4px 0 2px 0;">FCN</h4>
              <p style="font-size: 13px; color: #FFFFFF;">Fundamentals of Computer
Networks</p>
```

```
              <p style="font-size: 12px; color: #CBD5E1; margin-top: 6px;">Faculty:
Prof. Mehta</p>
              <p style="font-size: 12px; color: #10B981; margin-top: 4px;">Theory +
Lab</p>
            </div>

            <!-- 3. DSA -->
            <div style="flex: 1; min-width: 170px; background-color: #141E2E;
padding: 14px; border-radius: 12px; border: 1.5px solid #6B4EFD;">
              <p style="color: #EC9F05; font-weight: bold; font-size:
13px;">ITUE201</p>
              <h4 style="font-size: 16px; margin: 4px 0 2px 0;">DSA</h4>
              <p style="font-size: 13px; color: #FFFFFF;">Data Structures &amp;
Algorithms</p>
              <p style="font-size: 12px; color: #CBD5E1; margin-top: 6px;">Faculty:
Dr. Verma</p>
              <p style="font-size: 12px; color: #10B981; margin-top: 4px;">Theory +
Lab</p>
            </div>

            <!-- 4. DISCRETE MATHS -->
            <div style="flex: 1; min-width: 170px; background-color: #141E2E;
padding: 14px; border-radius: 12px; border: 1.5px solid #6B4EFD;">
              <p style="color: #EC9F05; font-weight: bold; font-size:
13px;">MATH201</p>
              <h4 style="font-size: 16px; margin: 4px 0 2px 0;">DISCRETE MATHS</h4>
              <p style="font-size: 13px; color: #FFFFFF;">Discrete Mathematics</p>
              <p style="font-size: 12px; color: #CBD5E1; margin-top: 6px;">Faculty:
Dr. Joshi</p>
              <p style="font-size: 12px; color: #10B981; margin-top:
4px;">Theory</p>
            </div>

            <!-- 5. JAVAOOP -->
            <div style="flex: 1; min-width: 170px; background-color: #141E2E;
padding: 14px; border-radius: 12px; border: 1.5px solid #6B4EFD;">
              <p style="color: #EC9F05; font-weight: bold; font-size:
13px;">ITUE202</p>
              <h4 style="font-size: 16px; margin: 4px 0 2px 0;">JAVAOOP</h4>
              <p style="font-size: 13px; color: #FFFFFF;">Java &amp; OOP
Concepts</p>
              <p style="font-size: 12px; color: #CBD5E1; margin-top: 6px;">Faculty:
Dr. Patel</p>
              <p style="font-size: 12px; color: #10B981; margin-top: 4px;">Theory +
Lab</p>
            </div>
          </div>
        </div>

      </div>

    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
```

```
    </footer>
  </div>

  <script src="script.js"></script>
  <script>
    // Practical Access Guard: Redirect to login if not authenticated
    checkDashboardAuth();
  </script>
</body>

</html>
```

HTML Code - profile.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Profile</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Logout</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="dashboard.html">Dashboard</a>
      <a href="profile.html">Profile</a>
    </nav>
  </div>

  <main align="center" style="max-width: 500px; margin: 20px auto; padding: 10px;">
    <div class="table1" style="padding: 30px;">
      <!-- Profile Picture Circle Box -->
      <div style="width: 110px; height: 110px; border-radius: 50%; background-
color: #6B4EFD; color: white; display: flex; align-items: center; justify-content:
center; margin: 0 auto; border: 3px solid #EC9F05; font-weight: bold; text-align:
center;">
        Profile<br>Picture
      </div>

      <hr style="border-color: #6B4EFD; margin: 20px 0;">
```

```
      <table width="100%" cellpadding="8" style="text-align: left; font-size:
16px;">
        <tr>
          <td width="35%" style="color: #EC9F05;"><strong>Student ID
:</strong></td>
          <td><strong>25DCS200</strong></td>
        </tr>
        <tr>
          <td style="color: #EC9F05;"><strong>Name :</strong></td>
          <td>John Doe</td>
        </tr>
        <tr>
          <td style="color: #EC9F05;"><strong>Email :</strong></td>
          <td>john.doe@studenthub.edu</td>
        </tr>
        <tr>
          <td style="color: #EC9F05;"><strong>Mobile :</strong></td>
          <td>+1 987 654 3210</td>
        </tr>
        <tr>
          <td style="color: #EC9F05;"><strong>Course :</strong></td>
          <td>B.Tech Computer Science &amp; Engg</td>
        </tr>
        <tr>
          <td style="color: #EC9F05;"><strong>Semester :</strong></td>
          <td>3rd Semester (Sem-3)</td>
        </tr>
        <tr>
          <td style="color: #EC9F05; vertical-align: top;"><strong>Subjects
:</strong></td>
          <td>WDF, FCN, DSA, DISCRETE MATHS, JAVAOOP</td>
        </tr>
      </table>
      <br>
      <button style="padding: 10px 25px; font-size: 16px;">Edit Profile</button>
    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

HTML Code - admin.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Admin Panel</title>
  <link rel="stylesheet" href="style.css">
```

```
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
height="45"
          style="vertical-align: middle; margin-right: 8px;"></a>
      <strong style="margin: 10px;">Admin Panel</strong>
    </div>
    <div class="headerlinks">
      <button onclick="handleLogout()">Logout</button>
    </div>
  </header>

  <div class="navlinks">
    <nav align="center">
      <a href="admin.html">Dashboard</a> |
      <a href="events.html">Events</a> |
      <a href="index.html">Home</a>
    </nav>
  </div>

  <main style="max-width: 1000px; margin: 20px auto; padding: 10px;">

    <!-- Stats Grid -->
    <div style="display: flex; gap: 15px; flex-wrap: wrap;">
      <div class="table1" style="flex: 1; text-align: center; min-width: 200px;
padding: 20px;">
        <h3 style="font-size: 22px; color: #EC9F05; margin: 0;">Students</h3>
        <h1 style="font-size: 42px; margin: 10px 0 0 0;">250</h1>
      </div>
      <div class="table1" style="flex: 1; text-align: center; min-width: 200px;
padding: 20px;">
        <h3 style="font-size: 22px; color: #EC9F05; margin: 0;">Events</h3>
        <h1 style="font-size: 42px; margin: 10px 0 0 0;">12</h1>
      </div>
      <div class="table1" style="flex: 1; text-align: center; min-width: 200px;
padding: 20px;">
        <h3 style="font-size: 22px; color: #EC9F05; margin: 0;">Feedback</h3>
        <h1 style="font-size: 42px; margin: 10px 0 0 0;">18</h1>
      </div>
      <div class="table1" style="flex: 1; text-align: center; min-width: 200px;
padding: 20px;">
        <h3 style="font-size: 22px; color: #EC9F05; margin: 0;">Notices</h3>
        <h1 style="font-size: 42px; margin: 10px 0 0 0;">7</h1>
      </div>
    </div>

    <br>

    <!-- Recent Activities Card -->
    <div class="table1" style="padding: 25px;">
      <h3 style="font-size: 26px; color: #EC9F05; margin-top: 0;">Recent
Activities</h3>
      <ul style="font-size: 18px; line-height: 1.8;">
        <li>Student Registered</li>
        <li>Event Added</li>
        <li>Feedback Submitted</li>
      </ul>
```

```
    </div>
  </main>
  <br>
  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a> |
        <a href="contact.html">Contact</a> |
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
  <script>
    // Protect Admin Panel from unauthorized access
    checkAdminAuth();
  </script>
</body>

</html>
```

Practical No. 3 - Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

Question: Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Ensure mobile-first responsiveness.

Key Questions

Q: How does the layout adapt for mobile, tablet, and desktop screens?

Answer: Flexbox allows items to wrap and share available space, while media queries change widths and the navigation layout for smaller screens. The same content can fit different screen sizes.

Q: Which layout technique is used and why?

Answer: Flexbox is used for horizontal groups such as the header, navigation, form grid, and cards. It is suitable because it aligns items and distributes space with less code.

Q: Are typography, spacing, colors, and contrast readable?

Answer: The stylesheet uses a dark background, light text, contrasting yellow buttons, consistent padding, and readable font sizes. These choices make the important content visible.

Q: Is the framework used consistently without unnecessary inline CSS?

Answer: The project mainly uses one external style.css file. Some page-specific inline styles are present, but the repeated site-wide styles are kept in the common stylesheet.

Analysis

A: The CSS file defines the common color palette, cards, buttons, forms, navigation, and responsive rules. Flexbox is used for groups of elements and media queries adjust the layout for smaller screens.

Interpretation

I: The interface is designed to remain usable on desktop and mobile screens. Centralizing the main styles makes the pages look like one portal instead of separate unrelated pages.

CSS Code - style.css

```
/* ================================================================================
   STUDENT HUB PORTAL - RESPONSIVE MOBILE-FIRST CSS (GRID & FLEXBOX)
   ================================================================================ */

/* --------------------------------------------------------------------------------
   1. BASE & RESET STYLES
   -------------------------------------------------------------------------------- */
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  background-color: #0B162C;
  color: #FFFFFF;
  font-family: system-ui, -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto,
sans-serif;
  line-height: 1.6;
  padding: 12px;
```

```
  min-height: 100vh;
  display: flex;
  flex-direction: column;
  transition: background-color 0.3s, color 0.3s;
}

main {
  flex: 1;
}

/* ------------------------------------------------------------------------
   2. LIGHT THEME STYLES (HIGH CONTRAST, CLEAN & SIMPLE)
   ------------------------------------------------------------------------ */
body.light-theme {
  background-color: #F1F5F9;
  color: #0F172A;
}

/* Light Theme Cards & Containers */
body.light-theme .box,
body.light-theme .boxes2,
body.light-theme .table1,
body.light-theme .footer,
body.light-theme .modal-box,
body.light-theme .slider-container,
body.light-theme .auth-card {
  background-color: #FFFFFF;
  color: #0F172A;
  border: 1px solid #E2E8F0;
  box-shadow: 0 4px 16px rgba(0, 0, 0, 0.06);
}

/* Light Theme Headings & Text */
body.light-theme h1,
body.light-theme h2,
body.light-theme h3,
body.light-theme h4,
body.light-theme strong {
  color: #0F172A;
}

/* Light Theme Subtitles, Paragraphs & Muted Text */
body.light-theme p,
body.light-theme [style*="color: #CBD5E1"],
body.light-theme [style*="color: #cbd5e1"] {
  color: #475569 !important;
}

/* Light Theme Accent Amber (Richer Amber for High Contrast on White) */
body.light-theme h2[style*="color: #EC9F05"],
body.light-theme h3[style*="color: #EC9F05"],
body.light-theme h4[style*="color: #EC9F05"],
body.light-theme strong[style*="color: #EC9F05"],
body.light-theme p[style*="color: #EC9F05"],
body.light-theme .stat-label {
  color: #B45309 !important;
}

/* Light Theme Stat Numbers */
body.light-theme .stat-num {
  color: #0F172A !important;
```

```
}

/* Light Theme Inner Course Cards & Note Boxes */
body.light-theme div[style*="background-color: #141E2E"],
body.light-theme div[style*="background-color: #141e2e"] {
  background-color: #F8FAFC !important;
  border-color: #CBD5E1 !important;
}

body.light-theme div[style*="background-color: #141E2E"] h4,
body.light-theme div[style*="background-color: #141E2E"] p {
  color: #0F172A !important;
}

body.light-theme div[style*="background-color: #141E2E"] p[style*="color: #EC9F05"]
{
  color: #B45309 !important;
}

/* Light Theme Event Badges */
body.light-theme .event-badge {
  background-color: #F8FAFC;
  border-color: #6B4EFD;
  color: #B45309;
}

/* Light Theme Tables */
body.light-theme .dash-table td {
  border-bottom: 1px solid #E2E8F0;
  color: #1E293B;
}

body.light-theme .dash-table th {
  color: #B45309;
  border-bottom: 2px solid #6B4EFD;
}

/* Light Theme Status Pills */
body.light-theme .dash-tag.confirmed {
  background-color: #DCFCE7;
  color: #15803D;
}

body.light-theme .dash-tag.enrolled {
  background-color: #EDE9FE;
  color: #6D28D9;
}

body.light-theme .dash-tag.notice-tag {
  background-color: #FEF3C7;
  color: #B45309;
}

/* Light Theme FAQ */
body.light-theme .faq-item {
  border-bottom-color: #E2E8F0;
}

body.light-theme .faq-btn {
  color: #B45309;
```

```
}

body.light-theme .faq-btn:hover {
  color: #92400E;
}

body.light-theme .faq-answer,
body.light-theme .faq-answer p {
  color: #334155 !important;
}

/* Light Theme Inputs & Form Controls */
body.light-theme input[type="text"],
body.light-theme input[type="email"],
body.light-theme input[type="password"],
body.light-theme input[type="number"],
body.light-theme input[type="tel"],
body.light-theme select,
body.light-theme textarea {
  background-color: #FFFFFF !important;
  color: #0F172A !important;
  border: 1.5px solid #CBD5E1 !important;
}

body.light-theme input:focus,
body.light-theme select:focus,
body.light-theme textarea:focus {
  background-color: #FFFFFF !important;
  border-color: #6B4EFD !important;
  box-shadow: 0 0 0 3px rgba(107, 78, 253, 0.18) !important;
}

body.light-theme label {
  color: #0F172A !important;
}

/* Light Theme Header & Auth Text */
body.light-theme .header strong,
body.light-theme .auth-card h2,
body.light-theme .auth-footer-text {
  color: #0F172A;
}

body.light-theme .t1links a {
  color: #0F172A;
}

body.light-theme .t1links a:hover {
  color: #B45309;
}

body.light-theme a {
  color: #B45309;
}

body.light-theme a:hover {
  color: #92400E;
}

/* Base Universal Input & Form Styling */
```

```
input[type="text"],
input[type="email"],
input[type="password"],
input[type="number"],
input[type="tel"],
select,
textarea {
  width: 100%;
  background-color: #0B162C;
  color: #FFFFFF;
  border: 1.5px solid #6B4EFD;
  border-radius: 12px;
  padding: 11px 15px;
  font-size: 15px;
  font-family: inherit;
  outline: none;
  transition: border-color 0.2s, box-shadow 0.2s;
  box-sizing: border-box;
}

input:focus,
select:focus,
textarea:focus {
  border-color: #EC9F05;
  box-shadow: 0 0 0 3px rgba(236, 159, 5, 0.2);
}

a {
  color: #EC9F05;
  text-decoration: none;
  transition: 0.2s ease;
}

a:hover {
  color: #d88e04;
  text-decoration: underline;
}

/* Base Universal Button Styling (Never render unstyled browser buttons) */
button {
  font-family: inherit;
  background-color: #EC9F05;
  color: #FFFFFF;
  border: none;
  border-radius: 20px;
  font-size: 15px;
  font-weight: 600;
  padding: 9px 20px;
  cursor: pointer;
  transition: all 0.2s ease;
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  outline: none;
}

button:hover {
  background-color: #d88e04;
  transform: translateY(-2px);
  box-shadow: 0 4px 12px rgba(236, 159, 5, 0.35);
```

```
}

button:active {
  transform: translateY(0);
}

button a {
  color: inherit;
  text-decoration: none;
}

button a:hover {
  text-decoration: none;
  color: inherit;
}

/* ------------------------------------------------------------------------
   2. LOGO & HEADER STYLES (FIXED SIZE ON ALL WEBSITES)
   ------------------------------------------------------------------------ */
.header img,
.logo,
header img {
  height: 48px !important;
  max-height: 48px !important;
  width: auto !important;
  max-width: 160px !important;
  object-fit: contain;
  vertical-align: middle;
  border-radius: 8px;
}

header {
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: space-between;
  gap: 15px;
  padding: 10px 0;
}

.header {
  display: flex;
  align-items: center;
  gap: 12px;
  font-family: serif;
  font-size: 24px;
}

.header strong {
  color: #FFFFFF;
}

.headerlinks {
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
  justify-content: center;
  align-items: center;
}
```

```css
.headerlinks button,
.headerlinks a button {
  background-color: #EC9F05;
  color: white;
  border: none;
  border-radius: 20px;
  font-size: 15px;
  font-weight: 600;
  padding: 8px 18px;
  cursor: pointer;
  transition: background-color 0.2s, transform 0.1s, box-shadow 0.2s;
}

.headerlinks button:hover {
  background-color: #d88e04;
  transform: translateY(-1px);
  box-shadow: 0 4px 12px rgba(236, 159, 5, 0.3);
}

.headerlinks a {
  color: white;
  text-decoration: none;
}

/* ------------------------------------------------------------------------
   3. NOTIFICATION BANNER
   ------------------------------------------------------------------------ */
.notification-banner {
  background-color: #6B4EFD;
  color: white;
  padding: 10px 18px;
  border-radius: 12px;
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 12px;
  font-size: 15px;
}

.banner-close {
  background: none;
  border: none;
  color: white;
  font-size: 20px;
  font-weight: bold;
  cursor: pointer;
  padding: 0 6px;
}

/* ------------------------------------------------------------------------
   4. NAVIGATION BAR & HAMBURGER (FLEXBOX)
   ------------------------------------------------------------------------ */
.hamburger {
  display: block;
  background-color: #6B4EFD;
  color: white;
  border: none;
  border-radius: 10px;
  font-size: 18px;
  font-weight: bold;
  padding: 8px 16px;
```

```
  cursor: pointer;
  margin: 0 auto;
}

.navlinks {
  background-color: #6B4EFD;
  border-radius: 15px;
  margin: 10px 0 20px 0;
  padding: 10px;
}

.navlinks nav {
  display: none;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  gap: 10px;
}

.navlinks nav.show {
  display: flex;
}

.navlinks a {
  font-family: sans-serif;
  color: #FFFFFF;
  font-size: 16px;
  font-weight: 500;
  text-decoration: none;
  padding: 6px 12px;
  border-radius: 8px;
  transition: all 0.2s ease;
}

.navlinks a:hover {
  color: #DAF264;
  background-color: rgba(255, 255, 255, 0.15);
  text-decoration: none;
}

/* ------------------------------------------------------------------------
   5. PERFECT LOGIN & REGISTRATION BOXES + FORM VALIDATION
   ------------------------------------------------------------------------ */
.auth-card {
  background-color: #212C37;
  border: 1px solid rgba(255, 255, 255, 0.08);
  border-radius: 24px;
  padding: 35px 28px;
  max-width: 440px;
  width: 100%;
  margin: 25px auto;
  box-shadow: 0 12px 30px rgba(0, 0, 0, 0.35);
  box-sizing: border-box;
}

.auth-card.register-card {
  max-width: 680px;
}

.auth-card h2 {
  font-size: 28px;
```

```
  color: #EC9F05;
  text-align: center;
  margin-bottom: 22px;
  letter-spacing: 0.5px;
}

.form-group {
  display: flex;
  flex-direction: column;
  gap: 6px;
  margin-bottom: 16px;
  text-align: left;
}

.form-group label {
  font-size: 15px;
  font-weight: 600;
  color: #E2E8F0;
}

.form-group input,
.form-group select,
.form-group textarea {
  width: 100%;
  background-color: #0B162C;
  color: #FFFFFF;
  border: 1.5px solid #6B4EFD;
  border-radius: 12px;
  padding: 11px 15px;
  font-size: 15px;
  outline: none;
  transition: border-color 0.2s, box-shadow 0.2s;
  box-sizing: border-box;
}

.form-group input:focus,
.form-group select:focus,
.form-group textarea:focus {
  border-color: #EC9F05;
  box-shadow: 0 0 0 3px rgba(236, 159, 5, 0.2);
}

/* Error & Success States */
.error-text {
  color: #FF5252;
  font-size: 13px;
  margin-top: 3px;
  display: block;
  min-height: 18px;
}

.input-error {
  border-color: #FF5252 !important;
  box-shadow: 0 0 0 2px rgba(255, 82, 82, 0.25) !important;
}

.input-success {
  border-color: #10B981 !important;
}

/* 2-Column Grid for Registration Form */
```

```css
.form-grid {
  display: grid;
  grid-template-columns: 1fr;
  gap: 0 20px;
}

.radio-group {
  display: flex;
  gap: 20px;
  align-items: center;
  padding: 10px 0 4px 0;
}

.radio-group label {
  display: flex;
  align-items: center;
  gap: 6px;
  cursor: pointer;
  font-weight: normal;
}

.radio-group input[type="radio"],
.form-options input[type="checkbox"] {
  width: 18px;
  height: 18px;
  accent-color: #EC9F05;
  cursor: pointer;
}

.form-options {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 20px;
  font-size: 14px;
}

.form-options.center-terms {
  flex-direction: column;
  align-items: center;
  justify-content: center;
  margin-top: 10px;
  text-align: center;
}

.form-options label {
  display: flex;
  align-items: center;
  gap: 8px;
  cursor: pointer;
}

.auth-btn {
  width: 100%;
  background-color: #EC9F05;
  color: white;
  border: none;
  border-radius: 30px;
  font-size: 17px;
  font-weight: bold;
  padding: 13px;
```

```
  cursor: pointer;
  transition: background-color 0.2s, transform 0.1s;
  box-shadow: 0 4px 12px rgba(236, 159, 5, 0.3);
}

.auth-btn:hover {
  background-color: #d88e04;
  transform: translateY(-2px);
}

.auth-footer-text {
  text-align: center;
  margin-top: 18px;
  font-size: 15px;
  color: #CBD5E1;
}

/* ------------------------------------------------------------------------
   6. HERO, CARDS & GRIDS (HOME & ABOUT)
   ------------------------------------------------------------------------ */
.main {
  background: linear-gradient(135deg, #6B4EFD 0%, #4427D6 60%, #2A1792 100%);
  border-radius: 20px;
  padding: 36px 24px;
  margin: 15px 0 25px 0;
  text-align: center;
  color: white;
  box-shadow: 0 10px 30px rgba(107, 78, 253, 0.35);
  position: relative;
  overflow: hidden;
}

.main h1 {
  font-size: 32px;
  font-weight: 800;
  margin-bottom: 8px;
  letter-spacing: -0.5px;
}

.main p {
  font-size: 17px;
  color: rgba(255, 255, 255, 0.9);
  max-width: 600px;
  margin: 0 auto;
}

/* Styled View Announcement Button */
.main button,
.btn-announcement {
  background-color: #EC9F05;
  color: #FFFFFF;
  border: 2px solid transparent;
  font-size: 15px;
  font-weight: 700;
  padding: 12px 28px;
  border-radius: 30px;
  cursor: pointer;
  box-shadow: 0 6px 18px rgba(0, 0, 0, 0.25);
  transition: all 0.25s ease;
  display: inline-flex;
  align-items: center;
```

```
  justify-content: center;
  gap: 8px;
  margin-top: 6px;
}

.main button:hover,
.btn-announcement:hover {
  background-color: #FFFFFF;
  color: #6B4EFD;
  transform: translateY(-3px) scale(1.03);
  box-shadow: 0 10px 25px rgba(255, 255, 255, 0.3);
}

.boxes-container {
  display: flex;
  flex-direction: column;
  gap: 15px;
  width: 100%;
}

.box,
.boxes2,
.table1 {
  background-color: #212C37;
  border-radius: 20px;
  border: 1px solid rgba(255, 255, 255, 0.08);
  padding: 24px;
  width: 100%;
  box-sizing: border-box;
  box-shadow: 0 6px 20px rgba(0, 0, 0, 0.2);
  transition: transform 0.2s ease, border-color 0.2s ease, background-color 0.25s
ease, color 0.25s ease;
}

.box:hover,
.boxes2:hover,
.table1:hover {
  border-color: rgba(107, 78, 253, 0.4);
}

.box {
  min-height: 220px;
  text-align: center;
}

.boxes2 {
  min-height: 180px;
  text-align: center;
}

.box h3,
.boxes2 p:first-child {
  color: #EC9F05;
  margin-bottom: 12px;
}

.box ul {
  list-style-position: inside;
  text-align: left;
  display: inline-block;
}
```

```
.t1,
.t1links {
  font-size: 17px;
  margin-bottom: 8px;
}

.t1links a {
  color: #FFFFFF;
}

.t1links a:hover {
  color: #EC9F05;
}

/* ------------------------------------------------------------------------
   7. COLLAPSIBLE FAQ & MODAL & SLIDER
   ------------------------------------------------------------------------ */
.faq-item {
  border-bottom: 1px solid rgba(107, 78, 253, 0.4);
  padding: 10px 0;
}

.faq-btn {
  width: 100%;
  background: none;
  color: #EC9F05;
  font-size: 18px;
  font-weight: bold;
  border: none;
  padding: 10px 0;
  cursor: pointer;
  display: flex;
  justify-content: space-between;
  align-items: center;
  text-align: left;
  border-radius: 0;
  box-shadow: none;
}

.faq-btn:hover {
  background: none;
  color: #d88e04;
  transform: none;
  box-shadow: none;
}

.faq-answer {
  display: none;
  padding: 8px 0 12px 0;
  font-size: 16px;
  color: #CBD5E1;
}

.modal {
  display: none;
  position: fixed;
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
```

```
  background: rgba(0, 0, 0, 0.75);
  justify-content: center;
  align-items: center;
  z-index: 1000;
  padding: 15px;
  backdrop-filter: blur(4px);
}

.modal-box {
  background-color: #212C37;
  color: white;
  padding: 32px 28px;
  border-radius: 24px;
  max-width: 480px;
  width: 100%;
  text-align: center;
  position: relative;
  border: 1.5px solid #6B4EFD;
  box-shadow: 0 16px 40px rgba(0, 0, 0, 0.5);
}

.modal-close {
  position: absolute;
  top: 12px;
  right: 16px;
  background: none;
  color: #94A3B8;
  font-size: 26px;
  border: none;
  cursor: pointer;
  padding: 4px 8px;
  border-radius: 50%;
  box-shadow: none;
}

.modal-close:hover {
  background-color: rgba(255, 255, 255, 0.1);
  color: #FFFFFF;
  transform: none;
  box-shadow: none;
}

.modal-box button:not(.modal-close) {
  background-color: #EC9F05;
  color: #FFFFFF;
  border-radius: 25px;
  padding: 10px 28px;
  font-size: 15px;
  font-weight: 700;
  border: none;
  cursor: pointer;
  margin-top: 10px;
  box-shadow: 0 4px 14px rgba(236, 159, 5, 0.35);
  transition: all 0.2s ease;
}

.modal-box button:not(.modal-close):hover {
  background-color: #d88e04;
  transform: translateY(-2px);
  box-shadow: 0 6px 18px rgba(236, 159, 5, 0.45);
}
```

```css
/* Modern Slider Container */
.slider-container {
  background-color: #212C37;
  border-radius: 22px;
  border: 1.5px solid rgba(107, 78, 253, 0.3);
  padding: 30px 24px;
  margin: 25px 0;
  text-align: center;
  box-shadow: 0 8px 24px rgba(0, 0, 0, 0.25);
  position: relative;
}

.slide {
  display: none;
  animation: fadeIn 0.4s ease-in-out;
}

@keyframes fadeIn {
  from { opacity: 0; transform: translateY(6px); }
  to { opacity: 1; transform: translateY(0); }
}

.slide h3 {
  color: #EC9F05;
  font-size: 24px;
  font-weight: 700;
  margin-bottom: 10px;
}

.slide p {
  font-size: 16px;
  color: #CBD5E1;
  max-width: 650px;
  margin: 0 auto;
  line-height: 1.5;
}

/* Prev and Next Slider Control Buttons */
.slider-controls {
  margin-top: 20px;
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 14px;
}

.slider-controls button,
.slider-btn {
  background: #141E2E;
  color: #EC9F05;
  border: 1.5px solid #6B4EFD;
  border-radius: 25px;
  font-size: 14px;
  font-weight: 700;
  padding: 8px 22px;
  cursor: pointer;
  transition: all 0.2s ease;
  box-shadow: 0 4px 10px rgba(0, 0, 0, 0.25);
  display: inline-flex;
  align-items: center;
```

```
  gap: 6px;
}

.slider-controls button:hover,
.slider-btn:hover {
  background-color: #6B4EFD;
  color: #FFFFFF;
  border-color: #6B4EFD;
  transform: translateY(-2px);
  box-shadow: 0 6px 16px rgba(107, 78, 253, 0.4);
}

.slider-controls button:active,
.slider-btn:active {
  transform: translateY(0);
}

/* Light Theme Adjustments for Slider & Hero */
body.light-theme .main button,
body.light-theme .btn-announcement {
  background-color: #EC9F05;
  color: #FFFFFF;
}

body.light-theme .main button:hover,
body.light-theme .btn-announcement:hover {
  background-color: #FFFFFF;
  color: #6B4EFD;
}

body.light-theme .slider-container {
  background-color: #FFFFFF;
  border-color: #E2E8F0;
  box-shadow: 0 8px 20px rgba(0, 0, 0, 0.08);
}

body.light-theme .slide p {
  color: #475569;
}

body.light-theme .slider-controls button,
body.light-theme .slider-btn {
  background: #F8FAFC;
  color: #6B4EFD;
  border-color: #CBD5E1;
}

body.light-theme .slider-controls button:hover,
body.light-theme .slider-btn:hover {
  background-color: #6B4EFD;
  color: #FFFFFF;
  border-color: #6B4EFD;
}

body.light-theme .modal-box {
  background-color: #FFFFFF;
  color: #1A202C;
  border-color: #6B4EFD;
}
```

```css
body.light-theme .modal-close {
  color: #64748B;
}

body.light-theme .modal-close:hover {
  background-color: #F1F5F9;
  color: #1A202C;
}

/* ------------------------------------------------------------------------
   8. FOOTER STYLES
   ------------------------------------------------------------------------ */
.footer {
  background-color: #212C37;
  border-radius: 25px;
  padding: 16px;
  margin-top: 35px;
  text-align: center;
}

.footer footer p {
  display: flex;
  flex-wrap: wrap;
  align-items: center;
  justify-content: center;
  gap: 8px;
  font-size: 15px;
}

.footer a {
  background-color: #EC9F05;
  color: white;
  border-radius: 15px;
  font-size: 14px;
  padding: 4px 12px;
  text-decoration: none;
}

/* ------------------------------------------------------------------------
   9. DESKTOP & TABLET RESPONSIVENESS
   ------------------------------------------------------------------------ */
@media (min-width: 768px) {
  body {
    padding: 20px 40px;
  }

  header {
    flex-direction: row;
    padding: 10px 10px;
  }

  .header {
    font-size: 30px;
  }

  .hamburger {
    display: none;
  }

  .navlinks nav {
```

```
    display: flex !important;
    flex-direction: row;
  }

  .main {
    max-width: 850px;
    margin: 20px auto;
    padding: 35px 20px;
  }

  .main h1 {
    font-size: 42px;
  }

  /* 2-Column Grid on Registration Form */
  .form-grid {
    grid-template-columns: 1fr 1fr;
  }

  /* Horizontal Grid for Boxes on Desktop */
  .boxes-container {
    flex-direction: row !important;
  }

  .box {
    width: 33.33% !important;
  }

  .boxes2 {
    width: 50% !important;
  }
}

/* ================================================================================
   10. STUDENT DASHBOARD STYLES (BROAD, CLEAN & SIMPLE)
   ================================================================================ */
.dash-wrapper {
  max-width: 1280px;
  width: 100%;
  margin: 15px auto;
  padding: 10px;
  box-sizing: border-box;
}

.dash-main-flex {
  display: flex;
  flex-direction: column;
  gap: 20px;
  align-items: stretch;
}

@media (min-width: 850px) {
  .dash-main-flex {
    flex-direction: row;
  }
}

.dash-sidebar-col {
  width: 100%;
  max-width: 250px;
```

```
  flex-shrink: 0;
  box-sizing: border-box;
}

@media (max-width: 849px) {
  .dash-sidebar-col {
    max-width: 100%;
  }
}

.dash-content-col {
  flex: 1;
  min-width: 0;
  display: flex;
  flex-direction: column;
  gap: 20px;
}

.dash-stats-flex {
  display: flex;
  flex-direction: row;
  gap: 15px;
  width: 100%;
}

.dash-cards-flex {
  display: flex;
  flex-direction: column;
  gap: 20px;
  width: 100%;
}

@media (min-width: 768px) {
  .dash-cards-flex {
    flex-direction: row;
  }
}

.dash-sidebar-menu {
  display: flex;
  flex-direction: column;
  gap: 10px;
  margin-top: 10px;
}

.dash-nav-btn {
  display: block;
  width: 100%;
  background-color: #6B4EFD;
  color: #FFFFFF;
  padding: 10px 16px;
  border-radius: 12px;
  font-size: 15px;
  font-weight: 600;
  text-decoration: none;
  text-align: center;
  transition: all 0.2s ease;
  box-sizing: border-box;
}

.dash-nav-btn:hover {
```

```
  background-color: #EC9F05;
  color: #FFFFFF;
  text-decoration: none;
  transform: translateY(-2px);
}

.dash-nav-btn.active {
  background-color: #EC9F05;
  color: #FFFFFF;
}

.dash-nav-btn.logout-btn {
  background-color: transparent;
  border: 1.5px solid #FF5252;
  color: #FF5252;
  margin-top: 8px;
}

.dash-nav-btn.logout-btn:hover {
  background-color: #FF5252;
  color: #FFFFFF;
}

.stat-num {
  font-size: 36px;
  font-weight: bold;
  color: #FFFFFF;
  margin: 6px 0;
}

.stat-label {
  font-size: 16px;
  color: #EC9F05;
  font-weight: 600;
}

.dash-table {
  width: 100%;
  border-collapse: collapse;
  margin-top: 10px;
  text-align: left;
}

.dash-table th {
  color: #EC9F05;
  padding: 10px 8px;
  border-bottom: 1.5px solid #6B4EFD;
  font-size: 15px;
}

.dash-table td {
  padding: 12px 8px;
  border-bottom: 1px solid rgba(255, 255, 255, 0.08);
  font-size: 14px;
}

.dash-tag {
  display: inline-block;
  padding: 3px 10px;
  border-radius: 12px;
```

```
  font-size: 12px;
  font-weight: bold;
}

.dash-tag.confirmed {
  background-color: rgba(16, 185, 129, 0.2);
  color: #10B981;
}

.dash-tag.enrolled {
  background-color: rgba(107, 78, 253, 0.25);
  color: #9D88FD;
}

.dash-tag.notice-tag {
  background-color: rgba(236, 159, 5, 0.2);
  color: #EC9F05;
}

body.light-theme .dash-nav-btn {
  background-color: #6B4EFD;
  color: #FFFFFF;
}

body.light-theme .dash-nav-btn:hover {
  background-color: #B45309;
}

body.light-theme .dash-nav-btn.active {
  background-color: #B45309;
  color: #FFFFFF;
}

body.light-theme .stat-num {
  color: #0F172A;
}

body.light-theme .dash-table td {
  border-bottom: 1px solid #E2E8F0;
  color: #1E293B;
}

body.light-theme .dash-table th {
  border-bottom-color: #6B4EFD;
  color: #B45309;
}

/* ============================================================================
   11. EVENTS PAGE STYLES (CLEAN & SIMPLE)
   ============================================================================ */
.events-search-box {
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 12px;
  flex-wrap: wrap;
}

.events-search-box input {
  flex: 1;
```

```
  max-width: 480px;
  margin: 0;
}

.event-card-item {
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 20px;
  padding: 20px 24px;
  margin-bottom: 16px;
  transition: transform 0.2s ease, border-color 0.2s ease;
  flex-wrap: wrap;
}

.event-card-item:hover {
  transform: translateY(-2px);
  border-color: #6B4EFD;
}

.event-badge {
  width: 85px;
  height: 70px;
  background-color: #0B162C;
  border: 2px solid #6B4EFD;
  border-radius: 12px;
  display: flex;
  align-items: center;
  justify-content: center;
  color: #EC9F05;
  font-weight: bold;
  font-size: 15px;
  flex-shrink: 0;
  letter-spacing: 0.5px;
}

body.light-theme .event-badge {
  background-color: #F8FAFC;
  border-color: #6B4EFD;
}

/* Event Filter, Search & Sort Toolbar */
.events-toolbar {
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 15px;
  flex-wrap: wrap;
}

.events-toolbar .search-box-wrap {
  flex: 1.5;
  min-width: 260px;
  display: flex;
  align-items: center;
  gap: 10px;
}

.events-toolbar .controls-group {
  display: flex;
  align-items: center;
```

```
  gap: 10px;
  flex-wrap: wrap;
  flex: 1;
  justify-content: flex-end;
}

.events-toolbar select {
  width: auto;
  min-width: 140px;
  padding: 10px 14px;
}

/* Pagination Styles */
.pagination-container {
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 8px;
  margin-top: 25px;
  flex-wrap: wrap;
}

.pagination-btn {
  background-color: #141E2E;
  color: #FFFFFF;
  border: 1.5px solid #6B4EFD;
  padding: 8px 16px;
  border-radius: 12px;
  font-size: 14px;
  font-weight: 600;
  cursor: pointer;
  transition: all 0.2s ease;
}

.pagination-btn:hover:not(:disabled) {
  background-color: #6B4EFD;
  color: #FFFFFF;
  transform: translateY(-2px);
}

.pagination-btn.active {
  background-color: #EC9F05;
  border-color: #EC9F05;
  color: #FFFFFF;
}

.pagination-btn:disabled {
  opacity: 0.35;
  cursor: not-allowed;
  transform: none;
}

body.light-theme .pagination-btn {
  background-color: #FFFFFF;
  color: #0F172A;
  border-color: #CBD5E1;
}

body.light-theme .pagination-btn:hover:not(:disabled) {
  background-color: #6B4EFD;
  color: #FFFFFF;
```

```
  border-color: #6B4EFD;
}

body.light-theme .pagination-btn.active {
  background-color: #B45309;
  border-color: #B45309;
  color: #FFFFFF;
}
```

Practical No. 4 - JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Question: Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

Key Questions

Q: How are DOM elements selected and modified?

Answer: The script uses getElementById, querySelector, and querySelectorAll. It changes class names, text, display styles, and the current slide to update the page.

Q: Are event listeners attached correctly?

Answer: The HTML buttons call functions for clicks, and a DOMContentLoaded listener starts the first slider. A window click handler also closes a modal when the user clicks outside it.

Q: Is localStorage used for remembering theme choice?

Answer: Yes. The selected theme is stored in localStorage, and the saved value is checked when the page loads so the preference can be restored.

Q: Does interactivity improve usability without breaking accessibility?

Answer: The features make the page easier to use by hiding repeated information, showing announcements, and supporting mobile navigation. Clear buttons and visible text keep the controls understandable.

Analysis

A: The JavaScript file contains small functions for theme switching, menu display, banner closing, FAQ expansion, modal display, and slider movement. Each function changes only the DOM elements needed for that feature.

Interpretation

I: The portal becomes interactive without a server. Users can control the menu, announcements, FAQ answers, theme, and slider directly in the browser.

JavaScript Code - script.js

```
/* ================================================================================
   STUDENT HUB PORTAL - SIMPLE DYNAMIC UI & FORM VALIDATION JAVASCRIPT
   ================================================================================ */

// 1. LIGHT / DARK THEME SWITCHER
function toggleTheme() {
  document.body.classList.toggle('light-theme');
  const isLight = document.body.classList.contains('light-theme');
  localStorage.setItem('theme', isLight ? 'light' : 'dark');
}

// Load saved theme on startup
if (localStorage.getItem('theme') === 'light') {
  document.body.classList.add('light-theme');
}

// 2. HAMBURGER MENU (MOBILE NAVIGATION)
```

```
function toggleMenu() {
  const menu = document.getElementById('nav-menu');
  if (menu) {
    menu.classList.toggle('show');
  }
}

// 3. NOTIFICATION BANNER
function closeBanner() {
  const banner = document.getElementById('notification-banner');
  if (banner) {
    banner.style.display = 'none';
  }
}

// 4. COLLAPSIBLE FAQ ACCORDION
function toggleFaq(btn) {
  const answer = btn.nextElementSibling;
  const icon = btn.querySelector('.faq-icon');

  if (answer.style.display === 'block') {
    answer.style.display = 'none';
    if (icon) icon.textContent = '+';
  } else {
    answer.style.display = 'block';
    if (icon) icon.textContent = '-';
  }
}

// 5. MODAL POPUP (OPEN / CLOSE)
function openModal(modalId) {
  const modal = document.getElementById(modalId);
  if (modal) {
    modal.style.display = 'flex';
  }
}

function closeModal(modalId) {
  const modal = document.getElementById(modalId);
  if (modal) {
    modal.style.display = 'none';
  }
}

// Close modal if user clicks outside the modal box
window.onclick = function(event) {
  if (event.target.classList.contains('modal')) {
    event.target.style.display = 'none';
  }
};

// 6. IMAGE / CONTENT SLIDER
let currentSlide = 0;

function showSlide(index) {
  const slides = document.querySelectorAll('.slide');
  if (slides.length === 0) return;

  if (index >= slides.length) {
    currentSlide = 0;
```

```
  } else if (index < 0) {
    currentSlide = slides.length - 1;
  } else {
    currentSlide = index;
  }

  slides.forEach((slide, i) => {
    slide.style.display = (i === currentSlide) ? 'block' : 'none';
  });
}

function changeSlide(step) {
  showSlide(currentSlide + step);
}

// Initialize first slide on page load
document.addEventListener('DOMContentLoaded', () => {
  showSlide(0);
});



// 7. STUDENT REGISTRATION FORM VALIDATION (REGEX)
function validateRegistration(event) {
  event.preventDefault(); // Stop normal form submission

  let isValid = true;

  // Helper functions for displaying error / success
  function setError(id, message) {
    const errorElem = document.getElementById(id + 'Error');
    const inputElem = document.getElementById(id);
    if (errorElem) errorElem.textContent = message;
    if (inputElem) {
      inputElem.classList.add('input-error');
      inputElem.classList.remove('input-success');
    }
    isValid = false;
  }

  function setSuccess(id) {
    const errorElem = document.getElementById(id + 'Error');
    const inputElem = document.getElementById(id);
    if (errorElem) errorElem.textContent = '';
    if (inputElem) {
      inputElem.classList.remove('input-error');
      inputElem.classList.add('input-success');
    }
  }

  // 1. Name Validation (Letters & spaces, 3-30 characters)
  const name = document.getElementById('name').value.trim();
  const nameRegex = /^[A-Za-z\s]{3,30}$/;
  if (!nameRegex.test(name)) {
    setError('name', 'Name must be 3-30 letters without numbers/symbols');
  } else {
    setSuccess('name');
  }

  // 2. Email Validation (Standard email pattern)
  const email = document.getElementById('email').value.trim();
```

```
  const emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
  if (!emailRegex.test(email)) {
    setError('email', 'Please enter a valid email address');
  } else {
    setSuccess('email');
  }

  // 3. Mobile Number Validation (10-digit mobile number)
  const mobile = document.getElementById('mobile').value.trim();
  const mobileRegex = /^[6-9]\d{9}$/;
  if (!mobileRegex.test(mobile)) {
    setError('mobile', 'Enter a valid 10-digit mobile number');
  } else {
    setSuccess('mobile');
  }

  // 4. Course Validation (Selection required)
  const course = document.getElementById('course').value;
  if (!course) {
    setError('course', 'Please select your course');
  } else {
    setSuccess('course');
  }

  // 5. Year Validation (1 to 4)
  const year = document.getElementById('year').value.trim();
  const yearRegex = /^[1-4]$/;
  if (!yearRegex.test(year)) {
    setError('year', 'Enter a valid year (1 to 4)');
  } else {
    setSuccess('year');
  }

  // 6. Gender Validation (Radio button checked)
  const genderChecked = document.querySelector('input[name="gender"]:checked');
  const genderError = document.getElementById('genderError');
  if (!genderChecked) {
    if (genderError) genderError.textContent = 'Please select your gender';
    isValid = false;
  } else {
    if (genderError) genderError.textContent = '';
  }

  // 7. Password Validation (At least 6 characters with letters and numbers)
  const password = document.getElementById('password').value;
  const passwordRegex = /^(?=.*[A-Za-z])(?=.*\d)[A-Za-z\d@$!%*#?&]{6,}$/;
  if (!passwordRegex.test(password)) {
    setError('password', 'Password must have at least 6 characters (letters &
numbers)');
  } else {
    setSuccess('password');
  }

  // 8. Confirm Password Validation (Must match password)
  const confirmPassword = document.getElementById('confirmPassword').value;
  if (confirmPassword === '' || confirmPassword !== password) {
    setError('confirmPassword', 'Passwords do not match');
  } else {
    setSuccess('confirmPassword');
  }
```

```
  // 9. Terms Acceptance (Checkbox must be checked)
  const terms = document.getElementById('terms');
  const termsError = document.getElementById('termsError');
  if (!terms.checked) {
    if (termsError) termsError.textContent = 'You must accept the Terms &
Conditions';
    isValid = false;
  } else {
    if (termsError) termsError.textContent = '';
  }

  // Final Action on Successful Validation
  if (isValid) {
    alert('🎉 Student Registration Successful! Redirecting to Dashboard...');
    window.location.href = 'dashboard.html';
  }

  return isValid;
}

// ============================================================================
// 8. PRACTICAL 6: FETCH API, DYNAMIC RENDERING, SEARCH, FILTER, SORT & PAGINATION
// ============================================================================
let allEvents = [];
let filteredEvents = [];
let currentPage = 1;
const itemsPerPage = 4;
let activeEventTitle = '';

// Fallback dataset in case local file:// security limits fetch()
const fallbackEvents = [
  {
    "id": 1,
    "title": "SPOURAL - 2026",
    "category": "SPORTS",
    "date": "2026-10-25",
    "displayDate": "October 25, 2026",
    "venue": "College Main Ground",
    "description": "Annual Inter-College Sports & Athletics Championship"
  },
  {
    "id": 2,
    "title": "Cyber Security Workshop",
    "category": "TECH",
    "date": "2026-11-10",
    "displayDate": "November 10, 2026",
    "venue": "Auditorium B",
    "description": "Hands-on Ethical Hacking & Network Defense Bootcamp"
  },
  {
    "id": 3,
    "title": "AI & Robotics Workshop",
    "category": "WORKSHOP",
    "date": "2026-11-22",
    "displayDate": "November 22, 2026",
    "venue": "Computer Lab 3",
    "description": "Practical Machine Learning, Arduino & Autonomous Bots Session"
  },
  {
    "id": 4,
    "title": "Tarang Cultural Fest",
    "category": "CULTURAL",
```

```
    "date": "2026-12-05",
    "displayDate": "December 05, 2026",
    "venue": "Open Air Amphitheater",
    "description": "Annual Youth Festival featuring Music, Drama, Dance & Art"
  },
  {
    "id": 5,
    "title": "WDF Web Development Hackathon",
    "category": "TECH",
    "date": "2026-12-15",
    "displayDate": "December 15, 2026",
    "venue": "IT Lab 2",
    "description": "24-hour Full-Stack Web App Development Challenge"
  },
  {
    "id": 6,
    "title": "DSA CodeSprint Contest",
    "category": "ACADEMIC",
    "date": "2026-12-20",
    "displayDate": "December 20, 2026",
    "venue": "Online Portal & Lab 1",
    "description": "Algorithmic Problem Solving & Competitive Coding Battle"
  },
  {
    "id": 7,
    "title": "Java OOP Project Exhibition",
    "category": "ACADEMIC",
    "date": "2026-12-28",
    "displayDate": "December 28, 2026",
    "venue": "Seminar Hall A",
    "description": "Semester 3 Java Object-Oriented Software Projects Showcase"
  },
  {
    "id": 8,
    "title": "Networking & FCN Hands-on Lab",
    "category": "WORKSHOP",
    "date": "2027-01-08",
    "displayDate": "January 08, 2027",
    "venue": "Cisco Networking Lab",
    "description": "Practical Packet Tracer, Subnetting & Network Simulation"
  }
];

// Load external JSON data via Fetch API
function loadEventsData() {
  const container = document.getElementById('events-list');
  if (!container) return; // Not on events page

  fetch('events.json')
    .then(response => {
      if (!response.ok) throw new Error('Network error loading events.json');
      return response.json();
    })
    .then(data => {
      allEvents = data;
      applyEventFilters();
    })
    .catch(err => {
      console.warn('Fetch fallback activated (e.g. file:// protocol):', err);
      allEvents = fallbackEvents;
      applyEventFilters();
    });
```

```
}

// Combined Search, Filter & Sort Logic
function applyEventFilters() {
  const searchInput = document.getElementById('search');
  const categorySelect = document.getElementById('categoryFilter');
  const sortSelect = document.getElementById('sortOrder');

  const searchTerm = searchInput ? searchInput.value.toLowerCase().trim() : '';
  const selectedCategory = categorySelect ? categorySelect.value : 'ALL';
  const sortOrder = sortSelect ? sortSelect.value : 'date-asc';

  // 1. Filter by Search Query & Category
  filteredEvents = allEvents.filter(event => {
    const matchesSearch =
      event.title.toLowerCase().includes(searchTerm) ||
      event.venue.toLowerCase().includes(searchTerm) ||
      event.description.toLowerCase().includes(searchTerm) ||
      event.category.toLowerCase().includes(searchTerm);

    const matchesCategory =
      selectedCategory === 'ALL' || event.category === selectedCategory;

    return matchesSearch && matchesCategory;
  });

  // 2. Sort Events
  filteredEvents.sort((a, b) => {
    if (sortOrder === 'name-asc') {
      return a.title.localeCompare(b.title);
    } else if (sortOrder === 'name-desc') {
      return b.title.localeCompare(a.title);
    } else if (sortOrder === 'date-desc') {
      return new Date(b.date) - new Date(a.date);
    } else {
      // date-asc (default)
      return new Date(a.date) - new Date(b.date);
    }
  });

  // Reset to first page whenever filter changes
  currentPage = 1;
  renderEventsPage();
}

// Dynamic Rendering with Pagination
function renderEventsPage() {
  const container = document.getElementById('events-list');
  const paginationContainer = document.getElementById('pagination-controls');
  if (!container) return;

  // Handle empty state
  if (filteredEvents.length === 0) {
    container.innerHTML = `
      <div class="table1" style="text-align: center; padding: 36px 20px;">
        <h3 style="color: #EC9F05; margin-bottom: 8px; font-size: 20px;">No events
found</h3>
        <p style="color: #CBD5E1; font-size: 15px;">No campus events match your
search query or selected filter.</p>
```

```
        <button onclick="resetEventFilters()" style="margin-top: 14px; padding: 8px
20px;">Reset Filters</button>
      </div>
    `;
    if (paginationContainer) paginationContainer.innerHTML = '';
    return;
  }

  // Calculate slice for current page
  const startIndex = (currentPage - 1) * itemsPerPage;
  const endIndex = startIndex + itemsPerPage;
  const pageItems = filteredEvents.slice(startIndex, endIndex);

  // Generate HTML for events
  let html = '';
  pageItems.forEach(event => {
    html += `
      <div class="table1 event-card-item">
        <div class="event-badge">${event.category}</div>
        <div style="flex: 1; min-width: 260px;">
          <h3 style="font-size: 22px; color: #EC9F05; margin-bottom:
6px;">${event.title}</h3>
          <p style="font-size: 15px; margin-bottom: 4px;">
            <strong>📅 Date:</strong> ${event.displayDate}  |  <strong>📍
Venue:</strong> ${event.venue}
          </p>
          <p style="font-size: 14px; color: #CBD5E1;">${event.description}</p>
        </div>
        <div style="flex-shrink: 0;">
          <button onclick="openRegisterModal('${event.title.replace(/'/g, "\\'")}',
'${event.displayDate}', '${event.venue}')" style="padding: 10px
24px;">Register</button>
        </div>
      </div>
    `;
  });

  container.innerHTML = html;
  renderPaginationButtons();
}

// Render Pagination Controls
function renderPaginationButtons() {
  const paginationContainer = document.getElementById('pagination-controls');
  if (!paginationContainer) return;

  const totalPages = Math.ceil(filteredEvents.length / itemsPerPage);
  if (totalPages <= 1) {
    paginationContainer.innerHTML = '';
    return;
  }

  let paginationHtml = '';

  // Previous Button
  paginationHtml += `
    <button class="pagination-btn" onclick="changeEventsPage(${currentPage - 1})"
${currentPage === 1 ? 'disabled' : ''}>
      &laquo; Prev
    </button>
  `;
```

```
  // Page Numbers
  for (let i = 1; i <= totalPages; i++) {
    paginationHtml += `
      <button class="pagination-btn ${i === currentPage ? 'active' : ''}"
onclick="changeEventsPage(${i})">
        ${i}
      </button>
    `;
  }

  // Next Button
  paginationHtml += `
    <button class="pagination-btn" onclick="changeEventsPage(${currentPage + 1})"
${currentPage === totalPages ? 'disabled' : ''}>
      Next &raquo;
    </button>
  `;

  paginationContainer.innerHTML = paginationHtml;
}

// Change Page Handler
function changeEventsPage(pageNumber) {
  const totalPages = Math.ceil(filteredEvents.length / itemsPerPage);
  if (pageNumber < 1 || pageNumber > totalPages) return;
  currentPage = pageNumber;
  renderEventsPage();
  // Smooth scroll back to top of events list
  const container = document.getElementById('events-toolbar-card');
  if (container) {
    container.scrollIntoView({ behavior: 'smooth', block: 'start' });
  }
}

// Reset Search & Filters
function resetEventFilters() {
  const searchInput = document.getElementById('search');
  const categorySelect = document.getElementById('categoryFilter');
  const sortSelect = document.getElementById('sortOrder');

  if (searchInput) searchInput.value = '';
  if (categorySelect) categorySelect.value = 'ALL';
  if (sortSelect) sortSelect.value = 'date-asc';

  applyEventFilters();
}

// Universal Registration Modal Handlers
function openRegisterModal(title, date, venue) {
  activeEventTitle = title;
  const titleEl = document.getElementById('modalEventTitle');
  const detailsEl = document.getElementById('modalEventDetails');

  if (titleEl) titleEl.textContent = 'Register for ' + title;
  if (detailsEl) detailsEl.innerHTML = `<strong>Date:</strong> ${date} &bull;
<strong>Venue:</strong> ${venue}<br><br>Confirm your participation to secure your
registration slot.`;

  openModal('dynamicEventModal');
```

```
}

function confirmEventRegistration() {
  alert('🎉 Successfully registered for ' + (activeEventTitle || 'the event') +
'!');
  closeModal('dynamicEventModal');
}

// Automatically load events if on events page
document.addEventListener('DOMContentLoaded', () => {
  loadEventsData();
});

// ========================================================================
// 9. AUTHENTICATION & ACCESS CONTROL (STUDENT & ADMIN CREDENTIALS)
// ========================================================================
const PORTAL_CREDENTIALS = {
  student: {
    identifiers: ['student@studenthub.com', '25dcs200', 'john.doe@studenthub.edu'],
    password: 'student123',
    role: 'student',
    name: 'John Doe',
    studentId: '25DCS200',
    course: 'B.Tech CSE - 3rd Sem'
  },
  admin: {
    identifiers: ['admin@studenthub.com', 'admin'],
    password: 'admin123',
    role: 'admin',
    name: 'Portal Administrator',
    studentId: 'ADMIN-01',
    course: 'Faculty / Administration'
  }
};

// Handle Login Form Submission
function handlePortalLogin(event) {
  if (event) event.preventDefault();

  const idInput = document.getElementById('email');
  const passInput = document.getElementById('password');
  const errorMsg = document.getElementById('login-error-msg');

  if (!idInput || !passInput) return;

  const enteredId = idInput.value.trim().toLowerCase();
  const enteredPass = passInput.value.trim();

  // 1. Check Student Credentials
  if (PORTAL_CREDENTIALS.student.identifiers.includes(enteredId) && enteredPass ===
PORTAL_CREDENTIALS.student.password) {
    sessionStorage.setItem('currentUser',
JSON.stringify(PORTAL_CREDENTIALS.student));
    alert('☑ Welcome, John Doe! Logged in as Student.');
    window.location.href = 'dashboard.html';
    return;
  }

  // 2. Check Admin Credentials
```

```
  if (PORTAL_CREDENTIALS.admin.identifiers.includes(enteredId) && enteredPass ===
PORTAL_CREDENTIALS.admin.password) {
    sessionStorage.setItem('currentUser',
JSON.stringify(PORTAL_CREDENTIALS.admin));
    alert('🌓 Welcome, Administrator! Full Dashboard and Admin Access Granted.');
    window.location.href = 'dashboard.html';
    return;
  }

  // 3. Failed Login
  if (errorMsg) {
    errorMsg.textContent = '❌ Invalid Credentials! Use 25DCS200 / student123 or
admin@studenthub.com / admin123.';
    errorMsg.style.display = 'block';
  } else {
    alert('❌ Invalid Credentials!\n\n• Student: 25DCS200 / student123\n• Admin:
admin@studenthub.com / admin123');
  }
}

// Quick Fill Demo Credentials
function fillCredentials(id, pass) {
  const idInput = document.getElementById('email');
  const passInput = document.getElementById('password');
  const errorMsg = document.getElementById('login-error-msg');

  if (idInput) idInput.value = id;
  if (passInput) passInput.value = pass;
  if (errorMsg) errorMsg.style.display = 'none';
}

// Protect Dashboard (Redirect if not logged in)
function checkDashboardAuth() {
  const sessionUser = sessionStorage.getItem('currentUser');
  if (!sessionUser) {
    alert('🔒 Access Denied! Please login with Student or Admin credentials to view
the dashboard.');
    window.location.href = 'login.html';
    return false;
  }

  const user = JSON.parse(sessionUser);

  // If Admin is viewing, show special admin access badge
  const adminBadge = document.getElementById('admin-access-indicator');
  if (adminBadge && user.role === 'admin') {
    adminBadge.style.display = 'block';
  }

  return true;
}

// Logout Handler
function handleLogout() {
  sessionStorage.removeItem('currentUser');
  alert('👋 You have been logged out successfully.');
  window.location.href = 'login.html';
}
```

```
// Protect Admin Panel (Only Admin Allowed)
function checkAdminAuth() {
  const sessionUser = sessionStorage.getItem('currentUser');
  if (!sessionUser) {
    alert('🔒 Admin Access Only! Please login with Administrator credentials.');
    window.location.href = 'login.html';
    return false;
  }

  const user = JSON.parse(sessionUser);
  if (user.role !== 'admin') {
    alert('⛔ Access Restricted! Only Administrators can access the Admin Panel.');
    window.location.href = 'dashboard.html';
    return false;
  }

  return true;
}
```

Practical No. 5 - Registration Form with Frontend Validation and User-Friendly Error Handling

Question: Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

Key Questions

Q: Are correct input types and attributes used?

Answer: Yes. The form uses text, email, tel, number, password, select, radio, and checkbox controls. Labels, placeholders, min and max values, and validation checks support correct input.

Q: Are validation errors displayed near the relevant fields?

Answer: Yes. Each important field has a nearby error element such as nameError or emailError. The validation function writes the message there and adds an input-error class.

Q: Is password strength checked?

Answer: Yes. A regular expression checks that the password has at least six characters, at least one letter, and at least one number. The confirm password is also compared with it.

Q: Is the form accessible using keyboard and screen-reader-friendly labels?

Answer: The controls are connected to visible labels using for and id values, and the radio buttons have a shared name. This gives keyboard and assistive-technology users understandable fields.

Analysis

A: The registration form collects the required student details and validates them before accepting the form. Invalid fields receive a message and an error class, while valid fields receive a success class.

Interpretation

I: Client-side validation gives immediate feedback and prevents incomplete or incorrectly formatted data from being submitted. Server-side validation would still be needed in a real application.

HTML Code - register.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Student Registration</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
```

```
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="login.html">Login</a>
    </nav>
  </div>

  <main>
    <!-- Perfect Student Registration Box -->
    <div class="auth-card register-card">
      <h2>STUDENT REGISTRATION</h2>

      <form id="registrationForm" onsubmit="return validateRegistration(event)">
        <div class="form-grid">
          <!-- 1. Full Name -->
          <div class="form-group">
            <label for="name">Full Name</label>
            <input type="text" id="name" placeholder="e.g. John Doe">
            <small class="error-text" id="nameError"></small>
          </div>

          <!-- 2. Email -->
          <div class="form-group">
            <label for="email">Email Address</label>
            <input type="email" id="email" placeholder="student@college.edu">
            <small class="error-text" id="emailError"></small>
          </div>

          <!-- 3. Mobile Number -->
          <div class="form-group">
            <label for="mobile">Mobile Number</label>
            <input type="tel" id="mobile" placeholder="10-digit mobile number">
            <small class="error-text" id="mobileError"></small>
          </div>

          <!-- 4. Course -->
          <div class="form-group">
            <label for="course">Course</label>
            <select id="course">
              <option value="">-- Select Course --</option>
              <option value="B.Tech Computer Science">B.Tech Computer
Science</option>
              <option value="B.Tech Information Technology">B.Tech Information
Technology</option>
              <option value="BCA">BCA</option>
              <option value="MCA">MCA</option>
              <option value="B.Sc IT">B.Sc IT</option>
            </select>
            <small class="error-text" id="courseError"></small>
          </div>
```

```
          <!-- 5. Year -->
          <div class="form-group">
            <label for="year">Year of Study</label>
            <input type="number" id="year" min="1" max="4" placeholder="Enter Year
(1 - 4)">
            <small class="error-text" id="yearError"></small>
          </div>

          <!-- 6. Gender -->
          <div class="form-group">
            <label>Gender</label>
            <div class="radio-group">
              <label for="male">
                <input type="radio" name="gender" id="male" value="male"> Male
              </label>
              <label for="female">
                <input type="radio" name="gender" id="female" value="female">
Female
              </label>
              <label for="other">
                <input type="radio" name="gender" id="other" value="other"> Other
              </label>
            </div>
            <small class="error-text" id="genderError"></small>
          </div>

          <!-- 7. Password -->
          <div class="form-group">
            <label for="password">Password</label>
            <input type="password" id="password" placeholder="Minimum 6 chars
(letters & numbers)">
            <small class="error-text" id="passwordError"></small>
          </div>

          <!-- 8. Confirm Password -->
          <div class="form-group">
            <label for="confirmPassword">Confirm Password</label>
            <input type="password" id="confirmPassword" placeholder="Re-type
password">
            <small class="error-text" id="confirmPasswordError"></small>
          </div>
        </div>

        <!-- 9. Terms Acceptance -->
        <div class="form-options center-terms">
          <label for="terms">
            <input type="checkbox" id="terms"> I accept the Terms &amp;amp; Conditions
          </label>
          <small class="error-text" id="termsError"></small>
        </div>

        <button type="submit" class="auth-btn">Register</button>

        <p class="auth-footer-text">
          Already registered? <a href="login.html">Login here</a>
        </p>
      </form>
    </div>
  </main>
```

```
  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

JavaScript Code - validation in script.js

```
/* ================================================================================
   STUDENT HUB PORTAL - SIMPLE DYNAMIC UI & FORM VALIDATION JAVASCRIPT
   ================================================================================ */

// 1. LIGHT / DARK THEME SWITCHER
function toggleTheme() {
  document.body.classList.toggle('light-theme');
  const isLight = document.body.classList.contains('light-theme');
  localStorage.setItem('theme', isLight ? 'light' : 'dark');
}

// Load saved theme on startup
if (localStorage.getItem('theme') === 'light') {
  document.body.classList.add('light-theme');
}

// 2. HAMBURGER MENU (MOBILE NAVIGATION)
function toggleMenu() {
  const menu = document.getElementById('nav-menu');
  if (menu) {
    menu.classList.toggle('show');
  }
}

// 3. NOTIFICATION BANNER
function closeBanner() {
  const banner = document.getElementById('notification-banner');
  if (banner) {
    banner.style.display = 'none';
  }
}

// 4. COLLAPSIBLE FAQ ACCORDION
function toggleFaq(btn) {
  const answer = btn.nextElementSibling;
  const icon = btn.querySelector('.faq-icon');

  if (answer.style.display === 'block') {
    answer.style.display = 'none';
    if (icon) icon.textContent = '+';
  } else {
    answer.style.display = 'block';
    if (icon) icon.textContent = '-';
  }
```

```
}

// 5. MODAL POPUP (OPEN / CLOSE)
function openModal(modalId) {
  const modal = document.getElementById(modalId);
  if (modal) {
    modal.style.display = 'flex';
  }
}

function closeModal(modalId) {
  const modal = document.getElementById(modalId);
  if (modal) {
    modal.style.display = 'none';
  }
}

// Close modal if user clicks outside the modal box
window.onclick = function(event) {
  if (event.target.classList.contains('modal')) {
    event.target.style.display = 'none';
  }
};

// 6. IMAGE / CONTENT SLIDER
let currentSlide = 0;

function showSlide(index) {
  const slides = document.querySelectorAll('.slide');
  if (slides.length === 0) return;

  if (index >= slides.length) {
    currentSlide = 0;
  } else if (index < 0) {
    currentSlide = slides.length - 1;
  } else {
    currentSlide = index;
  }

  slides.forEach((slide, i) => {
    slide.style.display = (i === currentSlide) ? 'block' : 'none';
  });
}

function changeSlide(step) {
  showSlide(currentSlide + step);
}

// Initialize first slide on page load
document.addEventListener('DOMContentLoaded', () => {
  showSlide(0);
});



// 7. STUDENT REGISTRATION FORM VALIDATION (REGEX)
function validateRegistration(event) {
  event.preventDefault(); // Stop normal form submission

  let isValid = true;
```

```
// Helper functions for displaying error / success
function setError(id, message) {
  const errorElem = document.getElementById(id + 'Error');
  const inputElem = document.getElementById(id);
  if (errorElem) errorElem.textContent = message;
  if (inputElem) {
    inputElem.classList.add('input-error');
    inputElem.classList.remove('input-success');
  }
  isValid = false;
}

function setSuccess(id) {
  const errorElem = document.getElementById(id + 'Error');
  const inputElem = document.getElementById(id);
  if (errorElem) errorElem.textContent = '';
  if (inputElem) {
    inputElem.classList.remove('input-error');
    inputElem.classList.add('input-success');
  }
}

// 1. Name Validation (Letters & spaces, 3-30 characters)
const name = document.getElementById('name').value.trim();
const nameRegex = /^[A-Za-z\s]{3,30}$/;
if (!nameRegex.test(name)) {
  setError('name', 'Name must be 3-30 letters without numbers/symbols');
} else {
  setSuccess('name');
}

// 2. Email Validation (Standard email pattern)
const email = document.getElementById('email').value.trim();
const emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
if (!emailRegex.test(email)) {
  setError('email', 'Please enter a valid email address');
} else {
  setSuccess('email');
}

// 3. Mobile Number Validation (10-digit mobile number)
const mobile = document.getElementById('mobile').value.trim();
const mobileRegex = /^[6-9]\d{9}$/;
if (!mobileRegex.test(mobile)) {
  setError('mobile', 'Enter a valid 10-digit mobile number');
} else {
  setSuccess('mobile');
}

// 4. Course Validation (Selection required)
const course = document.getElementById('course').value;
if (!course) {
  setError('course', 'Please select your course');
} else {
  setSuccess('course');
}

// 5. Year Validation (1 to 4)
const year = document.getElementById('year').value.trim();
const yearRegex = /^[1-4]$/;
```

```
  if (!yearRegex.test(year)) {
    setError('year', 'Enter a valid year (1 to 4)');
  } else {
    setSuccess('year');
  }

  // 6. Gender Validation (Radio button checked)
  const genderChecked = document.querySelector('input[name="gender"]:checked');
  const genderError = document.getElementById('genderError');
  if (!genderChecked) {
    if (genderError) genderError.textContent = 'Please select your gender';
    isValid = false;
  } else {
    if (genderError) genderError.textContent = '';
  }

  // 7. Password Validation (At least 6 characters with letters and numbers)
  const password = document.getElementById('password').value;
  const passwordRegex = /^(?=.*[A-Za-z])(?=.*\d)[A-Za-z\d@$!%*#?&]{6,}$/;
  if (!passwordRegex.test(password)) {
    setError('password', 'Password must have at least 6 characters (letters &
numbers)');
  } else {
    setSuccess('password');
  }

  // 8. Confirm Password Validation (Must match password)
  const confirmPassword = document.getElementById('confirmPassword').value;
  if (confirmPassword === '' || confirmPassword !== password) {
    setError('confirmPassword', 'Passwords do not match');
  } else {
    setSuccess('confirmPassword');
  }

  // 9. Terms Acceptance (Checkbox must be checked)
  const terms = document.getElementById('terms');
  const termsError = document.getElementById('termsError');
  if (!terms.checked) {
    if (termsError) termsError.textContent = 'You must accept the Terms &
Conditions';
    isValid = false;
  } else {
    if (termsError) termsError.textContent = '';
  }

  // Final Action on Successful Validation
  if (isValid) {
    alert('🎉 Student Registration Successful! Redirecting to Dashboard...');
    window.location.href = 'dashboard.html';
  }

  return isValid;
}

// ========================================================================
// 8. PRACTICAL 6: FETCH API, DYNAMIC RENDERING, SEARCH, FILTER, SORT & PAGINATION
// ========================================================================
let allEvents = [];
let filteredEvents = [];
let currentPage = 1;
const itemsPerPage = 4;
```

```
let activeEventTitle = '';

// Fallback dataset in case local file:// security limits fetch()
const fallbackEvents = [
  {
    "id": 1,
    "title": "SPOURAL - 2026",
    "category": "SPORTS",
    "date": "2026-10-25",
    "displayDate": "October 25, 2026",
    "venue": "College Main Ground",
    "description": "Annual Inter-College Sports & Athletics Championship"
  },
  {
    "id": 2,
    "title": "Cyber Security Workshop",
    "category": "TECH",
    "date": "2026-11-10",
    "displayDate": "November 10, 2026",
    "venue": "Auditorium B",
    "description": "Hands-on Ethical Hacking & Network Defense Bootcamp"
  },
  {
    "id": 3,
    "title": "AI & Robotics Workshop",
    "category": "WORKSHOP",
    "date": "2026-11-22",
    "displayDate": "November 22, 2026",
    "venue": "Computer Lab 3",
    "description": "Practical Machine Learning, Arduino & Autonomous Bots Session"
  },
  {
    "id": 4,
    "title": "Tarang Cultural Fest",
    "category": "CULTURAL",
    "date": "2026-12-05",
    "displayDate": "December 05, 2026",
    "venue": "Open Air Amphitheater",
    "description": "Annual Youth Festival featuring Music, Drama, Dance & Art"
  },
  {
    "id": 5,
    "title": "WDF Web Development Hackathon",
    "category": "TECH",
    "date": "2026-12-15",
    "displayDate": "December 15, 2026",
    "venue": "IT Lab 2",
    "description": "24-hour Full-Stack Web App Development Challenge"
  },
  {
    "id": 6,
    "title": "DSA CodeSprint Contest",
    "category": "ACADEMIC",
    "date": "2026-12-20",
    "displayDate": "December 20, 2026",
    "venue": "Online Portal & Lab 1",
    "description": "Algorithmic Problem Solving & Competitive Coding Battle"
  },
  {
    "id": 7,
    "title": "Java OOP Project Exhibition",
    "category": "ACADEMIC",
    "date": "2026-12-28",
```

```
    "displayDate": "December 28, 2026",
    "venue": "Seminar Hall A",
    "description": "Semester 3 Java Object-Oriented Software Projects Showcase"
  },
  {
    "id": 8,
    "title": "Networking & FCN Hands-on Lab",
    "category": "WORKSHOP",
    "date": "2027-01-08",
    "displayDate": "January 08, 2027",
    "venue": "Cisco Networking Lab",
    "description": "Practical Packet Tracer, Subnetting & Network Simulation"
  }
];

// Load external JSON data via Fetch API
function loadEventsData() {
  const container = document.getElementById('events-list');
  if (!container) return; // Not on events page

  fetch('events.json')
    .then(response => {
      if (!response.ok) throw new Error('Network error loading events.json');
      return response.json();
    })
    .then(data => {
      allEvents = data;
      applyEventFilters();
    })
    .catch(err => {
      console.warn('Fetch fallback activated (e.g. file:// protocol):', err);
      allEvents = fallbackEvents;
      applyEventFilters();
    });
}

// Combined Search, Filter & Sort Logic
function applyEventFilters() {
  const searchInput = document.getElementById('search');
  const categorySelect = document.getElementById('categoryFilter');
  const sortSelect = document.getElementById('sortOrder');

  const searchTerm = searchInput ? searchInput.value.toLowerCase().trim() : '';
  const selectedCategory = categorySelect ? categorySelect.value : 'ALL';
  const sortOrder = sortSelect ? sortSelect.value : 'date-asc';

  // 1. Filter by Search Query & Category
  filteredEvents = allEvents.filter(event => {
    const matchesSearch =
      event.title.toLowerCase().includes(searchTerm) ||
      event.venue.toLowerCase().includes(searchTerm) ||
      event.description.toLowerCase().includes(searchTerm) ||
      event.category.toLowerCase().includes(searchTerm);

    const matchesCategory =
      selectedCategory === 'ALL' || event.category === selectedCategory;

    return matchesSearch && matchesCategory;
  });

  // 2. Sort Events
```

```
  filteredEvents.sort((a, b) => {
    if (sortOrder === 'name-asc') {
      return a.title.localeCompare(b.title);
    } else if (sortOrder === 'name-desc') {
      return b.title.localeCompare(a.title);
    } else if (sortOrder === 'date-desc') {
      return new Date(b.date) - new Date(a.date);
    } else {
      // date-asc (default)
      return new Date(a.date) - new Date(b.date);
    }
  });

  // Reset to first page whenever filter changes
  currentPage = 1;
  renderEventsPage();
}

// Dynamic Rendering with Pagination
function renderEventsPage() {
  const container = document.getElementById('events-list');
  const paginationContainer = document.getElementById('pagination-controls');
  if (!container) return;

  // Handle empty state
  if (filteredEvents.length === 0) {
    container.innerHTML = `
      <div class="table1" style="text-align: center; padding: 36px 20px;">
        <h3 style="color: #EC9F05; margin-bottom: 8px; font-size: 20px;">No events
found</h3>
        <p style="color: #CBD5E1; font-size: 15px;">No campus events match your
search query or selected filter.</p>
        <button onclick="resetEventFilters()" style="margin-top: 14px; padding: 8px
20px;">Reset Filters</button>
      </div>
    `;
    if (paginationContainer) paginationContainer.innerHTML = '';
    return;
  }

  // Calculate slice for current page
  const startIndex = (currentPage - 1) * itemsPerPage;
  const endIndex = startIndex + itemsPerPage;
  const pageItems = filteredEvents.slice(startIndex, endIndex);

  // Generate HTML for events
  let html = '';
  pageItems.forEach(event => {
    html += `
      <div class="table1 event-card-item">
        <div class="event-badge">${event.category}</div>
        <div style="flex: 1; min-width: 260px;">
          <h3 style="font-size: 22px; color: #EC9F05; margin-bottom:
6px;">${event.title}</h3>
          <p style="font-size: 15px; margin-bottom: 4px;">
            <strong>📅 Date:</strong> ${event.displayDate}  |  <strong>📍
Venue:</strong> ${event.venue}
          </p>
          <p style="font-size: 14px; color: #CBD5E1;">${event.description}</p>
        </div>
        <div style="flex-shrink: 0;">
```

```
          <button onclick="openRegisterModal('${event.title.replace(/'/g, "\\'")}',
'${event.displayDate}', '${event.venue}')" style="padding: 10px
24px;">Register</button>
        </div>
      </div>
    `;
  });

  container.innerHTML = html;
  renderPaginationButtons();
}

// Render Pagination Controls
function renderPaginationButtons() {
  const paginationContainer = document.getElementById('pagination-controls');
  if (!paginationContainer) return;

  const totalPages = Math.ceil(filteredEvents.length / itemsPerPage);
  if (totalPages <= 1) {
    paginationContainer.innerHTML = '';
    return;
  }

  let paginationHtml = '';

  // Previous Button
  paginationHtml += `
    <button class="pagination-btn" onclick="changeEventsPage(${currentPage - 1})"
${currentPage === 1 ? 'disabled' : ''}>
      &laquo; Prev
    </button>
  `;

  // Page Numbers
  for (let i = 1; i <= totalPages; i++) {
    paginationHtml += `
      <button class="pagination-btn ${i === currentPage ? 'active' : ''}"
onclick="changeEventsPage(${i})">
        ${i}
      </button>
    `;
  }

  // Next Button
  paginationHtml += `
    <button class="pagination-btn" onclick="changeEventsPage(${currentPage + 1})"
${currentPage === totalPages ? 'disabled' : ''}>
      Next &raquo;
    </button>
  `;

  paginationContainer.innerHTML = paginationHtml;
}

// Change Page Handler
function changeEventsPage(pageNumber) {
  const totalPages = Math.ceil(filteredEvents.length / itemsPerPage);
  if (pageNumber < 1 || pageNumber > totalPages) return;
  currentPage = pageNumber;
  renderEventsPage();
```

```
  // Smooth scroll back to top of events list
  const container = document.getElementById('events-toolbar-card');
  if (container) {
    container.scrollIntoView({ behavior: 'smooth', block: 'start' });
  }
}

// Reset Search & Filters
function resetEventFilters() {
  const searchInput = document.getElementById('search');
  const categorySelect = document.getElementById('categoryFilter');
  const sortSelect = document.getElementById('sortOrder');

  if (searchInput) searchInput.value = '';
  if (categorySelect) categorySelect.value = 'ALL';
  if (sortSelect) sortSelect.value = 'date-asc';

  applyEventFilters();
}

// Universal Registration Modal Handlers
function openRegisterModal(title, date, venue) {
  activeEventTitle = title;
  const titleEl = document.getElementById('modalEventTitle');
  const detailsEl = document.getElementById('modalEventDetails');

  if (titleEl) titleEl.textContent = 'Register for ' + title;
  if (detailsEl) detailsEl.innerHTML = `<strong>Date:</strong> ${date} &bull;
<strong>Venue:</strong> ${venue}<br><br>Confirm your participation to secure your
registration slot.`;

  openModal('dynamicEventModal');
}

function confirmEventRegistration() {
  alert('🎉 Successfully registered for ' + (activeEventTitle || 'the event') +
'!');
  closeModal('dynamicEventModal');
}

// Automatically load events if on events page
document.addEventListener('DOMContentLoaded', () => {
  loadEventsData();
});

// ============================================================================
// 9. AUTHENTICATION & ACCESS CONTROL (STUDENT & ADMIN CREDENTIALS)
// ============================================================================
const PORTAL_CREDENTIALS = {
  student: {
    identifiers: ['student@studenthub.com', '25dcs200', 'john.doe@studenthub.edu'],
    password: 'student123',
    role: 'student',
    name: 'John Doe',
    studentId: '25DCS200',
    course: 'B.Tech CSE - 3rd Sem'
  },
  admin: {
    identifiers: ['admin@studenthub.com', 'admin'],
    password: 'admin123',
```

```
    role: 'admin',
    name: 'Portal Administrator',
    studentId: 'ADMIN-01',
    course: 'Faculty / Administration'
  }
};

// Handle Login Form Submission
function handlePortalLogin(event) {
  if (event) event.preventDefault();

  const idInput = document.getElementById('email');
  const passInput = document.getElementById('password');
  const errorMsg = document.getElementById('login-error-msg');

  if (!idInput || !passInput) return;

  const enteredId = idInput.value.trim().toLowerCase();
  const enteredPass = passInput.value.trim();

  // 1. Check Student Credentials
  if (PORTAL_CREDENTIALS.student.identifiers.includes(enteredId) && enteredPass ===
PORTAL_CREDENTIALS.student.password) {
    sessionStorage.setItem('currentUser',
JSON.stringify(PORTAL_CREDENTIALS.student));
    alert('☑ Welcome, John Doe! Logged in as Student.');
    window.location.href = 'dashboard.html';
    return;
  }

  // 2. Check Admin Credentials
  if (PORTAL_CREDENTIALS.admin.identifiers.includes(enteredId) && enteredPass ===
PORTAL_CREDENTIALS.admin.password) {
    sessionStorage.setItem('currentUser',
JSON.stringify(PORTAL_CREDENTIALS.admin));
    alert('◐ Welcome, Administrator! Full Dashboard and Admin Access Granted.');
    window.location.href = 'dashboard.html';
    return;
  }

  // 3. Failed Login
  if (errorMsg) {
    errorMsg.textContent = '✕ Invalid Credentials! Use 25DCS200 / student123 or
admin@studenthub.com / admin123.';
    errorMsg.style.display = 'block';
  } else {
    alert('✕ Invalid Credentials!\n\n• Student: 25DCS200 / student123\n• Admin:
admin@studenthub.com / admin123');
  }
}

// Quick Fill Demo Credentials
function fillCredentials(id, pass) {
  const idInput = document.getElementById('email');
  const passInput = document.getElementById('password');
  const errorMsg = document.getElementById('login-error-msg');

  if (idInput) idInput.value = id;
  if (passInput) passInput.value = pass;
```

```
  if (errorMsg) errorMsg.style.display = 'none';
}

// Protect Dashboard (Redirect if not logged in)
function checkDashboardAuth() {
  const sessionUser = sessionStorage.getItem('currentUser');
  if (!sessionUser) {
    alert('🔒 Access Denied! Please login with Student or Admin credentials to view
the dashboard.');
    window.location.href = 'login.html';
    return false;
  }

  const user = JSON.parse(sessionUser);

  // If Admin is viewing, show special admin access badge
  const adminBadge = document.getElementById('admin-access-indicator');
  if (adminBadge && user.role === 'admin') {
    adminBadge.style.display = 'block';
  }

  return true;
}

// Logout Handler
function handleLogout() {
  sessionStorage.removeItem('currentUser');
  alert('👋 You have been logged out successfully.');
  window.location.href = 'login.html';
}

// Protect Admin Panel (Only Admin Allowed)
function checkAdminAuth() {
  const sessionUser = sessionStorage.getItem('currentUser');
  if (!sessionUser) {
    alert('🔒 Admin Access Only! Please login with Administrator credentials.');
    window.location.href = 'login.html';
    return false;
  }

  const user = JSON.parse(sessionUser);
  if (user.role !== 'admin') {
    alert('⛔ Access Restricted! Only Administrators can access the Admin Panel.');
    window.location.href = 'dashboard.html';
    return false;
  }

  return true;
}
```

Practical No. 6 - Rendering External JSON Data using Fetch API, Search and Filter

Question: Display event lists, student profiles, notices, or FAQs from an external JSON file using Fetch API. Implement dynamic rendering, search, filter, sorting, and pagination.

Key Questions and Answers

Q: How is JSON fetched, parsed, and rendered?

Answer: The fetch function requests events.json and returns a response. The response is converted to JavaScript objects with response.json(), and the objects are inserted into the page as event cards.

Q: Which array methods are used for search, filter, sort, and pagination?

Answer: filter() selects matching events, sort() orders events by date, map() creates display markup, and slice() selects the records for the current page.

Q: How are loading and error states handled?

Answer: The page initially shows a loading message. If the request fails or the response is not successful, catch() displays an error message and clears the event list.

Q: How is modularity maintained in JavaScript files?

Answer: The code is divided into small functions such as loadEvents, addCategories, applyFilters, renderEvents, renderPagination, and formatDate. Each function has one clear responsibility.

Analysis

A: The application loads event records from events.json instead of writing every event directly in HTML. Search, category filtering, date sorting, and pagination are applied to the JavaScript array before the visible cards are updated.

Interpretation

I: This approach is useful for modern web applications because the same page can display changing data without rewriting the HTML. In a real project, the JSON file can later be replaced by an API endpoint.

Required Practical 6 Code

Testing Note: Run the files through a local server such as Live Server. Fetch may be blocked when the HTML file is opened directly with a file path in the browser.

HTML Code - events-fetch.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Events</title>
  <link rel="stylesheet" href="practical6.css">
</head>
<body>
  <main class="container">
    <h1>Student Hub Events</h1>
    <p>Events are loaded from an external JSON file using the Fetch API.</p>

    <section class="controls" aria-label="Event controls">
```

```
      <label for="searchInput">Search</label>
      <input id="searchInput" type="search" placeholder="Search by event or venue">
      <label for="categoryFilter">Category</label>
      <select id="categoryFilter">
        <option value="all">All categories</option>
      </select>
      <label for="sortSelect">Sort by date</label>
      <select id="sortSelect">
        <option value="asc">Oldest first</option>
        <option value="desc">Newest first</option>
      </select>
    </section>

    <p id="status" class="status" role="status">Loading events...</p>
    <section id="eventList" class="event-grid" aria-live="polite"></section>
    <nav id="pagination" class="pagination" aria-label="Event pages"></nav>
  </main>
  <script src="practical6.js"></script>
</body>
</html>
```

JSON Code - events.json

```
[
  {"id": 1, "title": "SPOURAL 2026", "date": "2026-10-25", "venue": "College Main
Ground", "category": "Sports"},
  {"id": 2, "title": "Cyber Security Workshop", "date": "2026-11-10", "venue":
"Auditorium B", "category": "Technical"},
  {"id": 3, "title": "Career Guidance Seminar", "date": "2026-11-22", "venue":
"Seminar Hall", "category": "Career"},
  {"id": 4, "title": "Student Hackathon", "date": "2026-12-05", "venue":
"Innovation Lab", "category": "Technical"},
  {"id": 5, "title": "Annual Cultural Night", "date": "2026-12-18", "venue": "Open
Air Theatre", "category": "Cultural"},
  {"id": 6, "title": "Resume Building Session", "date": "2027-01-08", "venue":
"Placement Cell", "category": "Career"}
]
```

JavaScript Code - practical6.js

```
const state = {
  events: [],
  filtered: [],
  page: 1,
  pageSize: 4
};

const list = document.getElementById("eventList");
const status = document.getElementById("status");
const searchInput = document.getElementById("searchInput");
const categoryFilter = document.getElementById("categoryFilter");
const sortSelect = document.getElementById("sortSelect");
const pagination = document.getElementById("pagination");

async function loadEvents() {
  try {
    const response = await fetch("events.json");
    if (!response.ok) throw new Error("Unable to load event data");
    state.events = await response.json();
    addCategories();
    applyFilters();
  } catch (error) {
    status.textContent = error.message;
    list.innerHTML = "";
```

```
  }
}

function addCategories() {
  const categories = [...new Set(state.events.map(event =>
event.category))].sort();
  categories.forEach(category => {
    const option = document.createElement("option");
    option.value = category;
    option.textContent = category;
    categoryFilter.appendChild(option);
  });
}

function applyFilters() {
  const term = searchInput.value.trim().toLowerCase();
  const category = categoryFilter.value;
  const direction = sortSelect.value === "asc" ? 1 : -1;

  state.filtered = state.events
    .filter(event => {
      const searchable = `${event.title} ${event.venue}`.toLowerCase();
      return searchable.includes(term) && (category === "all" || event.category ===
category);
    })
    .sort((a, b) => (new Date(a.date) - new Date(b.date)) * direction);

  state.page = 1;
  renderEvents();
  renderPagination();
}

function renderEvents() {
  const start = (state.page - 1) * state.pageSize;
  const pageItems = state.filtered.slice(start, start + state.pageSize);
  status.textContent = `${state.filtered.length} event(s) found`;
  list.innerHTML = pageItems.length ? pageItems.map(event => `
    <article class="event-card">
      <h2>${event.title}</h2>
      <p><strong>Date:</strong> ${formatDate(event.date)}</p>
      <p><strong>Venue:</strong> ${event.venue}</p>
      <p><strong>Category:</strong> ${event.category}</p>
    </article>
  `).join("") : "<p>No matching events found.</p>";
}

function renderPagination() {
  const totalPages = Math.ceil(state.filtered.length / state.pageSize);
  pagination.innerHTML = "";
  for (let page = 1; page <= totalPages; page += 1) {
    const button = document.createElement("button");
    button.textContent = page;
    button.setAttribute("aria-label", `Go to page ${page}`);
    button.className = page === state.page ? "active" : "";
    button.addEventListener("click", () => {
      state.page = page;
      renderEvents();
      renderPagination();
    });
    pagination.appendChild(button);
  }
```

```
}

function formatDate(value) {
  return new Date(value).toLocaleDateString("en-IN", {
    day: "2-digit", month: "short", year: "numeric"
  });
}

searchInput.addEventListener("input", applyFilters);
categoryFilter.addEventListener("change", applyFilters);
sortSelect.addEventListener("change", applyFilters);
loadEvents();
```

CSS Code - practical6.css

```
* { box-sizing: border-box; }
body { margin: 0; font-family: Arial, sans-serif; background: #0b162c; color:
#f5f7fb; }
.container { width: min(100% - 32px, 1000px); margin: 32px auto; }
h1 { color: #ec9f05; }
.controls { display: grid; grid-template-columns: auto 1fr auto 1fr auto 1fr; gap:
10px; align-items: center; padding: 16px; background: #152644; border-radius: 10px;
}
input, select { width: 100%; padding: 10px; border: 1px solid #7184a8; border-
radius: 6px; background: #ffffff; color: #17233b; }
.status { color: #b7c5dd; }
.event-grid { display: grid; grid-template-columns: repeat(2, 1fr); gap: 16px; }
.event-card { padding: 18px; border: 1px solid #52698e; border-radius: 10px;
background: #13213a; }
.event-card h2 { margin-top: 0; color: #ec9f05; font-size: 20px; }
.pagination { display: flex; gap: 8px; margin-top: 20px; }
.pagination button { padding: 8px 12px; border: 0; border-radius: 5px; cursor:
pointer; }
.pagination .active { background: #ec9f05; color: #17233b; font-weight: bold; }
@media (max-width: 720px) { .controls, .event-grid { grid-template-columns: 1fr; }
}
```

Complete Source Code Appendix

Included Files: The following appendix contains every code file currently present in the final folder: HTML, CSS, JavaScript, JSON, and Python source files. Generated Word/PDF files and image assets are not included as code.

Source Code - style.css

```
/* ================================================================================
   STUDENT HUB PORTAL - RESPONSIVE MOBILE-FIRST CSS (GRID & FLEXBOX)
   ================================================================================ */

/* --------------------------------------------------------------------------------
   1. BASE & RESET STYLES
   -------------------------------------------------------------------------------- */
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  background-color: #0B162C;
  color: #FFFFFF;
  font-family: system-ui, -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto,
sans-serif;
  line-height: 1.6;
  padding: 12px;
  min-height: 100vh;
  display: flex;
  flex-direction: column;
  transition: background-color 0.3s, color 0.3s;
}

main {
  flex: 1;
}

/* --------------------------------------------------------------------------------
   2. LIGHT THEME STYLES (HIGH CONTRAST, CLEAN & SIMPLE)
   -------------------------------------------------------------------------------- */
body.light-theme {
  background-color: #F1F5F9;
  color: #0F172A;
}

/* Light Theme Cards & Containers */
body.light-theme .box,
body.light-theme .boxes2,
body.light-theme .table1,
body.light-theme .footer,
body.light-theme .modal-box,
body.light-theme .slider-container,
body.light-theme .auth-card {
  background-color: #FFFFFF;
  color: #0F172A;
  border: 1px solid #E2E8F0;
  box-shadow: 0 4px 16px rgba(0, 0, 0, 0.06);
}
```

```css
/* Light Theme Headings & Text */
body.light-theme h1,
body.light-theme h2,
body.light-theme h3,
body.light-theme h4,
body.light-theme strong {
  color: #0F172A;
}

/* Light Theme Subtitles, Paragraphs & Muted Text */
body.light-theme p,
body.light-theme [style*="color: #CBD5E1"],
body.light-theme [style*="color: #cbd5e1"] {
  color: #475569 !important;
}

/* Light Theme Accent Amber (Richer Amber for High Contrast on White) */
body.light-theme h2[style*="color: #EC9F05"],
body.light-theme h3[style*="color: #EC9F05"],
body.light-theme h4[style*="color: #EC9F05"],
body.light-theme strong[style*="color: #EC9F05"],
body.light-theme p[style*="color: #EC9F05"],
body.light-theme .stat-label {
  color: #B45309 !important;
}

/* Light Theme Stat Numbers */
body.light-theme .stat-num {
  color: #0F172A !important;
}

/* Light Theme Inner Course Cards & Note Boxes */
body.light-theme div[style*="background-color: #141E2E"],
body.light-theme div[style*="background-color: #141e2e"] {
  background-color: #F8FAFC !important;
  border-color: #CBD5E1 !important;
}

body.light-theme div[style*="background-color: #141E2E"] h4,
body.light-theme div[style*="background-color: #141E2E"] p {
  color: #0F172A !important;
}

body.light-theme div[style*="background-color: #141E2E"] p[style*="color: #EC9F05"]
{
  color: #B45309 !important;
}

/* Light Theme Event Badges */
body.light-theme .event-badge {
  background-color: #F8FAFC;
  border-color: #6B4EFD;
  color: #B45309;
}

/* Light Theme Tables */
body.light-theme .dash-table td {
  border-bottom: 1px solid #E2E8F0;
  color: #1E293B;
}
```

```css
body.light-theme .dash-table th {
  color: #B45309;
  border-bottom: 2px solid #6B4EFD;
}

/* Light Theme Status Pills */
body.light-theme .dash-tag.confirmed {
  background-color: #DCFCE7;
  color: #15803D;
}

body.light-theme .dash-tag.enrolled {
  background-color: #EDE9FE;
  color: #6D28D9;
}

body.light-theme .dash-tag.notice-tag {
  background-color: #FEF3C7;
  color: #B45309;
}

/* Light Theme FAQ */
body.light-theme .faq-item {
  border-bottom-color: #E2E8F0;
}

body.light-theme .faq-btn {
  color: #B45309;
}

body.light-theme .faq-btn:hover {
  color: #92400E;
}

body.light-theme .faq-answer,
body.light-theme .faq-answer p {
  color: #334155 !important;
}

/* Light Theme Inputs & Form Controls */
body.light-theme input[type="text"],
body.light-theme input[type="email"],
body.light-theme input[type="password"],
body.light-theme input[type="number"],
body.light-theme input[type="tel"],
body.light-theme select,
body.light-theme textarea {
  background-color: #FFFFFF !important;
  color: #0F172A !important;
  border: 1.5px solid #CBD5E1 !important;
}

body.light-theme input:focus,
body.light-theme select:focus,
body.light-theme textarea:focus {
  background-color: #FFFFFF !important;
  border-color: #6B4EFD !important;
  box-shadow: 0 0 0 3px rgba(107, 78, 253, 0.18) !important;
}
```

```
body.light-theme label {
  color: #0F172A !important;
}

/* Light Theme Header & Auth Text */
body.light-theme .header strong,
body.light-theme .auth-card h2,
body.light-theme .auth-footer-text {
  color: #0F172A;
}

body.light-theme .t1links a {
  color: #0F172A;
}

body.light-theme .t1links a:hover {
  color: #B45309;
}

body.light-theme a {
  color: #B45309;
}

body.light-theme a:hover {
  color: #92400E;
}

/* Base Universal Input & Form Styling */
input[type="text"],
input[type="email"],
input[type="password"],
input[type="number"],
input[type="tel"],
select,
textarea {
  width: 100%;
  background-color: #0B162C;
  color: #FFFFFF;
  border: 1.5px solid #6B4EFD;
  border-radius: 12px;
  padding: 11px 15px;
  font-size: 15px;
  font-family: inherit;
  outline: none;
  transition: border-color 0.2s, box-shadow 0.2s;
  box-sizing: border-box;
}

input:focus,
select:focus,
textarea:focus {
  border-color: #EC9F05;
  box-shadow: 0 0 0 3px rgba(236, 159, 5, 0.2);
}

a {
  color: #EC9F05;
  text-decoration: none;
  transition: 0.2s ease;
```

```
}

a:hover {
  color: #d88e04;
  text-decoration: underline;
}

/* Base Universal Button Styling (Never render unstyled browser buttons) */
button {
  font-family: inherit;
  background-color: #EC9F05;
  color: #FFFFFF;
  border: none;
  border-radius: 20px;
  font-size: 15px;
  font-weight: 600;
  padding: 9px 20px;
  cursor: pointer;
  transition: all 0.2s ease;
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  outline: none;
}

button:hover {
  background-color: #d88e04;
  transform: translateY(-2px);
  box-shadow: 0 4px 12px rgba(236, 159, 5, 0.35);
}

button:active {
  transform: translateY(0);
}

button a {
  color: inherit;
  text-decoration: none;
}

button a:hover {
  text-decoration: none;
  color: inherit;
}

/* ------------------------------------------------------------------------
   2. LOGO & HEADER STYLES (FIXED SIZE ON ALL WEBSITES)
   ------------------------------------------------------------------------ */
.header img,
.logo,
header img {
  height: 48px !important;
  max-height: 48px !important;
  width: auto !important;
  max-width: 160px !important;
  object-fit: contain;
  vertical-align: middle;
  border-radius: 8px;
}
```

```css
header {
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: space-between;
  gap: 15px;
  padding: 10px 0;
}

.header {
  display: flex;
  align-items: center;
  gap: 12px;
  font-family: serif;
  font-size: 24px;
}

.header strong {
  color: #FFFFFF;
}

.headerlinks {
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
  justify-content: center;
  align-items: center;
}

.headerlinks button,
.headerlinks a button {
  background-color: #EC9F05;
  color: white;
  border: none;
  border-radius: 20px;
  font-size: 15px;
  font-weight: 600;
  padding: 8px 18px;
  cursor: pointer;
  transition: background-color 0.2s, transform 0.1s, box-shadow 0.2s;
}

.headerlinks button:hover {
  background-color: #d88e04;
  transform: translateY(-1px);
  box-shadow: 0 4px 12px rgba(236, 159, 5, 0.3);
}

.headerlinks a {
  color: white;
  text-decoration: none;
}

/* ------------------------------------------------------------------------
   3. NOTIFICATION BANNER
   ------------------------------------------------------------------------ */
.notification-banner {
  background-color: #6B4EFD;
  color: white;
  padding: 10px 18px;
  border-radius: 12px;
```

```
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 12px;
  font-size: 15px;
}

.banner-close {
  background: none;
  border: none;
  color: white;
  font-size: 20px;
  font-weight: bold;
  cursor: pointer;
  padding: 0 6px;
}

/* ------------------------------------------------------------------------
   4. NAVIGATION BAR & HAMBURGER (FLEXBOX)
   ------------------------------------------------------------------------ */
.hamburger {
  display: block;
  background-color: #6B4EFD;
  color: white;
  border: none;
  border-radius: 10px;
  font-size: 18px;
  font-weight: bold;
  padding: 8px 16px;
  cursor: pointer;
  margin: 0 auto;
}

.navlinks {
  background-color: #6B4EFD;
  border-radius: 15px;
  margin: 10px 0 20px 0;
  padding: 10px;
}

.navlinks nav {
  display: none;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  gap: 10px;
}

.navlinks nav.show {
  display: flex;
}

.navlinks a {
  font-family: sans-serif;
  color: #FFFFFF;
  font-size: 16px;
  font-weight: 500;
  text-decoration: none;
  padding: 6px 12px;
  border-radius: 8px;
  transition: all 0.2s ease;
```

```
}

.navlinks a:hover {
  color: #DAF264;
  background-color: rgba(255, 255, 255, 0.15);
  text-decoration: none;
}

/* ------------------------------------------------------------------------
   5. PERFECT LOGIN & REGISTRATION BOXES + FORM VALIDATION
   ------------------------------------------------------------------------ */
.auth-card {
  background-color: #212C37;
  border: 1px solid rgba(255, 255, 255, 0.08);
  border-radius: 24px;
  padding: 35px 28px;
  max-width: 440px;
  width: 100%;
  margin: 25px auto;
  box-shadow: 0 12px 30px rgba(0, 0, 0, 0.35);
  box-sizing: border-box;
}

.auth-card.register-card {
  max-width: 680px;
}

.auth-card h2 {
  font-size: 28px;
  color: #EC9F05;
  text-align: center;
  margin-bottom: 22px;
  letter-spacing: 0.5px;
}

.form-group {
  display: flex;
  flex-direction: column;
  gap: 6px;
  margin-bottom: 16px;
  text-align: left;
}

.form-group label {
  font-size: 15px;
  font-weight: 600;
  color: #E2E8F0;
}

.form-group input,
.form-group select,
.form-group textarea {
  width: 100%;
  background-color: #0B162C;
  color: #FFFFFF;
  border: 1.5px solid #6B4EFD;
  border-radius: 12px;
  padding: 11px 15px;
  font-size: 15px;
  outline: none;
  transition: border-color 0.2s, box-shadow 0.2s;
```

```
  box-sizing: border-box;
}

.form-group input:focus,
.form-group select:focus,
.form-group textarea:focus {
  border-color: #EC9F05;
  box-shadow: 0 0 0 3px rgba(236, 159, 5, 0.2);
}

/* Error & Success States */
.error-text {
  color: #FF5252;
  font-size: 13px;
  margin-top: 3px;
  display: block;
  min-height: 18px;
}

.input-error {
  border-color: #FF5252 !important;
  box-shadow: 0 0 0 2px rgba(255, 82, 82, 0.25) !important;
}

.input-success {
  border-color: #10B981 !important;
}

/* 2-Column Grid for Registration Form */
.form-grid {
  display: grid;
  grid-template-columns: 1fr;
  gap: 0 20px;
}

.radio-group {
  display: flex;
  gap: 20px;
  align-items: center;
  padding: 10px 0 4px 0;
}

.radio-group label {
  display: flex;
  align-items: center;
  gap: 6px;
  cursor: pointer;
  font-weight: normal;
}

.radio-group input[type="radio"],
.form-options input[type="checkbox"] {
  width: 18px;
  height: 18px;
  accent-color: #EC9F05;
  cursor: pointer;
}

.form-options {
  display: flex;
```

```
  justify-content: space-between;
  align-items: center;
  margin-bottom: 20px;
  font-size: 14px;
}

.form-options.center-terms {
  flex-direction: column;
  align-items: center;
  justify-content: center;
  margin-top: 10px;
  text-align: center;
}

.form-options label {
  display: flex;
  align-items: center;
  gap: 8px;
  cursor: pointer;
}

.auth-btn {
  width: 100%;
  background-color: #EC9F05;
  color: white;
  border: none;
  border-radius: 30px;
  font-size: 17px;
  font-weight: bold;
  padding: 13px;
  cursor: pointer;
  transition: background-color 0.2s, transform 0.1s;
  box-shadow: 0 4px 12px rgba(236, 159, 5, 0.3);
}

.auth-btn:hover {
  background-color: #d88e04;
  transform: translateY(-2px);
}

.auth-footer-text {
  text-align: center;
  margin-top: 18px;
  font-size: 15px;
  color: #CBD5E1;
}

/* ------------------------------------------------------------------------
   6. HERO, CARDS & GRIDS (HOME & ABOUT)
   ------------------------------------------------------------------------ */
.main {
  background: linear-gradient(135deg, #6B4EFD 0%, #4427D6 60%, #2A1792 100%);
  border-radius: 20px;
  padding: 36px 24px;
  margin: 15px 0 25px 0;
  text-align: center;
  color: white;
  box-shadow: 0 10px 30px rgba(107, 78, 253, 0.35);
  position: relative;
  overflow: hidden;
}
```

```css
.main h1 {
  font-size: 32px;
  font-weight: 800;
  margin-bottom: 8px;
  letter-spacing: -0.5px;
}

.main p {
  font-size: 17px;
  color: rgba(255, 255, 255, 0.9);
  max-width: 600px;
  margin: 0 auto;
}

/* Styled View Announcement Button */
.main button,
.btn-announcement {
  background-color: #EC9F05;
  color: #FFFFFF;
  border: 2px solid transparent;
  font-size: 15px;
  font-weight: 700;
  padding: 12px 28px;
  border-radius: 30px;
  cursor: pointer;
  box-shadow: 0 6px 18px rgba(0, 0, 0, 0.25);
  transition: all 0.25s ease;
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 8px;
  margin-top: 6px;
}

.main button:hover,
.btn-announcement:hover {
  background-color: #FFFFFF;
  color: #6B4EFD;
  transform: translateY(-3px) scale(1.03);
  box-shadow: 0 10px 25px rgba(255, 255, 255, 0.3);
}

.boxes-container {
  display: flex;
  flex-direction: column;
  gap: 15px;
  width: 100%;
}

.box,
.boxes2,
.table1 {
  background-color: #212C37;
  border-radius: 20px;
  border: 1px solid rgba(255, 255, 255, 0.08);
  padding: 24px;
  width: 100%;
  box-sizing: border-box;
  box-shadow: 0 6px 20px rgba(0, 0, 0, 0.2);
```

```
  transition: transform 0.2s ease, border-color 0.2s ease, background-color 0.25s
ease, color 0.25s ease;
}

.box:hover,
.boxes2:hover,
.table1:hover {
  border-color: rgba(107, 78, 253, 0.4);
}

.box {
  min-height: 220px;
  text-align: center;
}

.boxes2 {
  min-height: 180px;
  text-align: center;
}

.box h3,
.boxes2 p:first-child {
  color: #EC9F05;
  margin-bottom: 12px;
}

.box ul {
  list-style-position: inside;
  text-align: left;
  display: inline-block;
}

.t1,
.t1links {
  font-size: 17px;
  margin-bottom: 8px;
}

.t1links a {
  color: #FFFFFF;
}

.t1links a:hover {
  color: #EC9F05;
}

/* ------------------------------------------------------------------------
   7. COLLAPSIBLE FAQ & MODAL & SLIDER
   ------------------------------------------------------------------------ */
.faq-item {
  border-bottom: 1px solid rgba(107, 78, 253, 0.4);
  padding: 10px 0;
}

.faq-btn {
  width: 100%;
  background: none;
  color: #EC9F05;
  font-size: 18px;
  font-weight: bold;
```

```
  border: none;
  padding: 10px 0;
  cursor: pointer;
  display: flex;
  justify-content: space-between;
  align-items: center;
  text-align: left;
  border-radius: 0;
  box-shadow: none;
}

.faq-btn:hover {
  background: none;
  color: #d88e04;
  transform: none;
  box-shadow: none;
}

.faq-answer {
  display: none;
  padding: 8px 0 12px 0;
  font-size: 16px;
  color: #CBD5E1;
}

.modal {
  display: none;
  position: fixed;
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
  background: rgba(0, 0, 0, 0.75);
  justify-content: center;
  align-items: center;
  z-index: 1000;
  padding: 15px;
  backdrop-filter: blur(4px);
}

.modal-box {
  background-color: #212C37;
  color: white;
  padding: 32px 28px;
  border-radius: 24px;
  max-width: 480px;
  width: 100%;
  text-align: center;
  position: relative;
  border: 1.5px solid #6B4EFD;
  box-shadow: 0 16px 40px rgba(0, 0, 0, 0.5);
}

.modal-close {
  position: absolute;
  top: 12px;
  right: 16px;
  background: none;
  color: #94A3B8;
  font-size: 26px;
  border: none;
```

```
  cursor: pointer;
  padding: 4px 8px;
  border-radius: 50%;
  box-shadow: none;
}

.modal-close:hover {
  background-color: rgba(255, 255, 255, 0.1);
  color: #FFFFFF;
  transform: none;
  box-shadow: none;
}

.modal-box button:not(.modal-close) {
  background-color: #EC9F05;
  color: #FFFFFF;
  border-radius: 25px;
  padding: 10px 28px;
  font-size: 15px;
  font-weight: 700;
  border: none;
  cursor: pointer;
  margin-top: 10px;
  box-shadow: 0 4px 14px rgba(236, 159, 5, 0.35);
  transition: all 0.2s ease;
}

.modal-box button:not(.modal-close):hover {
  background-color: #d88e04;
  transform: translateY(-2px);
  box-shadow: 0 6px 18px rgba(236, 159, 5, 0.45);
}

/* Modern Slider Container */
.slider-container {
  background-color: #212C37;
  border-radius: 22px;
  border: 1.5px solid rgba(107, 78, 253, 0.3);
  padding: 30px 24px;
  margin: 25px 0;
  text-align: center;
  box-shadow: 0 8px 24px rgba(0, 0, 0, 0.25);
  position: relative;
}

.slide {
  display: none;
  animation: fadeIn 0.4s ease-in-out;
}

@keyframes fadeIn {
  from { opacity: 0; transform: translateY(6px); }
  to { opacity: 1; transform: translateY(0); }
}

.slide h3 {
  color: #EC9F05;
  font-size: 24px;
  font-weight: 700;
  margin-bottom: 10px;
}
```

```css
.slide p {
  font-size: 16px;
  color: #CBD5E1;
  max-width: 650px;
  margin: 0 auto;
  line-height: 1.5;
}

/* Prev and Next Slider Control Buttons */
.slider-controls {
  margin-top: 20px;
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 14px;
}

.slider-controls button,
.slider-btn {
  background: #141E2E;
  color: #EC9F05;
  border: 1.5px solid #6B4EFD;
  border-radius: 25px;
  font-size: 14px;
  font-weight: 700;
  padding: 8px 22px;
  cursor: pointer;
  transition: all 0.2s ease;
  box-shadow: 0 4px 10px rgba(0, 0, 0, 0.25);
  display: inline-flex;
  align-items: center;
  gap: 6px;
}

.slider-controls button:hover,
.slider-btn:hover {
  background-color: #6B4EFD;
  color: #FFFFFF;
  border-color: #6B4EFD;
  transform: translateY(-2px);
  box-shadow: 0 6px 16px rgba(107, 78, 253, 0.4);
}

.slider-controls button:active,
.slider-btn:active {
  transform: translateY(0);
}

/* Light Theme Adjustments for Slider & Hero */
body.light-theme .main button,
body.light-theme .btn-announcement {
  background-color: #EC9F05;
  color: #FFFFFF;
}

body.light-theme .main button:hover,
body.light-theme .btn-announcement:hover {
  background-color: #FFFFFF;
  color: #6B4EFD;
}
```

```css
body.light-theme .slider-container {
  background-color: #FFFFFF;
  border-color: #E2E8F0;
  box-shadow: 0 8px 20px rgba(0, 0, 0, 0.08);
}

body.light-theme .slide p {
  color: #475569;
}

body.light-theme .slider-controls button,
body.light-theme .slider-btn {
  background: #F8FAFC;
  color: #6B4EFD;
  border-color: #CBD5E1;
}

body.light-theme .slider-controls button:hover,
body.light-theme .slider-btn:hover {
  background-color: #6B4EFD;
  color: #FFFFFF;
  border-color: #6B4EFD;
}

body.light-theme .modal-box {
  background-color: #FFFFFF;
  color: #1A202C;
  border-color: #6B4EFD;
}

body.light-theme .modal-close {
  color: #64748B;
}

body.light-theme .modal-close:hover {
  background-color: #F1F5F9;
  color: #1A202C;
}

/* ------------------------------------------------------------------------
   8. FOOTER STYLES
   ------------------------------------------------------------------------ */
.footer {
  background-color: #212C37;
  border-radius: 25px;
  padding: 16px;
  margin-top: 35px;
  text-align: center;
}

.footer footer p {
  display: flex;
  flex-wrap: wrap;
  align-items: center;
  justify-content: center;
  gap: 8px;
  font-size: 15px;
}
```

```
.footer a {
  background-color: #EC9F05;
  color: white;
  border-radius: 15px;
  font-size: 14px;
  padding: 4px 12px;
  text-decoration: none;
}

/* ------------------------------------------------------------------------
   9. DESKTOP & TABLET RESPONSIVENESS
   ------------------------------------------------------------------------ */
@media (min-width: 768px) {
  body {
    padding: 20px 40px;
  }

  header {
    flex-direction: row;
    padding: 10px 10px;
  }

  .header {
    font-size: 30px;
  }

  .hamburger {
    display: none;
  }

  .navlinks nav {
    display: flex !important;
    flex-direction: row;
  }

  .main {
    max-width: 850px;
    margin: 20px auto;
    padding: 35px 20px;
  }

  .main h1 {
    font-size: 42px;
  }

  /* 2-Column Grid on Registration Form */
  .form-grid {
    grid-template-columns: 1fr 1fr;
  }

  /* Horizontal Grid for Boxes on Desktop */
  .boxes-container {
    flex-direction: row !important;
  }

  .box {
    width: 33.33% !important;
  }

  .boxes2 {
```

```
    width: 50% !important;
  }
}

/* ================================================================================
   10. STUDENT DASHBOARD STYLES (BROAD, CLEAN & SIMPLE)
   ================================================================================ */
.dash-wrapper {
  max-width: 1280px;
  width: 100%;
  margin: 15px auto;
  padding: 10px;
  box-sizing: border-box;
}

.dash-main-flex {
  display: flex;
  flex-direction: column;
  gap: 20px;
  align-items: stretch;
}

@media (min-width: 850px) {
  .dash-main-flex {
    flex-direction: row;
  }
}

.dash-sidebar-col {
  width: 100%;
  max-width: 250px;
  flex-shrink: 0;
  box-sizing: border-box;
}

@media (max-width: 849px) {
  .dash-sidebar-col {
    max-width: 100%;
  }
}

.dash-content-col {
  flex: 1;
  min-width: 0;
  display: flex;
  flex-direction: column;
  gap: 20px;
}

.dash-stats-flex {
  display: flex;
  flex-direction: row;
  gap: 15px;
  width: 100%;
}

.dash-cards-flex {
  display: flex;
  flex-direction: column;
  gap: 20px;
  width: 100%;
```

```
}

@media (min-width: 768px) {
  .dash-cards-flex {
    flex-direction: row;
  }
}

.dash-sidebar-menu {
  display: flex;
  flex-direction: column;
  gap: 10px;
  margin-top: 10px;
}

.dash-nav-btn {
  display: block;
  width: 100%;
  background-color: #6B4EFD;
  color: #FFFFFF;
  padding: 10px 16px;
  border-radius: 12px;
  font-size: 15px;
  font-weight: 600;
  text-decoration: none;
  text-align: center;
  transition: all 0.2s ease;
  box-sizing: border-box;
}

.dash-nav-btn:hover {
  background-color: #EC9F05;
  color: #FFFFFF;
  text-decoration: none;
  transform: translateY(-2px);
}

.dash-nav-btn.active {
  background-color: #EC9F05;
  color: #FFFFFF;
}

.dash-nav-btn.logout-btn {
  background-color: transparent;
  border: 1.5px solid #FF5252;
  color: #FF5252;
  margin-top: 8px;
}

.dash-nav-btn.logout-btn:hover {
  background-color: #FF5252;
  color: #FFFFFF;
}

.stat-num {
  font-size: 36px;
  font-weight: bold;
  color: #FFFFFF;
  margin: 6px 0;
}
```

```
.stat-label {
  font-size: 16px;
  color: #EC9F05;
  font-weight: 600;
}

.dash-table {
  width: 100%;
  border-collapse: collapse;
  margin-top: 10px;
  text-align: left;
}

.dash-table th {
  color: #EC9F05;
  padding: 10px 8px;
  border-bottom: 1.5px solid #6B4EFD;
  font-size: 15px;
}

.dash-table td {
  padding: 12px 8px;
  border-bottom: 1px solid rgba(255, 255, 255, 0.08);
  font-size: 14px;
}

.dash-tag {
  display: inline-block;
  padding: 3px 10px;
  border-radius: 12px;
  font-size: 12px;
  font-weight: bold;
}

.dash-tag.confirmed {
  background-color: rgba(16, 185, 129, 0.2);
  color: #10B981;
}

.dash-tag.enrolled {
  background-color: rgba(107, 78, 253, 0.25);
  color: #9D88FD;
}

.dash-tag.notice-tag {
  background-color: rgba(236, 159, 5, 0.2);
  color: #EC9F05;
}

body.light-theme .dash-nav-btn {
  background-color: #6B4EFD;
  color: #FFFFFF;
}

body.light-theme .dash-nav-btn:hover {
  background-color: #B45309;
}

body.light-theme .dash-nav-btn.active {
```

```
  background-color: #B45309;
  color: #FFFFFF;
}

body.light-theme .stat-num {
  color: #0F172A;
}

body.light-theme .dash-table td {
  border-bottom: 1px solid #E2E8F0;
  color: #1E293B;
}

body.light-theme .dash-table th {
  border-bottom-color: #6B4EFD;
  color: #B45309;
}

/* ================================================================================
   11. EVENTS PAGE STYLES (CLEAN & SIMPLE)
   ================================================================================ */
.events-search-box {
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 12px;
  flex-wrap: wrap;
}

.events-search-box input {
  flex: 1;
  max-width: 480px;
  margin: 0;
}

.event-card-item {
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 20px;
  padding: 20px 24px;
  margin-bottom: 16px;
  transition: transform 0.2s ease, border-color 0.2s ease;
  flex-wrap: wrap;
}

.event-card-item:hover {
  transform: translateY(-2px);
  border-color: #6B4EFD;
}

.event-badge {
  width: 85px;
  height: 70px;
  background-color: #0B162C;
  border: 2px solid #6B4EFD;
  border-radius: 12px;
  display: flex;
  align-items: center;
  justify-content: center;
  color: #EC9F05;
```

```
  font-weight: bold;
  font-size: 15px;
  flex-shrink: 0;
  letter-spacing: 0.5px;
}

body.light-theme .event-badge {
  background-color: #F8FAFC;
  border-color: #6B4EFD;
}

/* Event Filter, Search & Sort Toolbar */
.events-toolbar {
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 15px;
  flex-wrap: wrap;
}

.events-toolbar .search-box-wrap {
  flex: 1.5;
  min-width: 260px;
  display: flex;
  align-items: center;
  gap: 10px;
}

.events-toolbar .controls-group {
  display: flex;
  align-items: center;
  gap: 10px;
  flex-wrap: wrap;
  flex: 1;
  justify-content: flex-end;
}

.events-toolbar select {
  width: auto;
  min-width: 140px;
  padding: 10px 14px;
}

/* Pagination Styles */
.pagination-container {
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 8px;
  margin-top: 25px;
  flex-wrap: wrap;
}

.pagination-btn {
  background-color: #141E2E;
  color: #FFFFFF;
  border: 1.5px solid #6B4EFD;
  padding: 8px 16px;
  border-radius: 12px;
  font-size: 14px;
  font-weight: 600;
```

```
  cursor: pointer;
  transition: all 0.2s ease;
}

.pagination-btn:hover:not(:disabled) {
  background-color: #6B4EFD;
  color: #FFFFFF;
  transform: translateY(-2px);
}

.pagination-btn.active {
  background-color: #EC9F05;
  border-color: #EC9F05;
  color: #FFFFFF;
}

.pagination-btn:disabled {
  opacity: 0.35;
  cursor: not-allowed;
  transform: none;
}

body.light-theme .pagination-btn {
  background-color: #FFFFFF;
  color: #0F172A;
  border-color: #CBD5E1;
}

body.light-theme .pagination-btn:hover:not(:disabled) {
  background-color: #6B4EFD;
  color: #FFFFFF;
  border-color: #6B4EFD;
}

body.light-theme .pagination-btn.active {
  background-color: #B45309;
  border-color: #B45309;
  color: #FFFFFF;
}
```

Source Code - about.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - About</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>
  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
```

```
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main style="max-width: 900px; margin: 0 auto; text-align: center;">
    <h1 style="font-size: 32px; margin-bottom: 20px;">About Student Hub</h1>

    <div style="display: flex; gap: 20px;" class="boxes-container">
      <div class="boxes2">
        <p style="font-size: 26px; font-weight: bold;">Our Mission</p>
        <p style="font-size: 18px;">To eliminate communication gaps across campus
by delivering real-time updates and effortless management tools for every
student.</p>
      </div>
      <div class="boxes2">
        <p style="font-size: 26px; font-weight: bold;">Our Vision</p>
        <p style="font-size: 18px;">To build a modern, fully digital campus
ecosystem where information is accessible anywhere, anytime with zero operational
friction.</p>
      </div>
    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

Source Code - admin.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Admin Panel</title>
  <link rel="stylesheet" href="style.css">
</head>
```

```
<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
height="45"
          style="vertical-align: middle; margin-right: 8px;"></a>
      <strong style="margin: 10px;">Admin Panel</strong>
    </div>
    <div class="headerlinks">
      <button onclick="handleLogout()">Logout</button>
    </div>
  </header>

  <div class="navlinks">
    <nav align="center">
      <a href="admin.html">Dashboard</a> |
      <a href="events.html">Events</a> |
      <a href="index.html">Home</a>
    </nav>
  </div>

  <main style="max-width: 1000px; margin: 20px auto; padding: 10px;">

    <!-- Stats Grid -->
    <div style="display: flex; gap: 15px; flex-wrap: wrap;">
      <div class="table1" style="flex: 1; text-align: center; min-width: 200px;
padding: 20px;">
        <h3 style="font-size: 22px; color: #EC9F05; margin: 0;">Students</h3>
        <h1 style="font-size: 42px; margin: 10px 0 0 0;">250</h1>
      </div>
      <div class="table1" style="flex: 1; text-align: center; min-width: 200px;
padding: 20px;">
        <h3 style="font-size: 22px; color: #EC9F05; margin: 0;">Events</h3>
        <h1 style="font-size: 42px; margin: 10px 0 0 0;">12</h1>
      </div>
      <div class="table1" style="flex: 1; text-align: center; min-width: 200px;
padding: 20px;">
        <h3 style="font-size: 22px; color: #EC9F05; margin: 0;">Feedback</h3>
        <h1 style="font-size: 42px; margin: 10px 0 0 0;">18</h1>
      </div>
      <div class="table1" style="flex: 1; text-align: center; min-width: 200px;
padding: 20px;">
        <h3 style="font-size: 22px; color: #EC9F05; margin: 0;">Notices</h3>
        <h1 style="font-size: 42px; margin: 10px 0 0 0;">7</h1>
      </div>
    </div>

    <br>

    <!-- Recent Activities Card -->
    <div class="table1" style="padding: 25px;">
      <h3 style="font-size: 26px; color: #EC9F05; margin-top: 0;">Recent
Activities</h3>
      <ul style="font-size: 18px; line-height: 1.8;">
        <li>Student Registered</li>
        <li>Event Added</li>
        <li>Feedback Submitted</li>
      </ul>
    </div>
```

```
  </main>
  <br>
  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a> |
        <a href="contact.html">Contact</a> |
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
  <script>
    // Protect Admin Panel from unauthorized access
    checkAdminAuth();
  </script>
</body>

</html>
```

Source Code - contact.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Contact &amp;amp; Feedback</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html" style="background-color: rgba(255, 255, 255, 0.2);
color: #DAF264; font-weight: bold;">Contact</a>
    </nav>
  </div>

  <main style="max-width: 1150px; width: 100%; margin: 20px auto; padding: 10px;">
```

```
    <h1 style="font-size: 32px; text-align: center; margin-bottom: 24px;">Contact
Us &amp; Feedback</h1>

    <div class="dash-cards-flex" style="align-items: stretch; gap: 24px;">

      <!-- Contact Info Card -->
      <div class="table1" style="flex: 1; padding: 30px; display: flex; flex-
direction: column; justify-content: space-between;">
        <div>
          <h2 style="font-size: 24px; color: #EC9F05; margin-bottom: 10px;">Get in
Touch</h2>
          <p style="color: #CBD5E1; font-size: 15px; margin-bottom: 20px; line-
height: 1.5;">
            Have questions about college events, your student profile, or academic
schedules? Reach out to our campus help desk directly.
          </p>
          <hr style="border-color: rgba(107, 78, 253, 0.4); margin: 15px 0 20px
0;">

          <div style="display: flex; flex-direction: column; gap: 16px; font-size:
15px;">
            <p>
              <strong style="color: #EC9F05;">✉ Email :</strong><br>
              <span style="color: #CBD5E1;">support@studenthub.com</span>
            </p>
            <p>
              <strong style="color: #EC9F05;">📞 Phone :</strong><br>
              <span style="color: #CBD5E1;">+1 (234) 567-8900 / +1 (987) 654-
3210</span>
            </p>
            <p>
              <strong style="color: #EC9F05;">📍 Address :</strong><br>
              <span style="color: #CBD5E1;">Campus Center, Student Block 4, Tech
Campus</span>
            </p>
            <p>
              <strong style="color: #EC9F05;">🕒 Office Hours :</strong><br>
              <span style="color: #CBD5E1;">Monday &ndash; Friday: 9:00 AM &ndash;
5:00 PM</span>
            </p>
          </div>
        </div>

        <div style="margin-top: 25px; padding: 14px; background-color: #141E2E;
border-radius: 12px; border-left: 4px solid #EC9F05; font-size: 13px; color:
#CBD5E1;">
          💡 <strong>Tip:</strong> For urgent exam or admit card queries, please
contact the Examination Branch in Block 2.
        </div>
      </div>

      <!-- Feedback Form Card -->
      <div class="table1" id="feedback" style="flex: 1.2; padding: 30px;">
        <h2 style="font-size: 24px; color: #EC9F05; margin-bottom: 10px;">Send a
Message</h2>
        <p style="color: #CBD5E1; font-size: 15px; margin-bottom: 20px;">
          Fill out the form below and our portal support team will respond
promptly.
        </p>
```

```
        <form onsubmit="event.preventDefault(); alert('Thank you! Your message has
been submitted successfully.'); this.reset();">
          <div class="form-group">
            <label for="fb-name">Your Full Name</label>
            <input type="text" id="fb-name" placeholder="e.g. John Doe" required>
          </div>

          <div class="form-group">
            <label for="fb-email">Your Email Address</label>
            <input type="email" id="fb-email" placeholder="e.g.
student@college.edu" required>
          </div>

          <div class="form-group">
            <label for="fb-category">Category</label>
            <select id="fb-category">
              <option value="general">General Inquiry</option>
              <option value="events">College Events &amp;amp; Workshops</option>
              <option value="academic">Academic &amp;amp; Exam Schedule</option>
              <option value="technical">Portal Technical Feedback</option>
            </select>
          </div>

          <div class="form-group">
            <label for="fb-message">Message / Feedback</label>
            <textarea id="fb-message" rows="4" placeholder="Write your message
here..." required></textarea>
          </div>

          <div style="margin-top: 20px; text-align: center;">
            <button type="submit" style="padding: 12px 35px; font-size: 16px; font-
weight: bold;">Send Message 🚀</button>
          </div>
        </form>
      </div>

    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

Source Code - dashboard.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Dashboard</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- Top Notification Banner -->
  <div id="notification-banner" class="notification-banner">
    <span>📢 <strong>Notice:</strong> Semester Examination Schedule has been
published!</span>
    <button class="banner-close" onclick="closeBanner()">&times;</button>
  </div>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>

    <div class="headerlinks">
      <button onclick="toggleTheme()">🌓 Theme</button>
      <button onclick="handleLogout()">Logout</button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="dashboard.html" style="background-color: rgba(255, 255, 255, 0.2);
color: #DAF264; font-weight: bold;">Dashboard</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main class="dash-wrapper">
    <div class="dash-main-flex">

      <!-- Sidebar Navigation Column -->
      <div class="table1 dash-sidebar-col">
        <!-- Student Info Header -->
        <div style="text-align: center; padding-bottom: 15px; border-bottom: 1px
solid rgba(107, 78, 253, 0.4);">
          <div style="width: 60px; height: 60px; border-radius: 50%; background-
color: #6B4EFD; color: white; display: flex; align-items: center; justify-content:
center; margin: 0 auto 10px auto; font-size: 20px; font-weight: bold; border: 2px
solid #EC9F05;">
            JD
          </div>
          <h3 style="font-size: 19px; margin-bottom: 3px;">John Doe</h3>
          <p style="font-size: 13px; color: #CBD5E1;">B.Tech CSE - 3rd Sem</p>
          <p style="font-size: 13px; color: #EC9F05; font-weight: bold; margin-top:
2px;">ID: 25DCS200</p>
        </div>
```

```
        <h4 style="color: #EC9F05; margin: 16px 0 8px 0; text-align: center; text-
transform: uppercase; font-size: 12px; letter-spacing: 1px;">Navigation</h4>
        <div class="dash-sidebar-menu">
          <a href="index.html" class="dash-nav-btn">Home</a>
          <a href="profile.html" class="dash-nav-btn">My Profile</a>
          <a href="events.html" class="dash-nav-btn">Events</a>
          <a href="dashboard.html" class="dash-nav-btn active">Dashboard</a>
          <a href="contact.html" class="dash-nav-btn">Contact Admin</a>
          <button onclick="handleLogout()" class="dash-nav-btn logout-btn"
style="cursor: pointer; width: 100%;">Logout</button>
        </div>

        <div style="margin-top: 20px; padding-top: 15px; border-top: 1px solid
rgba(107, 78, 253, 0.4); text-align: center; font-size: 13px; color: #CBD5E1;">
          <p>Academic Year 2026-27</p>
          <p style="color: #10B981; font-weight: bold; margin-top: 3px;">● Active
Student</p>
        </div>
      </div>

      <!-- Main Dashboard Content Column -->
      <div class="dash-content-col">

        <!-- Administrator Mode Indicator (Shown only when logged in as admin) -->
        <div id="admin-access-indicator" style="display: none; background-color:
#6B4EFD; color: white; padding: 10px 18px; border-radius: 12px; font-size: 14px;
text-align: center; border: 1.5px solid #EC9F05;">
          🌓 <strong>Administrator Mode:</strong> Viewing student dashboard &bull;
<a href="admin.html" style="color: #DAF264; font-weight: bold; text-decoration:
underline;">Switch to Admin Panel &rarr;</a>
        </div>

        <!-- Welcome Title Card -->
        <div class="table1" style="padding: 20px 24px;">
          <h2 style="font-size: 26px; margin-bottom: 4px;">Welcome back, John
Doe</h2>
          <p style="color: #CBD5E1; font-size: 15px;">B.Tech Computer Science &amp;
Engineering &ndash; Semester 3 | Student Hub Portal</p>
        </div>

        <!-- 3 Clean Stat Cards (No Notices, No Attendance, No Credits) -->
        <div class="dash-stats-flex">
          <div class="table1" style="flex: 1; text-align: center; padding: 18px
10px;">
            <p class="stat-label">Total Events</p>
            <h1 class="stat-num">20</h1>
            <p style="font-size: 12px; color: #CBD5E1;">Campus Activities</p>
          </div>
          <div class="table1" style="flex: 1; text-align: center; padding: 18px
10px;">
            <p class="stat-label">Registered Events</p>
            <h1 class="stat-num">5</h1>
            <p style="font-size: 12px; color: #CBD5E1;">Confirmed Slots</p>
          </div>
          <div class="table1" style="flex: 1; text-align: center; padding: 18px
10px;">
            <p class="stat-label">Enrolled Subjects</p>
            <h1 class="stat-num">5</h1>
            <p style="font-size: 12px; color: #10B981; font-weight: bold;">Sem 3
Regular</p>
          </div>
```

```
        </div>

        <!-- Full-Width Registered Events Table Card (No Notices Card) -->
        <div class="table1" style="width: 100%; padding: 24px;">
          <div style="display: flex; justify-content: space-between; align-items:
center; margin-bottom: 14px; flex-wrap: wrap; gap: 8px;">
            <h3 style="color: #EC9F05;">My Registered Events</h3>
            <a href="events.html" style="font-size: 14px;">Browse All Events
&rarr;</a>
          </div>

          <table class="dash-table">
            <thead>
              <tr>
                <th>Event Name</th>
                <th>Category</th>
                <th>Date</th>
                <th>Venue</th>
                <th>Status</th>
              </tr>
            </thead>
            <tbody>
              <tr>
                <td><strong>WDF Full-Stack Workshop</strong></td>
                <td><span class="dash-tag enrolled">TECH</span></td>
                <td>Nov 15, 2026</td>
                <td>Lab 3</td>
                <td><span class="dash-tag confirmed">Confirmed</span></td>
              </tr>
              <tr>
                <td><strong>DSA Coding Challenge</strong></td>
                <td><span class="dash-tag notice-tag">ACADEMIC</span></td>
                <td>Oct 28, 2026</td>
                <td>Lab 1</td>
                <td><span class="dash-tag confirmed">Confirmed</span></td>
              </tr>
              <tr>
                <td><strong>Java OOP Software Seminar</strong></td>
                <td><span class="dash-tag enrolled">TECH</span></td>
                <td>Dec 05, 2026</td>
                <td>Audi B</td>
                <td><span class="dash-tag enrolled">Enrolled</span></td>
              </tr>
              <tr>
                <td><strong>SPOURAL - Sports Championship</strong></td>
                <td><span class="dash-tag notice-tag">SPORTS</span></td>
                <td>Oct 25, 2026</td>
                <td>Main Ground</td>
                <td><span class="dash-tag confirmed">Confirmed</span></td>
              </tr>
            </tbody>
          </table>
        </div>

        <!-- Academic Overview Card - Sem 3 CS Subjects -->
        <div class="table1" style="padding: 22px;">
          <div style="display: flex; justify-content: space-between; align-items:
center; margin-bottom: 14px; flex-wrap: wrap; gap: 8px;">
            <h3 style="color: #EC9F05;">Semester 3 Enrolled Subjects (Computer
Science)</h3>
            <span style="font-size: 13px; color: #10B981; font-weight: bold;">5
Subjects Enrolled</span>
```

```
          </div>

          <div style="display: flex; gap: 14px; flex-wrap: wrap;">
            <!-- 1. WDF -->
            <div style="flex: 1; min-width: 170px; background-color: #141E2E;
padding: 14px; border-radius: 12px; border: 1.5px solid #6B4EFD;">
              <p style="color: #EC9F05; font-weight: bold; font-size:
13px;">ITUE203</p>
              <h4 style="font-size: 16px; margin: 4px 0 2px 0;">WDF</h4>
              <p style="font-size: 13px; color: #FFFFFF;">Web Development
Frameworks</p>
              <p style="font-size: 12px; color: #CBD5E1; margin-top: 6px;">Faculty:
Prof. Sharma</p>
              <p style="font-size: 12px; color: #10B981; margin-top: 4px;">Theory +
Lab</p>
            </div>

            <!-- 2. FCN -->
            <div style="flex: 1; min-width: 170px; background-color: #141E2E;
padding: 14px; border-radius: 12px; border: 1.5px solid #6B4EFD;">
              <p style="color: #EC9F05; font-weight: bold; font-size:
13px;">ITUE204</p>
              <h4 style="font-size: 16px; margin: 4px 0 2px 0;">FCN</h4>
              <p style="font-size: 13px; color: #FFFFFF;">Fundamentals of Computer
Networks</p>
              <p style="font-size: 12px; color: #CBD5E1; margin-top: 6px;">Faculty:
Prof. Mehta</p>
              <p style="font-size: 12px; color: #10B981; margin-top: 4px;">Theory +
Lab</p>
            </div>

            <!-- 3. DSA -->
            <div style="flex: 1; min-width: 170px; background-color: #141E2E;
padding: 14px; border-radius: 12px; border: 1.5px solid #6B4EFD;">
              <p style="color: #EC9F05; font-weight: bold; font-size:
13px;">ITUE201</p>
              <h4 style="font-size: 16px; margin: 4px 0 2px 0;">DSA</h4>
              <p style="font-size: 13px; color: #FFFFFF;">Data Structures &amp;
Algorithms</p>
              <p style="font-size: 12px; color: #CBD5E1; margin-top: 6px;">Faculty:
Dr. Verma</p>
              <p style="font-size: 12px; color: #10B981; margin-top: 4px;">Theory +
Lab</p>
            </div>

            <!-- 4. DISCRETE MATHS -->
            <div style="flex: 1; min-width: 170px; background-color: #141E2E;
padding: 14px; border-radius: 12px; border: 1.5px solid #6B4EFD;">
              <p style="color: #EC9F05; font-weight: bold; font-size:
13px;">MATH201</p>
              <h4 style="font-size: 16px; margin: 4px 0 2px 0;">DISCRETE MATHS</h4>
              <p style="font-size: 13px; color: #FFFFFF;">Discrete Mathematics</p>
              <p style="font-size: 12px; color: #CBD5E1; margin-top: 6px;">Faculty:
Dr. Joshi</p>
              <p style="font-size: 12px; color: #10B981; margin-top:
4px;">Theory</p>
            </div>

            <!-- 5. JAVAOOP -->
            <div style="flex: 1; min-width: 170px; background-color: #141E2E;
padding: 14px; border-radius: 12px; border: 1.5px solid #6B4EFD;">
```

```
              <p style="color: #EC9F05; font-weight: bold; font-size:
13px;">ITUE202</p>
              <h4 style="font-size: 16px; margin: 4px 0 2px 0;">JAVAOOP</h4>
              <p style="font-size: 13px; color: #FFFFFF;">Java &amp; OOP
Concepts</p>
              <p style="font-size: 12px; color: #CBD5E1; margin-top: 6px;">Faculty:
Dr. Patel</p>
              <p style="font-size: 12px; color: #10B981; margin-top: 4px;">Theory +
Lab</p>
            </div>
          </div>
        </div>

      </div>

    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
  <script>
    // Practical Access Guard: Redirect to login if not authenticated
    checkDashboardAuth();
  </script>
</body>

</html>
```

Source Code - events.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Events (Practical 6)</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Logout</a></button>
    </div>
```

```
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html" style="background-color: rgba(255, 255, 255, 0.2);
color: #DAF264; font-weight: bold;">Events</a>
      <a href="dashboard.html">Dashboard</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main style="max-width: 1150px; width: 100%; margin: 20px auto; padding: 10px;">
    <h1 style="font-size: 32px; text-align: center; margin-bottom: 6px;">College
Events</h1>
    <p style="text-align: center; color: #CBD5E1; margin-bottom: 24px; font-size:
15px;">
      Explore upcoming campus workshops, sports competitions, and cultural
festivals (Fetched dynamically from external JSON)
    </p>

    <!-- Practical 6: Search, Category Filter, and Sorting Controls Toolbar -->
    <div class="table1" id="events-toolbar-card" style="padding: 18px 24px; margin-
bottom: 24px;">
      <div class="events-toolbar">

        <!-- Live Search Input -->
        <div class="search-box-wrap">
          <label for="search" style="font-weight: bold; color: #EC9F05; white-
space: nowrap;">Search:</label>
          <input type="text" id="search" oninput="applyEventFilters()"
placeholder="Search by title, venue, category...">
        </div>

        <!-- Filter, Sort and Reset Controls -->
        <div class="controls-group">
          <!-- Category Filter -->
          <label for="categoryFilter" style="font-weight: bold; color: #EC9F05;
font-size: 14px; white-space: nowrap;">Category:</label>
          <select id="categoryFilter" onchange="applyEventFilters()">
            <option value="ALL">All Categories</option>
            <option value="SPORTS">Sports</option>
            <option value="TECH">Technology</option>
            <option value="WORKSHOP">Workshops</option>
            <option value="CULTURAL">Cultural</option>
            <option value="ACADEMIC">Academic</option>
          </select>

          <!-- Sorting Dropdown -->
          <label for="sortOrder" style="font-weight: bold; color: #EC9F05; font-
size: 14px; white-space: nowrap;">Sort by:</label>
          <select id="sortOrder" onchange="applyEventFilters()">
            <option value="date-asc">Date (Earliest First)</option>
            <option value="date-desc">Date (Latest First)</option>
            <option value="name-asc">Name (A &rarr; Z)</option>
            <option value="name-desc">Name (Z &rarr; A)</option>
          </select>
```

```
          <!-- Reset Filter Button -->
          <button onclick="resetEventFilters()" style="padding: 9px 18px; font-
size: 14px;">Reset</button>
        </div>

      </div>
    </div>

    <!-- Dynamic Events List Container (Populated via Fetch API) -->
    <div id="events-list">
      <!-- Dynamically generated event cards appear here -->
    </div>

    <!-- Practical 6 Pagination Controls -->
    <div id="pagination-controls" class="pagination-container">
      <!-- Dynamically generated page buttons appear here -->
    </div>

  </main>

  <!-- Universal Event Registration Modal -->
  <div id="dynamicEventModal" class="modal">
    <div class="modal-box">
      <button class="modal-close"
onclick="closeModal('dynamicEventModal')">&times;</button>
      <h2 id="modalEventTitle" style="color: #EC9F05; margin-bottom: 12px; font-
size: 22px;">Register for Event</h2>
      <p id="modalEventDetails" style="margin-bottom: 20px; color: #CBD5E1; font-
size: 15px;"></p>
      <button onclick="confirmEventRegistration()" style="padding: 10px 28px; font-
size: 15px;">Confirm Registration</button>
    </div>
  </div>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

Source Code - faq.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - FAQ</title>
  <link rel="stylesheet" href="style.css">
</head>
```

```
<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🎨 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="dashboard.html">Dashboard</a>
      <a href="faq.html" style="background-color: rgba(255, 255, 255, 0.2); color:
#DAF264; font-weight: bold;">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main style="max-width: 1150px; width: 100%; margin: 20px auto; padding: 10px;">
    <h1 style="font-size: 32px; text-align: center; margin-bottom: 6px;">Frequently
Asked Questions</h1>
    <p style="text-align: center; color: #CBD5E1; margin-bottom: 24px; font-size:
15px;">
      Quick answers to common questions about Student Hub Portal, events, and
academics
    </p>

    <!-- 2-Column FAQ Layout -->
    <div class="dash-cards-flex" style="align-items: stretch; gap: 24px;">

      <!-- Category 1: Portal & Student Account -->
      <div class="table1" style="flex: 1; padding: 26px;">
        <h2 style="font-size: 22px; color: #EC9F05; margin-bottom: 12px; border-
bottom: 1.5px solid #6B4EFD; padding-bottom: 8px;">
          🎓 Portal &amp;amp; Student Account
        </h2>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. How do I register as a student?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>Click on the 'Register' button in the top menu, fill in your details
(Name, Email, Course, Semester), accept the terms, and submit your
registration.</p>
          </div>
        </div>

        <div class="faq-item">
```

```
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. How do I log in to my dashboard?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>Go to the Login page, enter your registered email address and
password, then click the 'Login' button to open your student dashboard.</p>
          </div>
        </div>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. What is my Student ID format?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>For 3rd Semester Computer Science students, IDs follow the format
<strong>25DCS200</strong>, visible in your profile and dashboard.</p>
          </div>
        </div>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. Where can I see my enrolled subjects?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>Your Dashboard features a dedicated Sem 3 section listing your 5
registered subjects: WDF, FCN, DSA, Discrete Maths, and Java OOP.</p>
          </div>
        </div>
      </div>

      <!-- Category 2: Events & Campus Life -->
      <div class="table1" style="flex: 1; padding: 26px;">
        <h2 style="font-size: 22px; color: #EC9F05; margin-bottom: 12px; border-
bottom: 1.5px solid #6B4EFD; padding-bottom: 8px;">
          🖩 Events &amp;amp; Campus Notices
        </h2>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. How can I join college events?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>Navigate to the Events page, browse available workshops or
competitions, and click 'Register' for your selected event to reserve your entry
slot.</p>
          </div>
        </div>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. Where do I find exam dates and notices?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>Official circulars, exam dates, and assignment deadlines are
published on the Home page notice board and on your Dashboard.</p>
          </div>
```

```
        </div>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. Are workshop seats limited?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>Yes, hands-on technical workshops (like Cyber Security, AI &amp;
Robotics) have limited lab capacity and are allocated first-come, first-served.</p>
          </div>
        </div>

        <div class="faq-item">
          <button class="faq-btn" onclick="toggleFaq(this)">
            <span>Q. How do I contact the college admin?</span>
            <span class="faq-icon">+</span>
          </button>
          <div class="faq-answer">
            <p>You can use the Contact page form or send an email to
<strong>support@studenthub.com</strong> to get in touch directly with the
administration.</p>
          </div>
        </div>
      </div>

    </div>

    <!-- Bottom Help Card -->
    <div class="table1" style="padding: 22px; text-align: center; margin-top:
24px;">
      <h3 style="color: #EC9F05; margin-bottom: 6px;">Still have questions?</h3>
      <p style="color: #CBD5E1; margin-bottom: 16px; font-size: 15px;">
        Can't find the answer you are looking for? Reach out to our campus support
team.
      </p>
      <a href="contact.html" style="display: inline-block; background-color:
#EC9F05; color: white; padding: 10px 24px; border-radius: 20px; font-weight: bold;
text-decoration: none;">Contact Support &rarr;</a>
    </div>

  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

Source Code - index.html

```
<!DOCTYPE html>
```

```
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Home</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- 1. Notification Banner -->
  <div id="notification-banner" class="notification-banner">
    <span>📢 <strong>Notice:</strong> Tech Fest 2026 registration is now
open!</span>
    <button class="banner-close" onclick="closeBanner()">&times;</button>
  </div>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <!-- 2. Light / Dark Theme Switcher -->
      <button onclick="toggleTheme()">🌓 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <!-- 3. Hamburger Menu & Navigation -->
  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="contact.html">Contact</a>
    </nav>
  </div>

  <main>
    <div class="main">
      <h1>Student Hub Portal</h1>
      <p>Manage Students &amp;amp; College Events Easily</p>
      <br>
      <!-- 4. Modal Popup Trigger -->
      <button onclick="openModal('welcomeModal')" class="btn-announcement">📢 View
Announcement</button>
    </div>

    <!-- 5. Content / Announcement Slider -->
    <div class="slider-container">
      <div class="slide">
        <h3>🎉 Welcome New Students!</h3>
        <p>Orientation program starts this Monday at 10:00 AM in the
Auditorium.</p>
```

```
      </div>
      <div class="slide">
        <h3>⏳ Annual Sports Meet 2026</h3>
        <p>Register your sports teams before October 20th to participate.</p>
      </div>
      <div class="slide">
        <h3>💡 Hackathon Submissions Open</h3>
        <p>Submit your software project ideas on the dashboard before Friday.</p>
      </div>
      <div class="slider-controls">
        <button class="slider-btn" onclick="changeSlide(-1)">❮ Prev</button>
        <button class="slider-btn" onclick="changeSlide(1)">Next ❯</button>
      </div>
    </div>

    <br>

    <div style="display: flex; gap: 20px;" class="boxes-container">
      <div class="box">
        <h3 style="font-size:24px; text-decoration: underline;">Latest Notice</h3>
        <ul>
          <li class="t1">Semester Exam Schedule Released</li>
          <li class="t1">Holiday on Friday</li>
          <li class="t1">Library Books Return Date</li>
        </ul>
      </div>

      <div class="box">
        <h3 style="font-size:24px; text-decoration: underline;">Upcoming
Events</h3>
        <ul>
          <li class="t1">Tech Fest - Oct 25</li>
          <li class="t1">AI Workshop - Nov 10</li>
          <li class="t1">Career Seminar - Dec 02</li>
        </ul>
      </div>

      <div class="box">
        <h3 style="font-size:24px; text-decoration: underline;">Quick Links</h3>
        <ul>
          <li class="t1links"><a href="dashboard.html">Student Portal</a></li>
          <li class="t1links"><a href="events.html">Events</a></li>
          <li class="t1links"><a href="faq.html">FAQs</a></li>
        </ul>
      </div>
    </div>
  </main>

  <!-- 6. Modal Popup Structure -->
  <div id="welcomeModal" class="modal">
    <div class="modal-box">
      <button class="modal-close"
onclick="closeModal('welcomeModal')">&times;</button>
      <h2 style="color: #EC9F05; margin-bottom: 10px;">Important Announcement</h2>
      <p style="margin-bottom: 15px;">Welcome to the Student Hub Portal! Check the
Events section for upcoming workshops and register for campus activities.</p>
      <button onclick="closeModal('welcomeModal')">Got it!</button>
    </div>
  </div>
```

```
  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <!-- JavaScript File -->
  <script src="script.js"></script>
</body>

</html>
```

Source Code - login.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Login</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🌓 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="register.html">Register</a>
    </nav>
  </div>

  <main>
    <!-- Login Box -->
    <div class="auth-card">
      <h2>LOGIN</h2>
```

```
      <div id="login-error-msg" style="color: #FF5252; font-size: 13px; margin-
bottom: 14px; text-align: center; display: none; background: rgba(255, 82, 82,
0.15); padding: 8px; border-radius: 8px;"></div>

      <form onsubmit="handlePortalLogin(event)">
        <div class="form-group">
          <label for="email">Student ID or Email</label>
          <input type="text" id="email" required placeholder="e.g. 25DCS200 or
admin@studenthub.com">
        </div>

        <div class="form-group">
          <label for="password">Password</label>
          <input type="password" id="password" required placeholder="Enter your
password">
        </div>

        <div class="form-options">
          <label for="remember">
            <input type="checkbox" id="remember"> Remember Me
          </label>
          <a href="#">Forgot Password?</a>
        </div>

        <button type="submit" class="auth-btn">Login &rarr;</button>

        <!-- Credentials Helper Box -->
        <div style="margin-top: 18px; padding: 12px 14px; background-color:
#141E2E; border-radius: 12px; border-left: 3px solid #EC9F05; font-size: 12px;
color: #CBD5E1; text-align: left;">
          <strong style="color: #EC9F05; font-size: 13px;">🔑 Login
Credentials:</strong>

          <div style="display: flex; justify-content: space-between; align-items:
center; margin-top: 8px;">
            <span>&bull; <strong>Student:</strong> <code>25DCS200</code> /
<code>student123</code></span>
            <button type="button" onclick="fillCredentials('25DCS200',
'student123')" style="padding: 3px 8px; font-size: 11px; border-radius: 6px;
background-color: #6B4EFD;">Fill</button>
          </div>

          <div style="display: flex; justify-content: space-between; align-items:
center; margin-top: 6px;">
            <span>&bull; <strong>Admin:</strong> <code>admin@studenthub.com</code>
/ <code>admin123</code></span>
            <button type="button" onclick="fillCredentials('admin@studenthub.com',
'admin123')" style="padding: 3px 8px; font-size: 11px; border-radius: 6px;
background-color: #6B4EFD;">Fill</button>
          </div>
        </div>

        <p class="auth-footer-text">
          Don't have an account? <a href="register.html">Register here</a>
        </p>
      </form>
    </div>
  </main>

  <div class="footer">
```

```
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

Source Code - profile.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Profile</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🌓 Theme</button>
      <button><a href="login.html">Logout</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="dashboard.html">Dashboard</a>
      <a href="profile.html">Profile</a>
    </nav>
  </div>

  <main align="center" style="max-width: 500px; margin: 20px auto; padding: 10px;">
    <div class="table1" style="padding: 30px;">
      <!-- Profile Picture Circle Box -->
      <div style="width: 110px; height: 110px; border-radius: 50%; background-
color: #6B4EFD; color: white; display: flex; align-items: center; justify-content:
center; margin: 0 auto; border: 3px solid #EC9F05; font-weight: bold; text-align:
center;">
        Profile<br>Picture
      </div>
```

```
      <hr style="border-color: #6B4EFD; margin: 20px 0;">

      <table width="100%" cellpadding="8" style="text-align: left; font-size:
16px;">
        <tr>
          <td width="35%" style="color: #EC9F05;"><strong>Student ID
:</strong></td>
          <td><strong>25DCS200</strong></td>
        </tr>
        <tr>
          <td style="color: #EC9F05;"><strong>Name :</strong></td>
          <td>John Doe</td>
        </tr>
        <tr>
          <td style="color: #EC9F05;"><strong>Email :</strong></td>
          <td>john.doe@studenthub.edu</td>
        </tr>
        <tr>
          <td style="color: #EC9F05;"><strong>Mobile :</strong></td>
          <td>+1 987 654 3210</td>
        </tr>
        <tr>
          <td style="color: #EC9F05;"><strong>Course :</strong></td>
          <td>B.Tech Computer Science &amp;amp; Engg</td>
        </tr>
        <tr>
          <td style="color: #EC9F05;"><strong>Semester :</strong></td>
          <td>3rd Semester (Sem-3)</td>
        </tr>
        <tr>
          <td style="color: #EC9F05; vertical-align: top;"><strong>Subjects
:</strong></td>
          <td>WDF, FCN, DSA, DISCRETE MATHS, JAVAOOP</td>
        </tr>
      </table>
      <br>
      <button style="padding: 10px 25px; font-size: 16px;">Edit Profile</button>
    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

Source Code - register.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Portal - Student Registration</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <div class="header">
      <a href="index.html"><img src="logo.jpg" alt="Student Hub Portal Logo"
class="logo"></a>
      <strong>Student Hub Portal</strong>
    </div>
    <div class="headerlinks">
      <button onclick="toggleTheme()">🌓 Theme</button>
      <button><a href="login.html">Login</a></button>
      <button><a href="register.html">Register</a></button>
    </div>
  </header>

  <div class="navlinks">
    <button class="hamburger" onclick="toggleMenu()">☰ Menu</button>
    <nav id="nav-menu" align="center">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="events.html">Events</a>
      <a href="faq.html">FAQ</a>
      <a href="login.html">Login</a>
    </nav>
  </div>

  <main>
    <!-- Perfect Student Registration Box -->
    <div class="auth-card register-card">
      <h2>STUDENT REGISTRATION</h2>

      <form id="registrationForm" onsubmit="return validateRegistration(event)">
        <div class="form-grid">
          <!-- 1. Full Name -->
          <div class="form-group">
            <label for="name">Full Name</label>
            <input type="text" id="name" placeholder="e.g. John Doe">
            <small class="error-text" id="nameError"></small>
          </div>

          <!-- 2. Email -->
          <div class="form-group">
            <label for="email">Email Address</label>
            <input type="email" id="email" placeholder="student@college.edu">
            <small class="error-text" id="emailError"></small>
          </div>

          <!-- 3. Mobile Number -->
          <div class="form-group">
            <label for="mobile">Mobile Number</label>
            <input type="tel" id="mobile" placeholder="10-digit mobile number">
            <small class="error-text" id="mobileError"></small>
          </div>

          <!-- 4. Course -->
```

```
          <div class="form-group">
            <label for="course">Course</label>
            <select id="course">
              <option value="">-- Select Course --</option>
              <option value="B.Tech Computer Science">B.Tech Computer
Science</option>
              <option value="B.Tech Information Technology">B.Tech Information
Technology</option>
              <option value="BCA">BCA</option>
              <option value="MCA">MCA</option>
              <option value="B.Sc IT">B.Sc IT</option>
            </select>
            <small class="error-text" id="courseError"></small>
          </div>

          <!-- 5. Year -->
          <div class="form-group">
            <label for="year">Year of Study</label>
            <input type="number" id="year" min="1" max="4" placeholder="Enter Year
(1 - 4)">
            <small class="error-text" id="yearError"></small>
          </div>

          <!-- 6. Gender -->
          <div class="form-group">
            <label>Gender</label>
            <div class="radio-group">
              <label for="male">
                <input type="radio" name="gender" id="male" value="male"> Male
              </label>
              <label for="female">
                <input type="radio" name="gender" id="female" value="female">
Female
              </label>
              <label for="other">
                <input type="radio" name="gender" id="other" value="other"> Other
              </label>
            </div>
            <small class="error-text" id="genderError"></small>
          </div>

          <!-- 7. Password -->
          <div class="form-group">
            <label for="password">Password</label>
            <input type="password" id="password" placeholder="Minimum 6 chars
(letters & numbers)">
            <small class="error-text" id="passwordError"></small>
          </div>

          <!-- 8. Confirm Password -->
          <div class="form-group">
            <label for="confirmPassword">Confirm Password</label>
            <input type="password" id="confirmPassword" placeholder="Re-type
password">
            <small class="error-text" id="confirmPasswordError"></small>
          </div>
        </div>

        <!-- 9. Terms Acceptance -->
        <div class="form-options center-terms">
          <label for="terms">
```

```
            <input type="checkbox" id="terms"> I accept the Terms &amp;amp; Conditions
          </label>
          <small class="error-text" id="termsError"></small>
        </div>

        <button type="submit" class="auth-btn">Register</button>

        <p class="auth-footer-text">
          Already registered? <a href="login.html">Login here</a>
        </p>
      </form>
    </div>
  </main>

  <div class="footer">
    <footer align="center">
      <p>
        <a href="about.html">About</a>
        <a href="contact.html">Contact</a>
        Copyright &copy; Student Hub Portal
      </p>
    </footer>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

Source Code - script.js

```
/* ================================================================================
   STUDENT HUB PORTAL - SIMPLE DYNAMIC UI & FORM VALIDATION JAVASCRIPT
   ================================================================================ */

// 1. LIGHT / DARK THEME SWITCHER
function toggleTheme() {
  document.body.classList.toggle('light-theme');
  const isLight = document.body.classList.contains('light-theme');
  localStorage.setItem('theme', isLight ? 'light' : 'dark');
}

// Load saved theme on startup
if (localStorage.getItem('theme') === 'light') {
  document.body.classList.add('light-theme');
}

// 2. HAMBURGER MENU (MOBILE NAVIGATION)
function toggleMenu() {
  const menu = document.getElementById('nav-menu');
  if (menu) {
    menu.classList.toggle('show');
  }
}

// 3. NOTIFICATION BANNER
function closeBanner() {
  const banner = document.getElementById('notification-banner');
  if (banner) {
    banner.style.display = 'none';
  }
```

```
}

// 4. COLLAPSIBLE FAQ ACCORDION
function toggleFaq(btn) {
  const answer = btn.nextElementSibling;
  const icon = btn.querySelector('.faq-icon');

  if (answer.style.display === 'block') {
    answer.style.display = 'none';
    if (icon) icon.textContent = '+';
  } else {
    answer.style.display = 'block';
    if (icon) icon.textContent = '-';
  }
}

// 5. MODAL POPUP (OPEN / CLOSE)
function openModal(modalId) {
  const modal = document.getElementById(modalId);
  if (modal) {
    modal.style.display = 'flex';
  }
}

function closeModal(modalId) {
  const modal = document.getElementById(modalId);
  if (modal) {
    modal.style.display = 'none';
  }
}

// Close modal if user clicks outside the modal box
window.onclick = function(event) {
  if (event.target.classList.contains('modal')) {
    event.target.style.display = 'none';
  }
};

// 6. IMAGE / CONTENT SLIDER
let currentSlide = 0;

function showSlide(index) {
  const slides = document.querySelectorAll('.slide');
  if (slides.length === 0) return;

  if (index >= slides.length) {
    currentSlide = 0;
  } else if (index < 0) {
    currentSlide = slides.length - 1;
  } else {
    currentSlide = index;
  }

  slides.forEach((slide, i) => {
    slide.style.display = (i === currentSlide) ? 'block' : 'none';
  });
}

function changeSlide(step) {
  showSlide(currentSlide + step);
```

```
}

// Initialize first slide on page load
document.addEventListener('DOMContentLoaded', () => {
  showSlide(0);
});



// 7. STUDENT REGISTRATION FORM VALIDATION (REGEX)
function validateRegistration(event) {
  event.preventDefault(); // Stop normal form submission

  let isValid = true;

  // Helper functions for displaying error / success
  function setError(id, message) {
    const errorElem = document.getElementById(id + 'Error');
    const inputElem = document.getElementById(id);
    if (errorElem) errorElem.textContent = message;
    if (inputElem) {
      inputElem.classList.add('input-error');
      inputElem.classList.remove('input-success');
    }
    isValid = false;
  }

  function setSuccess(id) {
    const errorElem = document.getElementById(id + 'Error');
    const inputElem = document.getElementById(id);
    if (errorElem) errorElem.textContent = '';
    if (inputElem) {
      inputElem.classList.remove('input-error');
      inputElem.classList.add('input-success');
    }
  }

  // 1. Name Validation (Letters & spaces, 3-30 characters)
  const name = document.getElementById('name').value.trim();
  const nameRegex = /^[A-Za-z\s]{3,30}$/;
  if (!nameRegex.test(name)) {
    setError('name', 'Name must be 3-30 letters without numbers/symbols');
  } else {
    setSuccess('name');
  }

  // 2. Email Validation (Standard email pattern)
  const email = document.getElementById('email').value.trim();
  const emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
  if (!emailRegex.test(email)) {
    setError('email', 'Please enter a valid email address');
  } else {
    setSuccess('email');
  }

  // 3. Mobile Number Validation (10-digit mobile number)
  const mobile = document.getElementById('mobile').value.trim();
  const mobileRegex = /^[6-9]\d{9}$/;
  if (!mobileRegex.test(mobile)) {
    setError('mobile', 'Enter a valid 10-digit mobile number');
  } else {
```

```
    setSuccess('mobile');
  }

  // 4. Course Validation (Selection required)
  const course = document.getElementById('course').value;
  if (!course) {
    setError('course', 'Please select your course');
  } else {
    setSuccess('course');
  }

  // 5. Year Validation (1 to 4)
  const year = document.getElementById('year').value.trim();
  const yearRegex = /^[1-4]$/;
  if (!yearRegex.test(year)) {
    setError('year', 'Enter a valid year (1 to 4)');
  } else {
    setSuccess('year');
  }

  // 6. Gender Validation (Radio button checked)
  const genderChecked = document.querySelector('input[name="gender"]:checked');
  const genderError = document.getElementById('genderError');
  if (!genderChecked) {
    if (genderError) genderError.textContent = 'Please select your gender';
    isValid = false;
  } else {
    if (genderError) genderError.textContent = '';
  }

  // 7. Password Validation (At least 6 characters with letters and numbers)
  const password = document.getElementById('password').value;
  const passwordRegex = /^(?=.*[A-Za-z])(?=.*\d)[A-Za-z\d@$!%*#?&]{6,}$/;
  if (!passwordRegex.test(password)) {
    setError('password', 'Password must have at least 6 characters (letters &
numbers)');
  } else {
    setSuccess('password');
  }

  // 8. Confirm Password Validation (Must match password)
  const confirmPassword = document.getElementById('confirmPassword').value;
  if (confirmPassword === '' || confirmPassword !== password) {
    setError('confirmPassword', 'Passwords do not match');
  } else {
    setSuccess('confirmPassword');
  }

  // 9. Terms Acceptance (Checkbox must be checked)
  const terms = document.getElementById('terms');
  const termsError = document.getElementById('termsError');
  if (!terms.checked) {
    if (termsError) termsError.textContent = 'You must accept the Terms &
Conditions';
    isValid = false;
  } else {
    if (termsError) termsError.textContent = '';
  }

  // Final Action on Successful Validation
  if (isValid) {
```

```
    alert('🎉 Student Registration Successful! Redirecting to Dashboard...');
    window.location.href = 'dashboard.html';
  }

  return isValid;
}

// ============================================================================
// 8. PRACTICAL 6: FETCH API, DYNAMIC RENDERING, SEARCH, FILTER, SORT & PAGINATION
// ============================================================================
let allEvents = [];
let filteredEvents = [];
let currentPage = 1;
const itemsPerPage = 4;
let activeEventTitle = '';

// Fallback dataset in case local file:// security limits fetch()
const fallbackEvents = [
  {
    "id": 1,
    "title": "SPOURAL - 2026",
    "category": "SPORTS",
    "date": "2026-10-25",
    "displayDate": "October 25, 2026",
    "venue": "College Main Ground",
    "description": "Annual Inter-College Sports & Athletics Championship"
  },
  {
    "id": 2,
    "title": "Cyber Security Workshop",
    "category": "TECH",
    "date": "2026-11-10",
    "displayDate": "November 10, 2026",
    "venue": "Auditorium B",
    "description": "Hands-on Ethical Hacking & Network Defense Bootcamp"
  },
  {
    "id": 3,
    "title": "AI & Robotics Workshop",
    "category": "WORKSHOP",
    "date": "2026-11-22",
    "displayDate": "November 22, 2026",
    "venue": "Computer Lab 3",
    "description": "Practical Machine Learning, Arduino & Autonomous Bots Session"
  },
  {
    "id": 4,
    "title": "Tarang Cultural Fest",
    "category": "CULTURAL",
    "date": "2026-12-05",
    "displayDate": "December 05, 2026",
    "venue": "Open Air Amphitheater",
    "description": "Annual Youth Festival featuring Music, Drama, Dance & Art"
  },
  {
    "id": 5,
    "title": "WDF Web Development Hackathon",
    "category": "TECH",
    "date": "2026-12-15",
    "displayDate": "December 15, 2026",
    "venue": "IT Lab 2",
    "description": "24-hour Full-Stack Web App Development Challenge"
```

```
  },
  {
    "id": 6,
    "title": "DSA CodeSprint Contest",
    "category": "ACADEMIC",
    "date": "2026-12-20",
    "displayDate": "December 20, 2026",
    "venue": "Online Portal & Lab 1",
    "description": "Algorithmic Problem Solving & Competitive Coding Battle"
  },
  {
    "id": 7,
    "title": "Java OOP Project Exhibition",
    "category": "ACADEMIC",
    "date": "2026-12-28",
    "displayDate": "December 28, 2026",
    "venue": "Seminar Hall A",
    "description": "Semester 3 Java Object-Oriented Software Projects Showcase"
  },
  {
    "id": 8,
    "title": "Networking & FCN Hands-on Lab",
    "category": "WORKSHOP",
    "date": "2027-01-08",
    "displayDate": "January 08, 2027",
    "venue": "Cisco Networking Lab",
    "description": "Practical Packet Tracer, Subnetting & Network Simulation"
  }
];

// Load external JSON data via Fetch API
function loadEventsData() {
  const container = document.getElementById('events-list');
  if (!container) return; // Not on events page

  fetch('events.json')
    .then(response => {
      if (!response.ok) throw new Error('Network error loading events.json');
      return response.json();
    })
    .then(data => {
      allEvents = data;
      applyEventFilters();
    })
    .catch(err => {
      console.warn('Fetch fallback activated (e.g. file:// protocol):', err);
      allEvents = fallbackEvents;
      applyEventFilters();
    });
}

// Combined Search, Filter & Sort Logic
function applyEventFilters() {
  const searchInput = document.getElementById('search');
  const categorySelect = document.getElementById('categoryFilter');
  const sortSelect = document.getElementById('sortOrder');

  const searchTerm = searchInput ? searchInput.value.toLowerCase().trim() : '';
  const selectedCategory = categorySelect ? categorySelect.value : 'ALL';
  const sortOrder = sortSelect ? sortSelect.value : 'date-asc';
```

```
  // 1. Filter by Search Query & Category
  filteredEvents = allEvents.filter(event => {
    const matchesSearch =
      event.title.toLowerCase().includes(searchTerm) ||
      event.venue.toLowerCase().includes(searchTerm) ||
      event.description.toLowerCase().includes(searchTerm) ||
      event.category.toLowerCase().includes(searchTerm);

    const matchesCategory =
      selectedCategory === 'ALL' || event.category === selectedCategory;

    return matchesSearch && matchesCategory;
  });

  // 2. Sort Events
  filteredEvents.sort((a, b) => {
    if (sortOrder === 'name-asc') {
      return a.title.localeCompare(b.title);
    } else if (sortOrder === 'name-desc') {
      return b.title.localeCompare(a.title);
    } else if (sortOrder === 'date-desc') {
      return new Date(b.date) - new Date(a.date);
    } else {
      // date-asc (default)
      return new Date(a.date) - new Date(b.date);
    }
  });

  // Reset to first page whenever filter changes
  currentPage = 1;
  renderEventsPage();
}

// Dynamic Rendering with Pagination
function renderEventsPage() {
  const container = document.getElementById('events-list');
  const paginationContainer = document.getElementById('pagination-controls');
  if (!container) return;

  // Handle empty state
  if (filteredEvents.length === 0) {
    container.innerHTML = `
      <div class="table1" style="text-align: center; padding: 36px 20px;">
        <h3 style="color: #EC9F05; margin-bottom: 8px; font-size: 20px;">No events
found</h3>
        <p style="color: #CBD5E1; font-size: 15px;">No campus events match your
search query or selected filter.</p>
        <button onclick="resetEventFilters()" style="margin-top: 14px; padding: 8px
20px;">Reset Filters</button>
      </div>
    `;
    if (paginationContainer) paginationContainer.innerHTML = '';
    return;
  }

  // Calculate slice for current page
  const startIndex = (currentPage - 1) * itemsPerPage;
  const endIndex = startIndex + itemsPerPage;
  const pageItems = filteredEvents.slice(startIndex, endIndex);

  // Generate HTML for events
```

```
  let html = '';
  pageItems.forEach(event => {
    html += `
      <div class="table1 event-card-item">
        <div class="event-badge">${event.category}</div>
        <div style="flex: 1; min-width: 260px;">
          <h3 style="font-size: 22px; color: #EC9F05; margin-bottom:
6px;">${event.title}</h3>
          <p style="font-size: 15px; margin-bottom: 4px;">
            <strong>📅 Date:</strong> ${event.displayDate}  |  <strong>📍
Venue:</strong> ${event.venue}
          </p>
          <p style="font-size: 14px; color: #CBD5E1;">${event.description}</p>
        </div>
        <div style="flex-shrink: 0;">
          <button onclick="openRegisterModal('${event.title.replace(/'/g, "\\'")}',
'${event.displayDate}', '${event.venue}')" style="padding: 10px
24px;">Register</button>
        </div>
      </div>
    `;
  });

  container.innerHTML = html;
  renderPaginationButtons();
}

// Render Pagination Controls
function renderPaginationButtons() {
  const paginationContainer = document.getElementById('pagination-controls');
  if (!paginationContainer) return;

  const totalPages = Math.ceil(filteredEvents.length / itemsPerPage);
  if (totalPages <= 1) {
    paginationContainer.innerHTML = '';
    return;
  }

  let paginationHtml = '';

  // Previous Button
  paginationHtml += `
    <button class="pagination-btn" onclick="changeEventsPage(${currentPage - 1})"
${currentPage === 1 ? 'disabled' : ''}>
      &laquo; Prev
    </button>
  `;

  // Page Numbers
  for (let i = 1; i <= totalPages; i++) {
    paginationHtml += `
      <button class="pagination-btn ${i === currentPage ? 'active' : ''}"
onclick="changeEventsPage(${i})">
        ${i}
      </button>
    `;
  }

  // Next Button
  paginationHtml += `
```

```
    <button class="pagination-btn" onclick="changeEventsPage(${currentPage + 1})"
${currentPage === totalPages ? 'disabled' : ''}>
      Next &raquo;
    </button>
  `;

  paginationContainer.innerHTML = paginationHtml;
}

// Change Page Handler
function changeEventsPage(pageNumber) {
  const totalPages = Math.ceil(filteredEvents.length / itemsPerPage);
  if (pageNumber < 1 || pageNumber > totalPages) return;
  currentPage = pageNumber;
  renderEventsPage();
  // Smooth scroll back to top of events list
  const container = document.getElementById('events-toolbar-card');
  if (container) {
    container.scrollIntoView({ behavior: 'smooth', block: 'start' });
  }
}

// Reset Search & Filters
function resetEventFilters() {
  const searchInput = document.getElementById('search');
  const categorySelect = document.getElementById('categoryFilter');
  const sortSelect = document.getElementById('sortOrder');

  if (searchInput) searchInput.value = '';
  if (categorySelect) categorySelect.value = 'ALL';
  if (sortSelect) sortSelect.value = 'date-asc';

  applyEventFilters();
}

// Universal Registration Modal Handlers
function openRegisterModal(title, date, venue) {
  activeEventTitle = title;
  const titleEl = document.getElementById('modalEventTitle');
  const detailsEl = document.getElementById('modalEventDetails');

  if (titleEl) titleEl.textContent = 'Register for ' + title;
  if (detailsEl) detailsEl.innerHTML = `<strong>Date:</strong> ${date} &bull;
<strong>Venue:</strong> ${venue}<br><br>Confirm your participation to secure your
registration slot.`;

  openModal('dynamicEventModal');
}

function confirmEventRegistration() {
  alert('🎉 Successfully registered for ' + (activeEventTitle || 'the event') +
'!');
  closeModal('dynamicEventModal');
}

// Automatically load events if on events page
document.addEventListener('DOMContentLoaded', () => {
  loadEventsData();
});
```

```
// ============================================================================
// 9. AUTHENTICATION & ACCESS CONTROL (STUDENT & ADMIN CREDENTIALS)
// ============================================================================
const PORTAL_CREDENTIALS = {
  student: {
    identifiers: ['student@studenthub.com', '25dcs200', 'john.doe@studenthub.edu'],
    password: 'student123',
    role: 'student',
    name: 'John Doe',
    studentId: '25DCS200',
    course: 'B.Tech CSE - 3rd Sem'
  },
  admin: {
    identifiers: ['admin@studenthub.com', 'admin'],
    password: 'admin123',
    role: 'admin',
    name: 'Portal Administrator',
    studentId: 'ADMIN-01',
    course: 'Faculty / Administration'
  }
};

// Handle Login Form Submission
function handlePortalLogin(event) {
  if (event) event.preventDefault();

  const idInput = document.getElementById('email');
  const passInput = document.getElementById('password');
  const errorMsg = document.getElementById('login-error-msg');

  if (!idInput || !passInput) return;

  const enteredId = idInput.value.trim().toLowerCase();
  const enteredPass = passInput.value.trim();

  // 1. Check Student Credentials
  if (PORTAL_CREDENTIALS.student.identifiers.includes(enteredId) && enteredPass ===
PORTAL_CREDENTIALS.student.password) {
    sessionStorage.setItem('currentUser',
JSON.stringify(PORTAL_CREDENTIALS.student));
    alert('☑ Welcome, John Doe! Logged in as Student.');
    window.location.href = 'dashboard.html';
    return;
  }

  // 2. Check Admin Credentials
  if (PORTAL_CREDENTIALS.admin.identifiers.includes(enteredId) && enteredPass ===
PORTAL_CREDENTIALS.admin.password) {
    sessionStorage.setItem('currentUser',
JSON.stringify(PORTAL_CREDENTIALS.admin));
    alert('◐ Welcome, Administrator! Full Dashboard and Admin Access Granted.');
    window.location.href = 'dashboard.html';
    return;
  }

  // 3. Failed Login
  if (errorMsg) {
    errorMsg.textContent = '✕ Invalid Credentials! Use 25DCS200 / student123 or
admin@studenthub.com / admin123.';
```

```
    errorMsg.style.display = 'block';
  } else {
    alert('❌ Invalid Credentials!\n\n• Student: 25DCS200 / student123\n• Admin:
admin@studenthub.com / admin123');
  }
}

// Quick Fill Demo Credentials
function fillCredentials(id, pass) {
  const idInput = document.getElementById('email');
  const passInput = document.getElementById('password');
  const errorMsg = document.getElementById('login-error-msg');

  if (idInput) idInput.value = id;
  if (passInput) passInput.value = pass;
  if (errorMsg) errorMsg.style.display = 'none';
}

// Protect Dashboard (Redirect if not logged in)
function checkDashboardAuth() {
  const sessionUser = sessionStorage.getItem('currentUser');
  if (!sessionUser) {
    alert('🔒 Access Denied! Please login with Student or Admin credentials to view
the dashboard.');
    window.location.href = 'login.html';
    return false;
  }

  const user = JSON.parse(sessionUser);

  // If Admin is viewing, show special admin access badge
  const adminBadge = document.getElementById('admin-access-indicator');
  if (adminBadge && user.role === 'admin') {
    adminBadge.style.display = 'block';
  }

  return true;
}

// Logout Handler
function handleLogout() {
  sessionStorage.removeItem('currentUser');
  alert('👋 You have been logged out successfully.');
  window.location.href = 'login.html';
}

// Protect Admin Panel (Only Admin Allowed)
function checkAdminAuth() {
  const sessionUser = sessionStorage.getItem('currentUser');
  if (!sessionUser) {
    alert('🔒 Admin Access Only! Please login with Administrator credentials.');
    window.location.href = 'login.html';
    return false;
  }

  const user = JSON.parse(sessionUser);
  if (user.role !== 'admin') {
    alert('⛔ Access Restricted! Only Administrators can access the Admin Panel.');
    window.location.href = 'dashboard.html';
```

```
    return false;
  }

  return true;
}
```

Source Code - events.json

```
[
  {
    "id": 1,
    "title": "SPOURAL – 2026",
    "category": "SPORTS",
    "date": "2026-10-25",
    "displayDate": "October 25, 2026",
    "venue": "College Main Ground",
    "description": "Annual Inter-College Sports & Athletics Championship"
  },
  {
    "id": 2,
    "title": "Cyber Security Workshop",
    "category": "TECH",
    "date": "2026-11-10",
    "displayDate": "November 10, 2026",
    "venue": "Auditorium B",
    "description": "Hands-on Ethical Hacking & Network Defense Bootcamp"
  },
  {
    "id": 3,
    "title": "AI & Robotics Workshop",
    "category": "WORKSHOP",
    "date": "2026-11-22",
    "displayDate": "November 22, 2026",
    "venue": "Computer Lab 3",
    "description": "Practical Machine Learning, Arduino & Autonomous Bots Session"
  },
  {
    "id": 4,
    "title": "Tarang Cultural Fest",
    "category": "CULTURAL",
    "date": "2026-12-05",
    "displayDate": "December 05, 2026",
    "venue": "Open Air Amphitheater",
    "description": "Annual Youth Festival featuring Music, Drama, Dance & Art"
  },
  {
    "id": 5,
    "title": "WDF Web Development Hackathon",
    "category": "TECH",
    "date": "2026-12-15",
    "displayDate": "December 15, 2026",
    "venue": "IT Lab 2",
    "description": "24-hour Full-Stack Web App Development Challenge"
  },
  {
    "id": 6,
    "title": "DSA CodeSprint Contest",
    "category": "ACADEMIC",
    "date": "2026-12-20",
    "displayDate": "December 20, 2026",
    "venue": "Online Portal & Lab 1",
    "description": "Algorithmic Problem Solving & Competitive Coding Battle"
  },
  {
```

```
    "id": 7,
    "title": "Java OOP Project Exhibition",
    "category": "ACADEMIC",
    "date": "2026-12-28",
    "displayDate": "December 28, 2026",
    "venue": "Seminar Hall A",
    "description": "Semester 3 Java Object-Oriented Software Projects Showcase"
  },
  {
    "id": 8,
    "title": "Networking & FCN Hands-on Lab",
    "category": "WORKSHOP",
    "date": "2027-01-08",
    "displayDate": "January 08, 2027",
    "venue": "Cisco Networking Lab",
    "description": "Practical Packet Tracer, Subnetting & Network Simulation"
  }
]
```

Source Code - build_practical1_6_all.py

```
from pathlib import Path
import importlib.util

from docx import Document
from docx.enum.text import WD_ALIGN_PARAGRAPH
from docx.shared import Pt, RGBColor


ROOT = Path(__file__).resolve().parent
OUT = ROOT / "Student_Hub_Practical_File_1_to_6_All_Code.docx"


def load_module(name, path):
    spec = importlib.util.spec_from_file_location(name, path)
    module = importlib.util.module_from_spec(spec)
    spec.loader.exec_module(module)
    return module


base = load_module("base_builder", ROOT / "build_practical_file.py")
p6 = load_module("practical6_builder", ROOT / "build_practical6.py")


def title_page(doc):
    p = doc.add_paragraph(style="Title")
    base.style_paragraph(p, before=100, after=8, line=1.0,
align=WD_ALIGN_PARAGRAPH.CENTER)
    r = p.add_run("Student Hub Portal")
    base.set_run_font(r, size=25, bold=True, color="000000")

    p = doc.add_paragraph()
    base.style_paragraph(p, after=8, line=1.0, align=WD_ALIGN_PARAGRAPH.CENTER)
    r = p.add_run("Practical File 1 to 6")
    base.set_run_font(r, size=20, bold=True, color="1F4E79")

    p = doc.add_paragraph()
    base.style_paragraph(p, after=14, line=1.0, align=WD_ALIGN_PARAGRAPH.CENTER)
    r = p.add_run("HTML CSS JavaScript JSON and supporting source code")
    base.set_run_font(r, size=12, color="555555")
```

```
    p = doc.add_paragraph()
    base.style_paragraph(p, after=0, line=1.0, align=WD_ALIGN_PARAGRAPH.CENTER)
    r = p.add_run("Questions answers analysis interpretation and complete code")
    base.set_run_font(r, size=11, color="555555")
    doc.add_page_break()


def add_practical6(doc):
    base.add_heading(doc, "Practical No. 6 - Rendering External JSON Data using
Fetch API, Search and Filter")
    base.add_body(doc, "Question", "Display event lists, student profiles, notices,
or FAQs from an external JSON file using Fetch API. Implement dynamic rendering,
search, filter, sorting, and pagination.")
    base.add_subheading(doc, "Key Questions and Answers")
    questions = [
        ("How is JSON fetched, parsed, and rendered?", "The fetch function requests
events.json and returns a response. The response is converted to JavaScript objects
with response.json(), and the objects are inserted into the page as event cards."),
        ("Which array methods are used for search, filter, sort, and pagination?",
"filter() selects matching events, sort() orders events by date, map() creates
display markup, and slice() selects the records for the current page."),
        ("How are loading and error states handled?", "The page initially shows a
loading message. If the request fails or the response is not successful, catch()
displays an error message and clears the event list."),
        ("How is modularity maintained in JavaScript files?", "The code is divided
into small functions such as loadEvents, addCategories, applyFilters, renderEvents,
renderPagination, and formatDate. Each function has one clear responsibility."),
    ]
    for question, answer in questions:
        base.add_body(doc, "Q", question)
        base.add_body(doc, "Answer", answer)
    base.add_subheading(doc, "Analysis")
    base.add_body(doc, "A", "The application loads event records from events.json
instead of writing every event directly in HTML. Search, category filtering, date
sorting, and pagination are applied to the JavaScript array before the visible
cards are updated.")
    base.add_subheading(doc, "Interpretation")
    base.add_body(doc, "I", "This approach is useful for modern web applications
because the same page can display changing data without rewriting the HTML. In a
real project, the JSON file can later be replaced by an API endpoint.")
    base.add_subheading(doc, "Required Practical 6 Code")
    base.add_body(doc, "Testing Note", "Run the files through a local server such
as Live Server. Fetch may be blocked when the HTML file is opened directly with a
file path in the browser.")
    for label, code in [
        ("HTML Code - events-fetch.html", p6.HTML),
        ("JSON Code - events.json", p6.JSON_DATA),
        ("JavaScript Code - practical6.js", p6.JS),
        ("CSS Code - practical6.css", p6.CSS),
    ]:
        base.add_code_heading(doc, label)
        base.add_code_block(doc, code)


def add_code_appendix(doc):
    doc.add_page_break()
    base.add_heading(doc, "Complete Source Code Appendix")
    base.add_body(doc, "Included Files", "The following appendix contains every
code file currently present in the final folder: HTML, CSS, JavaScript, JSON, and
```

```
    Python source files. Generated Word/PDF files and image assets are not included as
    code.")
        extensions = {".html", ".css", ".js", ".json", ".py"}
        files = sorted((p for p in ROOT.iterdir() if p.is_file() and p.suffix.lower()
    in extensions), key=lambda p: (p.suffix.lower(), p.name.lower()))
        for path in files:
            base.add_code_heading(doc, f"Source Code - {path.name}")
            base.add_code_block(doc, path.read_text(encoding="utf-8"))


    def main():
        doc = Document()
        base.configure_document(doc)
        title_page(doc)
        for item_index, item in enumerate(base.practical_data_upto_5()):
            if item_index:
                doc.add_page_break()
            base.add_practical(doc, *item)
        doc.add_page_break()
        add_practical6(doc)
        add_code_appendix(doc)
        doc.save(OUT)
        print(f"Saved {OUT}")


    if __name__ == "__main__":
        main()
```

Source Code - build_practical6.py

```
    from pathlib import Path
    import re

    from docx import Document
    from docx.enum.text import WD_ALIGN_PARAGRAPH
    from docx.oxml.ns import qn
    from docx.shared import Inches, Pt, RGBColor


    ROOT = Path(__file__).resolve().parent
    OUT = ROOT / "Student_Hub_Practical_6_Fetch_API_Search_Filter.docx"


    def font(run, name="Calibri", size=11, bold=False, color="000000"):
        run.font.name = name
        run._element.rPr.rFonts.set(qn("w:ascii"), name)
        run._element.rPr.rFonts.set(qn("w:hAnsi"), name)
        run.font.size = Pt(size)
        run.bold = bold
        run.font.color.rgb = RGBColor.from_string(color)


    def para(p, before=0, after=4, line=1.15, align=None):
        f = p.paragraph_format
        f.space_before = Pt(before)
        f.space_after = Pt(after)
        f.line_spacing = line
        if align is not None:
            p.alignment = align
```

```
def heading(doc, text, level=1):
    p = doc.add_paragraph()
    para(p, before=12 if level == 1 else 8, after=5, line=1.05)
    r = p.add_run(text)
    font(r, size=16 if level == 1 else 12.5, bold=True, color="1F4E79")


def labeled(doc, label, text):
    p = doc.add_paragraph()
    para(p, after=4)
    r = p.add_run(label + ": ")
    font(r, bold=True, color="1F4E79")
    r = p.add_run(text)
    font(r)


def code_color(line):
    stripped = line.strip()
    if not stripped:
        return "666666"
    if stripped.startswith("//") or stripped.startswith("/*") or
stripped.startswith("*") or stripped.startswith("<!--"):
        return "008000"
    if stripped.startswith("{") or stripped.startswith("}") or
stripped.startswith('"'):
        return "A31515"
    if
re.search(r"<(/?)(html|head|body|main|section|div|h1|h2|p|input|button|script|link|
select|option|form|label|title)", stripped):
        return "0000CC"
    if
re.search(r"\b(function|const|let|return|async|await|if|else|for|from|import|class)
\b", stripped):
        return "7F0055"
    if ":" in stripped and stripped.endswith(";"):
        return "006666"
    return "333333"


def code_block(doc, title, code):
    p = doc.add_paragraph()
    para(p, before=8, after=3, line=1.0)
    r = p.add_run(title)
    font(r, size=11, bold=True, color="1F4E79")
    for line in code.rstrip().splitlines():
        p = doc.add_paragraph()
        para(p, after=0, line=1.0)
        p.paragraph_format.left_indent = Inches(0.15)
        r = p.add_run(line)
        font(r, name="Courier New", size=9, color=code_color(line))


HTML = '''<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Hub Events</title>
  <link rel="stylesheet" href="practical6.css">
```

```
</head>
<body>
  <main class="container">
    <h1>Student Hub Events</h1>
    <p>Events are loaded from an external JSON file using the Fetch API.</p>

    <section class="controls" aria-label="Event controls">
      <label for="searchInput">Search</label>
      <input id="searchInput" type="search" placeholder="Search by event or venue">
      <label for="categoryFilter">Category</label>
      <select id="categoryFilter">
        <option value="all">All categories</option>
      </select>
      <label for="sortSelect">Sort by date</label>
      <select id="sortSelect">
        <option value="asc">Oldest first</option>
        <option value="desc">Newest first</option>
      </select>
    </section>

    <p id="status" class="status" role="status">Loading events...</p>
    <section id="eventList" class="event-grid" aria-live="polite"></section>
    <nav id="pagination" class="pagination" aria-label="Event pages"></nav>
  </main>
  <script src="practical6.js"></script>
</body>
</html>'''


JSON_DATA = '''[
  {"id": 1, "title": "SPOURAL 2026", "date": "2026-10-25", "venue": "College Main
Ground", "category": "Sports"},
  {"id": 2, "title": "Cyber Security Workshop", "date": "2026-11-10", "venue":
"Auditorium B", "category": "Technical"},
  {"id": 3, "title": "Career Guidance Seminar", "date": "2026-11-22", "venue":
"Seminar Hall", "category": "Career"},
  {"id": 4, "title": "Student Hackathon", "date": "2026-12-05", "venue":
"Innovation Lab", "category": "Technical"},
  {"id": 5, "title": "Annual Cultural Night", "date": "2026-12-18", "venue": "Open
Air Theatre", "category": "Cultural"},
  {"id": 6, "title": "Resume Building Session", "date": "2027-01-08", "venue":
"Placement Cell", "category": "Career"}
]'''


JS = '''const state = {
  events: [],
  filtered: [],
  page: 1,
  pageSize: 4
};

const list = document.getElementById("eventList");
const status = document.getElementById("status");
const searchInput = document.getElementById("searchInput");
const categoryFilter = document.getElementById("categoryFilter");
const sortSelect = document.getElementById("sortSelect");
const pagination = document.getElementById("pagination");

async function loadEvents() {
  try {
```

```
    const response = await fetch("events.json");
    if (!response.ok) throw new Error("Unable to load event data");
    state.events = await response.json();
    addCategories();
    applyFilters();
  } catch (error) {
    status.textContent = error.message;
    list.innerHTML = "";
  }
}

function addCategories() {
  const categories = [...new Set(state.events.map(event =>
event.category))].sort();
  categories.forEach(category => {
    const option = document.createElement("option");
    option.value = category;
    option.textContent = category;
    categoryFilter.appendChild(option);
  });
}

function applyFilters() {
  const term = searchInput.value.trim().toLowerCase();
  const category = categoryFilter.value;
  const direction = sortSelect.value === "asc" ? 1 : -1;

  state.filtered = state.events
    .filter(event => {
      const searchable = `${event.title} ${event.venue}`.toLowerCase();
      return searchable.includes(term) && (category === "all" || event.category ===
category);
    })
    .sort((a, b) => (new Date(a.date) - new Date(b.date)) * direction);

  state.page = 1;
  renderEvents();
  renderPagination();
}

function renderEvents() {
  const start = (state.page - 1) * state.pageSize;
  const pageItems = state.filtered.slice(start, start + state.pageSize);
  status.textContent = `${state.filtered.length} event(s) found`;
  list.innerHTML = pageItems.length ? pageItems.map(event => `
    <article class="event-card">
      <h2>${event.title}</h2>
      <p><strong>Date:</strong> ${formatDate(event.date)}</p>
      <p><strong>Venue:</strong> ${event.venue}</p>
      <p><strong>Category:</strong> ${event.category}</p>
    </article>
  `).join("") : "<p>No matching events found.</p>";
}

function renderPagination() {
  const totalPages = Math.ceil(state.filtered.length / state.pageSize);
  pagination.innerHTML = "";
  for (let page = 1; page <= totalPages; page += 1) {
    const button = document.createElement("button");
    button.textContent = page;
    button.setAttribute("aria-label", `Go to page ${page}`);
```

```
    button.className = page === state.page ? "active" : "";
    button.addEventListener("click", () => {
      state.page = page;
      renderEvents();
      renderPagination();
    });
    pagination.appendChild(button);
  }
}

function formatDate(value) {
  return new Date(value).toLocaleDateString("en-IN", {
    day: "2-digit", month: "short", year: "numeric"
  });
}

searchInput.addEventListener("input", applyFilters);
categoryFilter.addEventListener("change", applyFilters);
sortSelect.addEventListener("change", applyFilters);
loadEvents();'''



CSS = '''* { box-sizing: border-box; }
body { margin: 0; font-family: Arial, sans-serif; background: #0b162c; color:
#f5f7fb; }
.container { width: min(100% - 32px, 1000px); margin: 32px auto; }
h1 { color: #ec9f05; }
.controls { display: grid; grid-template-columns: auto 1fr auto 1fr auto 1fr; gap:
10px; align-items: center; padding: 16px; background: #152644; border-radius: 10px;
}
input, select { width: 100%; padding: 10px; border: 1px solid #7184a8; border-
radius: 6px; background: #ffffff; color: #17233b; }
.status { color: #b7c5dd; }
.event-grid { display: grid; grid-template-columns: repeat(2, 1fr); gap: 16px; }
.event-card { padding: 18px; border: 1px solid #52698e; border-radius: 10px;
background: #13213a; }
.event-card h2 { margin-top: 0; color: #ec9f05; font-size: 20px; }
.pagination { display: flex; gap: 8px; margin-top: 20px; }
.pagination button { padding: 8px 12px; border: 0; border-radius: 5px; cursor:
pointer; }
.pagination .active { background: #ec9f05; color: #17233b; font-weight: bold; }
@media (max-width: 720px) { .controls, .event-grid { grid-template-columns: 1fr; }
}
'''



def main():
    doc = Document()
    section = doc.sections[0]
    section.top_margin = Inches(0.8)
    section.bottom_margin = Inches(0.8)
    section.left_margin = Inches(0.85)
    section.right_margin = Inches(0.85)

    title = doc.add_paragraph(style="Title")
    para(title, before=0, after=4, align=WD_ALIGN_PARAGRAPH.CENTER)
    title_run = title.add_run("Practical 6 Fetch API Search and Filter")
    font(title_run, size=24, bold=True, color="000000")
    sub = doc.add_paragraph()
    para(sub, after=14, align=WD_ALIGN_PARAGRAPH.CENTER)
    r = sub.add_run("Student Hub Portal")
```

```
    font(r, size=13, color="2E74B5")

    heading(doc, "Question")
    labeled(doc, "Practical 6", "Rendering External JSON Data using Fetch API,
Search and Filter")
    labeled(doc, "Problem Definition", "Display event lists, student profiles,
notices, or FAQs from an external JSON file using the Fetch API. Implement dynamic
rendering, search, filter, sorting, and pagination.")

    heading(doc, "Key Questions and Answers", 2)
    questions = [
        ("How is JSON fetched, parsed, and rendered?", "The fetch function requests
events.json and returns a response. The response is converted to JavaScript objects
with response.json(), and the objects are inserted into the page as event cards."),
        ("Which array methods are used for search, filter, sort, and pagination?",
"filter() selects matching events, sort() orders events by date, map() creates
display markup, and slice() selects the records for the current page."),
        ("How are loading and error states handled?", "The page initially shows a
loading message. If the request fails or the response is not successful, catch()
displays an error message and clears the event list."),
        ("How is modularity maintained in JavaScript files?", "The code is divided
into small functions such as loadEvents, addCategories, applyFilters, renderEvents,
renderPagination, and formatDate. Each function has one clear responsibility."),
    ]
    for q, a in questions:
        labeled(doc, "Q", q)
        labeled(doc, "Answer", a)

    heading(doc, "Analysis", 2)
    labeled(doc, "A", "The application loads event records from events.json instead
of writing every event directly in HTML. Search, category filtering, date sorting,
and pagination are applied to the JavaScript array before the visible cards are
updated.")
    heading(doc, "Interpretation", 2)
    labeled(doc, "I", "This approach is useful for modern web applications because
the same page can display changing data without rewriting the HTML. In a real
project, the JSON file can later be replaced by an API endpoint.")

    heading(doc, "Code Required", 2)
    labeled(doc, "Testing Note", "Run the files through a local server such as Live
Server. Fetch may be blocked when the HTML file is opened directly with a file path
in the browser.")
    code_block(doc, "HTML Code - events-fetch.html", HTML)
    code_block(doc, "JSON Code - events.json", JSON_DATA)
    code_block(doc, "JavaScript Code - practical6.js", JS)
    code_block(doc, "CSS Code - practical6.css", CSS)

    doc.save(OUT)
    print(f"Saved {OUT}")


if __name__ == "__main__":
    main()
```

Source Code - build_practical_file.py

```
from pathlib import Path

from docx import Document
from docx.enum.text import WD_ALIGN_PARAGRAPH
from docx.oxml.ns import qn
```

```
from docx.shared import Inches, Pt, RGBColor


ROOT = Path(__file__).resolve().parent
OUT_DOCX = ROOT / "Student_Hub_Practical_File_Upto_5.docx"


def read_text(name: str) -> str:
    return (ROOT / name).read_text(encoding="utf-8")


def set_run_font(run, name="Calibri", size=11, bold=None, italic=None, color=None):
    run.font.name = name
    run._element.rPr.rFonts.set(qn("w:ascii"), name)
    run._element.rPr.rFonts.set(qn("w:hAnsi"), name)
    run.font.size = Pt(size)
    if bold is not None:
        run.bold = bold
    if italic is not None:
        run.italic = italic
    if color is not None:
        run.font.color.rgb = RGBColor.from_string(color)


def style_paragraph(paragraph, before=0, after=0, line=1.15, align=None):
    fmt = paragraph.paragraph_format
    fmt.space_before = Pt(before)
    fmt.space_after = Pt(after)
    fmt.line_spacing = line
    if align is not None:
        paragraph.alignment = align


def add_title_page(doc: Document):
    p = doc.add_paragraph()
    style_paragraph(p, before=120, after=8, line=1.0,
align=WD_ALIGN_PARAGRAPH.CENTER)
    r = p.add_run("Student Hub Portal")
    set_run_font(r, size=24, bold=True, color="1F1F1F")

    p = doc.add_paragraph()
    style_paragraph(p, before=0, after=6, line=1.0,
align=WD_ALIGN_PARAGRAPH.CENTER)
    r = p.add_run("Practical File")
    set_run_font(r, size=18, bold=True, color="2E74B5")

    p = doc.add_paragraph()
    style_paragraph(p, before=0, after=14, line=1.0,
align=WD_ALIGN_PARAGRAPH.CENTER)
    r = p.add_run("Practicals 1 to 5 - HTML, CSS and JavaScript")
    set_run_font(r, size=12.5, color="4D4D4D")

    p = doc.add_paragraph()
    style_paragraph(p, before=0, after=16, line=1.0,
align=WD_ALIGN_PARAGRAPH.CENTER)
    r = p.add_run("Practical-wise write-up with questions, answers, analysis,
interpretation, and source code.")
    set_run_font(r, size=11, color="4D4D4D")
```

```
    p = doc.add_paragraph()
    style_paragraph(p, before=0, after=0, line=1.0,
align=WD_ALIGN_PARAGRAPH.CENTER)
    r = p.add_run("Student Hub Portal Source Code")
    set_run_font(r, size=10.5, italic=True, color="666666")

    doc.add_page_break()


def add_heading(doc: Document, text: str):
    p = doc.add_paragraph()
    style_paragraph(p, before=6, after=6, line=1.1)
    r = p.add_run(text)
    set_run_font(r, size=15, bold=True, color="2E74B5")


def add_subheading(doc: Document, text: str):
    p = doc.add_paragraph()
    style_paragraph(p, before=4, after=2, line=1.05)
    r = p.add_run(text)
    set_run_font(r, size=11.5, bold=True, color="1F4D78")


def add_body(doc: Document, label: str, text: str):
    p = doc.add_paragraph()
    style_paragraph(p, before=0, after=3, line=1.15)
    r1 = p.add_run(f"{label}: ")
    set_run_font(r1, size=11, bold=True, color="000000")
    r2 = p.add_run(text)
    set_run_font(r2, size=11, color="000000")


def add_code_heading(doc: Document, text: str):
    p = doc.add_paragraph()
    style_paragraph(p, before=6, after=3, line=1.0)
    r = p.add_run(text)
    set_run_font(r, size=11, bold=True, color="1F4D78")


def add_code_block(doc: Document, code: str):
    for line in code.rstrip("\n").splitlines():
        p = doc.add_paragraph()
        style_paragraph(p, before=0, after=0, line=1.0)
        p.paragraph_format.left_indent = Inches(0.18)
        p.paragraph_format.right_indent = Inches(0.05)
        r = p.add_run(line.rstrip())
        stripped = line.strip()
        if stripped.startswith("<!--") or stripped.startswith("//") or
stripped.startswith("/*") or stripped.startswith("*"):
            color = "008000"
        elif stripped.startswith("<"):
            color = "0000CC"
        elif ":" in stripped and stripped.endswith(";"):
            color = "006666"
        elif any(word in stripped.split() for word in ("const", "let", "function",
"return", "if", "else")):
            color = "7F0055"
        else:
            color = "333333"
        set_run_font(r, name="Courier New", size=9, color=color)
```

```
def add_practical(doc: Document, no: int, title: str, question: str, key_questions,
analysis, interpretation, sources):
    add_heading(doc, f"Practical No. {no} - {title}")
    add_body(doc, "Question", question)

    add_subheading(doc, "Key Questions")
    for item in key_questions:
        add_body(doc, "Q", item["question"])
        add_body(doc, "Answer", item["answer"])

    add_subheading(doc, "Analysis")
    add_body(doc, "A", analysis)

    add_subheading(doc, "Interpretation")
    add_body(doc, "I", interpretation)

    for label, code in sources:
        add_code_heading(doc, label)
        add_code_block(doc, code)



def configure_document(doc: Document):
    section = doc.sections[0]
    section.page_width = Inches(8.5)
    section.page_height = Inches(11)
    section.top_margin = Inches(1)
    section.bottom_margin = Inches(1)
    section.left_margin = Inches(1)
    section.right_margin = Inches(1)
    section.header_distance = Inches(0.45)
    section.footer_distance = Inches(0.45)

    normal = doc.styles["Normal"]
    normal.font.name = "Calibri"
    normal.font.size = Pt(11)



def practical_data():
    style_css = read_text("style.css")
    return [
        (
            1,
            "Home Page (index.html)",
            "Create the home page of the Student Hub Portal using HTML and an
external CSS file so the page shows a header, navigation bar, hero area, notice
section, quick links, and footer.",
            [
                {
                    "question": "How does the header help the user identify the
portal quickly?",
                    "answer": "The header shows the logo and portal name at the
top, so the user immediately knows which site they are viewing.",
                },
                {
                    "question": "What navigation links make the home page easy to
explore?",
                    "answer": "The menu links to Home, About, Events, FAQ, Contact,
and Feedback, which gives direct access to the main sections of the portal.",
```

```
                },
                {
                    "question": "How do the notice, events, and quick links
sections organize the content?",
                    "answer": "They split the main content into three clear cards,
so updates, upcoming activities, and shortcuts are easy to read and find.",
                },
            ],
            "This page uses a clear visual hierarchy: a branded header, centered
hero message, three content cards, and a footer. The external stylesheet keeps the
layout, colors, buttons, and spacing consistent.",
            "The home page acts as the landing page for the portal. It should give
a fast overview and direct the user toward login, registration, events, and other
major sections.",
            [
                ("HTML Code - index.html", read_text("index.html")),
                ("CSS Code - style.css", style_css),
            ],
        ),
        (
            2,
            "About Page (about.html)",
            "Create the About page to present the mission and vision of Student Hub
Portal in a simple, readable layout.",
            [
                {
                    "question": "What is the main purpose of the About page?",
                    "answer": "The About page explains what the portal is for and
gives the user a short introduction to the project.",
                },
                {
                    "question": "How are the mission and vision separated for
clarity?",
                    "answer": "They are placed in separate boxes, which keeps each
idea distinct and easy to compare.",
                },
                {
                    "question": "Why is a two-card layout useful for this
content?",
                    "answer": "A two-card layout keeps the page balanced and lets
the reader focus on one message at a time.",
                },
            ],
            "The About page is written in a simple card layout so the mission and
vision can be read independently. The structure is short, direct, and easy to
scan.",
            "The page communicates what the portal stands for and what it aims to
achieve. It supports trust and gives the project a clear identity.",
            [("HTML Code - about.html", read_text("about.html"))],
        ),
        (
            3,
            "Events Page (events.html)",
            "Create the Events page so students can search events, view event
details, and register from a clean table layout.",
            [
                {
                    "question": "What information does each event card provide?",
                    "answer": "Each event card gives the event name, date, venue,
category, and a register button, so the user sees all important details at once.",
                },
                {
                    "question": "How does the search box support event discovery?",
```

```
                    "answer": "It lets the student type a keyword and quickly look
for a specific event instead of scanning the full list manually.",
                },
                {
                    "question": "Why is a table layout suitable for this page?",
                    "answer": "A table keeps the event details aligned in rows and
columns, which makes the page neat and easy to compare.",
                },
            ],
            "The Events page combines a search field with a structured event
listing. Each row gives the event title, date, venue, category, and a register
button for quick action.",
            "This page is meant for browsing and action. The design helps users
find an event and register without needing to open multiple pages.",
            [("HTML Code - events.html", read_text("events.html"))],
        ),
        (
            4,
            "FAQ Page (faq.html)",
            "Create the FAQ page and answer the common questions students may ask
about registration, login, events, and contact options.",
            [
                {
                    "question": "Which student problems are answered on this
page?",
                    "answer": "The page answers common doubts about registration,
login, joining events, and contacting the admin team.",
                },
                {
                    "question": "How does the FAQ reduce the need for support
messages?",
                    "answer": "It gives simple answers to repeated questions, so
students can solve basic issues without sending a new query every time.",
                },
                {
                    "question": "Why are the questions written in a direct
format?",
                    "answer": "Direct questions match the way students usually ask
for help, so the answers are easier to understand quickly.",
                },
            ],
            "The FAQ page uses repeated question-answer blocks separated by
horizontal lines. This gives the page a simple reading path and keeps the answers
easy to locate.",
            "The page works as a support section. It helps students solve common
doubts on their own and keeps the portal easier to use.",
            [("HTML Code - faq.html", read_text("faq.html"))],
        ),
        (
            5,
            "Contact & Feedback Page (contact.html)",
            "Create the Contact and Feedback page so users can find contact details
and submit feedback through a form.",
            [
                {
                    "question": "What contact details are shown first on the
page?",
                    "answer": "The page shows the email address, phone number, and
campus address first so the user can contact support quickly.",
                },
                {
                    "question": "How does the feedback form encourage user
response?",
```

```
                    "answer": "The form is short and simple, so the user can send a
message without spending too much time.",
                },
                {
                    "question": "Why are contact details and feedback kept in
separate sections?",
                    "answer": "Separating them keeps the page organized and makes
it clear whether the user is reading information or submitting a response.",
                },
            ],
            "The page is split into a contact information card and a feedback form
card. This separation helps the user either read details quickly or take action by
sending a message.",
            "The page serves both communication and collection purposes. It gives
the user a way to reach support and to share suggestions in a structured form.",
            [("HTML Code - contact.html", read_text("contact.html"))],
        ),
        (
            6,
            "Login Page (login.html)",
            "Create a login page for the Student Hub Portal with fields for email
and password, a remember-me option, a login button, and helpful navigation links.",
            [
                {
                    "question": "What input fields are required to sign in?",
                    "answer": "The user must enter an email address and password
before submitting the login form.",
                },
                {
                    "question": "How does the remember-me option improve
usability?",
                    "answer": "It allows the user to stay signed in on the device,
which makes future logins faster and easier.",
                },
                {
                    "question": "Why is there a forgot-password link below the
form?",
                    "answer": "The link helps users recover access if they forget
their password, which makes the page more practical.",
                },
            ],
            "The login form is kept compact so the user can enter credentials
quickly. The checkbox and support links make the page feel complete without
becoming crowded.",
            "The page focuses on authentication. It gives users a direct path into
the dashboard while keeping the form simple and familiar.",
            [("HTML Code - login.html", read_text("login.html"))],
        ),
        (
            7,
            "Student Registration Page (register.html)",
            "Create a student registration form that collects personal details,
course information, gender, password confirmation, and terms acceptance before
submission.",
            [
                {
                    "question": "Which student details are collected in the form?",
                    "answer": "The form collects name, email, mobile number,
course, semester, gender, password, and confirm password.",
                },
                {
                    "question": "Why are password and confirm-password fields both
included?",
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                    "answer": "Both fields are included to reduce typing mistakes
and make sure the student enters the correct password.",
                },
                {
                    "question": "How does the terms checkbox make the form more
complete?",
                    "answer": "The checkbox confirms that the user agrees to the
terms and conditions before creating the account.",
                },
            ],
            "The registration page contains a larger form because it collects
several student details. The fields are arranged in a straight order so the form
remains easy to fill in.",
            "This page captures the information needed to create a student account.
The design supports accuracy by asking for all required details before
submission.",
            [("HTML Code - register.html", read_text("register.html"))],
        ),
        (
            8,
            "Dashboard Page (dashboard.html)",
            "Create the student dashboard page to display menu options, summary
cards, upcoming events, and important notices after login.",
            [
                {
                    "question": "What summary information is shown on the
dashboard?",
                    "answer": "The dashboard shows the total events count,
registered count, upcoming event, and latest notice.",
                },
                {
                    "question": "How does the sidebar menu help navigation?",
                    "answer": "The sidebar gives quick buttons for Home, Profile,
Events, Notices, and Logout, so movement inside the portal is simple.",
                },
                {
                    "question": "Why is the dashboard split into multiple boxes?",
                    "answer": "Separate boxes make the information easier to scan
and help the important items stand out clearly.",
                },
            ],
            "The dashboard mixes navigation and live summary blocks. It is designed
like a control panel so the user can move around the portal and see important
updates at a glance.",
            "The dashboard is the main post-login workspace. It gives the student a
fast view of the most important information and action points.",
            [("HTML Code - dashboard.html", read_text("dashboard.html"))],
        ),
        (
            9,
            "Profile Page (profile.html)",
            "Create the profile page to show the student photo area, personal
details, and an Edit Profile button in a centered layout.",
            [
                {
                    "question": "What student data is shown on the profile page?",
                    "answer": "The page shows the student's name, email, mobile
number, course, and semester.",
                },
                {
                    "question": "How does the profile picture placeholder support
the layout?",
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```
                    "answer": "It gives a visible profile area even when no image
is uploaded, so the page still looks complete.",
                },
                {
                    "question": "Why is an edit button useful on this page?",
                    "answer": "The edit button lets the user update profile
information, which makes the page more useful and interactive.",
                },
            ],
            "The profile page presents a centered information card with personal
details and a clear action button. The layout is simple, which makes the profile
feel organized and readable.",
            "The page represents the student identity inside the portal. It
provides a quick summary of the user and allows profile updates when needed.",
            [("HTML Code - profile.html", read_text("profile.html"))],
        ),
        (
            10,
            "Admin Panel (admin.html)",
            "Create the admin panel page to show summary counters, recent
activities, and navigation for administration tasks.",
            [
                {
                    "question": "Which portal statistics are shown for the admin?",
                    "answer": "The panel shows the number of students, events,
feedback entries, and notices.",
                },
                {
                    "question": "How do summary counters help with monitoring?",
                    "answer": "They give a quick numerical view of portal activity,
which helps the admin understand the current status at a glance.",
                },
                {
                    "question": "Why is a recent activity list useful for
administration?",
                    "answer": "It highlights the latest actions on the portal, so
the admin can track changes and activity easily.",
                },
            ],
            "The admin panel uses compact summary boxes and a simple activity list.
This helps the administrator track the portal status without needing a complex
interface.",
            "The page is intended for monitoring and control. It gives a quick
operational overview and supports administrative decision-making.",
            [("HTML Code - admin.html", read_text("admin.html"))],
        ),
        (
            11,
            "Common Stylesheet (style.css)",
            "Create the external stylesheet used across the Student Hub Portal
pages so the navigation, buttons, cards, forms, and footer share one consistent
visual style.",
            [
                {
                    "question": "Which shared visual elements are controlled by the
stylesheet?",
                    "answer": "The stylesheet controls the body colors, header,
navigation, main cards, buttons, forms, links, and footer styles.",
                },
                {
                    "question": "How does the stylesheet keep the site design
consistent?",
```

```
                    "answer": "It applies the same fonts, colors, spacing, and
button style across all pages, so the project looks unified.",
                },
                {
                    "question": "Why is using one CSS file better than repeating
styles in each page?",
                    "answer": "One CSS file is easier to manage, reduces
duplication, and makes future edits much faster.",
                },
            ],
            "The stylesheet defines the site color palette, layout spacing, button
behavior, card styling, and form appearance. It gives all pages the same visual
identity.",
            "The CSS file is the design backbone of the project. It reduces
repetition, keeps styling consistent, and makes future changes easier to manage.",
            [("CSS Code - style.css", style_css)],
        ),
    ]



def practical_data_upto_5():
    style_css = read_text("style.css")
    script_js = read_text("script.js")
    html_files = [
        "index.html", "about.html", "events.html", "faq.html", "contact.html",
        "login.html", "register.html", "dashboard.html", "profile.html",
"admin.html",
    ]
    semantic_code = [(f"HTML Code - {name}", read_text(name)) for name in
html_files]
    return [
        (
            1,
            "Project Initiation, Requirement Analysis, Sitemap, Wireframe, and
GitHub Setup",
            "Initiate the semester-long StudentHub portal by identifying the
problem scope, user roles, key modules, navigation flow, and minimum 10 pages.
Create a sitemap, low-fidelity wireframe, project folder structure, README file,
and GitHub repository.",
            [
                {"question": "What is URL and parts of URL?", "answer": "A URL is
the address used to open a resource on the internet. Its main parts are the
protocol such as https, domain name, path, query string, and fragment."},
                {"question": "How does HTML file processed in web browser?",
"answer": "The browser reads the HTML file, builds the DOM tree, loads linked CSS
and JavaScript files, and then displays the final page on the screen."},
                {"question": "How will page navigation flow managed among all HTML
pages?", "answer": "Navigation is managed with anchor tags and relative links such
as index.html, about.html, events.html, and register.html. A common navigation bar
keeps the flow consistent."},
                {"question": "How will GitHub commits be maintained after each
practical?", "answer": "After completing and testing each practical, the changed
files should be added, committed with a clear message, and pushed to the GitHub
repository."},
            ],
            "The Student Hub project is divided into clear pages for home,
information, events, support, authentication, student details, and administration.
Separate HTML, CSS, and JavaScript files keep the work organized.",
            "A sitemap and wireframe give the project a clear direction before
coding starts. This reduces confusion about page connections and makes later GitHub
commits easier to understand.",
```

```
            [("Project File Structure",
"index.html\nabout.html\nevents.html\nfaq.html\ncontact.html\nlogin.html\nregister.
html\ndashboard.html\nprofile.html\nadmin.html\nstyle.css\nscript.js")],
        ),
        (
            2,
            "Semantic HTML5 Pages with Accessibility-Ready Structure",
            "Develop static HTML5 skeletons for at least 10 StudentHub pages such
as Home, About, Register, Login, Dashboard, Events, Profile, Contact, Admin, FAQ,
and Feedback. Use semantic tags and accessibility-friendly structure.",
            [
                {"question": "Are semantic tags such as header, nav, main, section,
article, aside, and footer used properly?", "answer": "Yes. The pages use header
for branding, nav for links, main for page content, sections or cards for grouped
information, and footer for common bottom links."},
                {"question": "Are form labels, headings, image alt text, and
navigation links accessible?", "answer": "The form fields have labels, the pages
use headings, the logo has alt text, and navigation is provided through normal
links. These features help keyboard and screen-reader users understand the page."},
                {"question": "Is the page structure consistent across all pages?",
"answer": "Yes. The pages follow a common pattern with a header, navigation menu,
main content area, footer, and linked CSS and JavaScript files."},
            ],
            "The project has ten linked HTML pages with a common layout. Semantic
elements make the structure easier for browsers, developers, and assistive
technologies to understand.",
            "The pages form a consistent static website foundation. A user can move
from the home page to information, events, registration, login, and support pages
without losing the common site structure.",
            semantic_code,
        ),
        (
            3,
            "Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind",
            "Design responsive layouts for Home, About, Registration, Dashboard,
and Events pages using CSS Grid and Flexbox. Ensure mobile-first responsiveness.",
            [
                {"question": "How does the layout adapt for mobile, tablet, and
desktop screens?", "answer": "Flexbox allows items to wrap and share available
space, while media queries change widths and the navigation layout for smaller
screens. The same content can fit different screen sizes."},
                {"question": "Which layout technique is used and why?", "answer":
"Flexbox is used for horizontal groups such as the header, navigation, form grid,
and cards. It is suitable because it aligns items and distributes space with less
code."},
                {"question": "Are typography, spacing, colors, and contrast
readable?", "answer": "The stylesheet uses a dark background, light text,
contrasting yellow buttons, consistent padding, and readable font sizes. These
choices make the important content visible."},
                {"question": "Is the framework used consistently without
unnecessary inline CSS?", "answer": "The project mainly uses one external style.css
file. Some page-specific inline styles are present, but the repeated site-wide
styles are kept in the common stylesheet."},
            ],
            "The CSS file defines the common color palette, cards, buttons, forms,
navigation, and responsive rules. Flexbox is used for groups of elements and media
queries adjust the layout for smaller screens.",
            "The interface is designed to remain usable on desktop and mobile
screens. Centralizing the main styles makes the pages look like one portal instead
of separate unrelated pages.",
            [("CSS Code - style.css", style_css)],
        ),
        (
```

```
            4,
            "JavaScript DOM Manipulation, Event Handling, and UI Interactivity",
            "Add dynamic UI components such as collapsible FAQ, modal popup,
image/content slider, notification banner, hamburger menu, and light/dark theme
switcher using JavaScript.",
            [
                {"question": "How are DOM elements selected and modified?",
"answer": "The script uses getElementById, querySelector, and querySelectorAll. It
changes class names, text, display styles, and the current slide to update the
page."},
                {"question": "Are event listeners attached correctly?", "answer":
"The HTML buttons call functions for clicks, and a DOMContentLoaded listener starts
the first slider. A window click handler also closes a modal when the user clicks
outside it."},
                {"question": "Is localStorage used for remembering theme choice?",
"answer": "Yes. The selected theme is stored in localStorage, and the saved value
is checked when the page loads so the preference can be restored."},
                {"question": "Does interactivity improve usability without breaking
accessibility?", "answer": "The features make the page easier to use by hiding
repeated information, showing announcements, and supporting mobile navigation.
Clear buttons and visible text keep the controls understandable."},
            ],
            "The JavaScript file contains small functions for theme switching, menu
display, banner closing, FAQ expansion, modal display, and slider movement. Each
function changes only the DOM elements needed for that feature.",
            "The portal becomes interactive without a server. Users can control the
menu, announcements, FAQ answers, theme, and slider directly in the browser.",
            [("JavaScript Code - script.js", script_js)],
        ),
        (
            5,
            "Registration Form with Frontend Validation and User-Friendly Error
Handling",
            "Create a student registration form with HTML5 input types and
JavaScript validation for name, email, mobile number, password, confirm password,
course, year, gender, and terms acceptance. Use Regular Expression for
validations.",
            [
                {"question": "Are correct input types and attributes used?",
"answer": "Yes. The form uses text, email, tel, number, password, select, radio,
and checkbox controls. Labels, placeholders, min and max values, and validation
checks support correct input."},
                {"question": "Are validation errors displayed near the relevant
fields?", "answer": "Yes. Each important field has a nearby error element such as
nameError or emailError. The validation function writes the message there and adds
an input-error class."},
                {"question": "Is password strength checked?", "answer": "Yes. A
regular expression checks that the password has at least six characters, at least
one letter, and at least one number. The confirm password is also compared with
it."},
                {"question": "Is the form accessible using keyboard and screen-
reader-friendly labels?", "answer": "The controls are connected to visible labels
using for and id values, and the radio buttons have a shared name. This gives
keyboard and assistive-technology users understandable fields."},
            ],
            "The registration form collects the required student details and
validates them before accepting the form. Invalid fields receive a message and an
error class, while valid fields receive a success class.",
            "Client-side validation gives immediate feedback and prevents
incomplete or incorrectly formatted data from being submitted. Server-side
validation would still be needed in a real application.",
            [("HTML Code - register.html", read_text("register.html")),
("JavaScript Code - validation in script.js", script_js)],
```

```
        ),
    ]


def main():
    doc = Document()
    configure_document(doc)
    add_title_page(doc)

    for i, item in enumerate(practical_data_upto_5()):
        if i:
            doc.add_page_break()
        add_practical(doc, *item)

    doc.save(OUT_DOCX)
    print(f"Saved {OUT_DOCX}")


if __name__ == "__main__":
    main()
```